unicef Office of Research-Innocenti

unicef for every child

Multidimensional Child Poverty in Montenegro

Understanding the complex realities of children in poverty using a mixed-method approach

This report was prepared by Alessandro Carraro, Maja Gavrilovic, Marija Novkovic, Stevan Stanisic, Danilo Smolovic with contributions from the UNICEF Montenegro Country Office team. The report was edited by Alison Raphael.

The UNICEF Office of Research – Innocenti is UNICEF's dedicated research centre. It undertakes research on emerging or current issues in order to inform the strategic direction, policies and programmes of UNICEF and its partners, shape global debates on child rights and development, and inform the global research and policy agenda for all children, and particularly for the most vulnerable.

The findings, interpretations and conclusions expressed in this paper are those of the authors and do not necessarily reflect the policies or views of UNICEF.
This paper has been reviewed by an internal UNICEF panel consisting of staff with subject-matter or methodological expertise.

For readers wishing to cite this document, we suggest the following form:

Carraro A., Gavrilovic M., Novkovic M., Stanisic S., Smolovic D., (2020). Multidimensional Child Poverty in Montenegro – Understanding the complex realities of children in poverty using a mixed-methods approach, UNICEF Office of Research – Innocenti and UNICEF Montenegro

No conflicts of interest were reported by the authors.

Cover photo: © UNICEF Montenegro – Dusko Miljanic

Design and Layout: Pixella Ltd.

Acknowledgements

The report was realized in close cooperation between researchers at the UNICEF Office of Research – Innocenti in Florence and the UNICEF Montenegro Office.

The research team would like to thank the following colleagues and partners for their contribution to the research design and without whom it would not have been possible to complete this study: Olivera Komar and the De Facto consultancy team who collected data, Gwyther Rees, Kat Chzhen, and Lucia Ferrone as peer reviewers, as well as the Statistical Office of Montenegro - MONSTAT, the Ministry of Finance and Social Welfare, the Ministry of Education, Science, Culture and Sports, the Ministry of Health, the Ministry of Justice and Minority Rights, the Institute for Public Health, the Institute for Social and Child Protection, Ombudsperson's Office, the Parliament (Parliamentary Committee on Human Rights and Freedoms & the Parliamentary Committee on Health, Labour and Social Welfare), the University of Montenegro- Faculty of Political Sciences, the Red Cross, NGO Nardos, NGO Parents, UNDP, and UNHCR.

Most of all, we owe a special debt of gratitude to the interviewed children, parents, caregivers and social service providers.

Content

Tables, figures and boxes

Tables

Figures

Boxes

Acronyms

CRC	Convention on the Rights of the Child
ECD	Early childhood development
EU	European Union
FGD	Focus group discussion
GDP	Gross domestic product
MICS	Multiple indicator cluster survey
MM-MODA	Mixed methods-MODA
MODA	Multiple overlapping deprivation analysis
MONSTAT	National Statistical Office of Montenegro
NGOs	Non-governmental organizations
NSSD	National strategy for sustainable development
PISA	Programme for International Student Assessment
SDGs	Sustainable Development Goals
SILC	Survey on income and living conditions
UN	United Nations
UNICEF	United Nations Children's Fund
UNHCR	United Nations High Commission for Refugees

FOREWORD

The world cannot afford a lost generation of youth, their lives set back by COVID-19 and their voices stifled by a lack of participation. Let us do far more to tap their talents as we tackle the pandemic and chart a recovery that leads to a more peaceful, sustainable and equitable future for all.

UN Secretary-General António Guterres, 2020

Child poverty is pervasive – affecting the least-developed as well as middle- and high-income countries. Even the unparalleled prosperity of the 21st century has not erased it, and the impact of COVID-19 is making it worse[1]. Growing up in poverty is a violation of children's rights, stifling children's growth, development and health and harming their learning potential and long-term prospects. The consequences of child poverty are felt at the individual level, but also affect the future of societies that rely on citizens' participation and contribution to productivity.

Now more than ever - as COVID-19 disrupts lives and livelihoods and slows economic growth - tackling child poverty is an urgent priority. The present study, Multidimensional Child Poverty in Montenegro, provides the knowledge required to formulate an appropriate national policy response. It reveals how child poverty and exclusion differ from deprivations faced by adults. It explains how children of different ages experience poverty – and what is needed to address these needs. It highlights existing poverty profiles in Montenegro and how different dimensions of deprivation combine and interlink. Understanding these issues is crucial to designing an effective response to child poverty – both to guarantee children's human rights and to lay the groundwork for sustainable development. The study concludes with policy recommendations to guide the response by the government and its development partners and provides concrete suggestions on how best to invest in children.

The methodology used for the study is based on the multiple overlapping deprivation analysis (MODA) methodology developed by UNICEF's Global Office of Research - 'Innocenti'. MODA is a tool that envisions poverty as a multi-faceted phenomenon – well beyond monetary poverty and including other rights dimensions such as access to education, health and protection. The decision to analyse multidimensional child poverty in Montenegro resulted from discussions with representatives of the government, parliament, civil society and international organizations, who jointly defined the concept and its different dimensions in the context of Montenegro. For the first time, the original MODA methodology was enriched with qualitative insights, giving voices to children and their parents.

The study includes touching personal accounts of how children and adolescents experience and cope with poverty in their daily lives, providing a clear picture of how it feels to be a poor child in Montenegro. The research also confirms that the stigma related to poverty has devastating effects on the lives of poor children – it motivates their decisions to drop out of school, to avoid denigration by peers and teachers and to give up on cherished aspirations.

The MODA demonstrates that young children are the most deprived age group in Montenegro. The early years are crucial for a child – a time when 90 per cent of the brain develops and they acquire the ability to form meaningful human relationships. If young children in poverty fail to receive the support they need to reach their full potential, their early struggles are likely to continue throughout their lifetimes. When this pattern is repeated with their children, a vicious circle of inter-generational poverty continues. To break the cycle, strategic investments in children are required, especially during the decisive early years. Investing in children is crucial to breaking the cycle of poverty and also represents the soundest economic investment in sustainable, inclusive and equitable economic growth for Montenegro's aging society.

By providing hard evidence and enhancing understanding of child poverty across Montenegrin society, this study can mobilize action, stimulate public debate to inform decisions on tackling multi-dimensional child poverty and deprivations, that can help children and families to overcome the debilitating impacts of the COVID-19 pandemic.

UNICEF Montenegro will continue to work with the government, parliament and partners from the private sector, civil society and the media, to address child poverty in Montenegro in all its dimensions – for the benefit of healthy, happy, well protected and educated children and a prosperous Montenegro.

This comprehensive analysis of multidimensional child poverty in Montenegro would not have been possible without the partnership of the Government of Montenegro and MONSTAT, as well as other stakeholders. We also acknowledge the excellent contribution made by the consultancy team that worked in the field to collect qualitative data.

Juan Santander,
Representative,
UNICEF Montenegro

Gunilla Olsson, Director,
UNICEF Office of Research
Innocenti, December 2020

1 This study was based on data gathered prior to the outbreak of COVID-19 and does not explore the pandemic's impact on child poverty.

As much as it is the most beautiful thing to write about a happy and satisfied childhood and youth, it is the most difficult thing for every person to consider the challenges that the young generations face and to find ways to overcome them. However, that is one of the main obligations of all of us who have the opportunity to shape the future from a decision-making position.

The wellbeing and development of children and young people is a key stimulus and the main driving force of any society, and the success of any society – including ours – rests on a continuous commitment to children and young people and on working with them. Therefore, we will strive for our young people to have optimal living conditions (education, security, peace, health and respect for human rights and freedoms) in which they will develop their potential and, successfully and in a quality way, become happy members of their families, local communities and society as a whole.

We are aware that states, including Montenegro, face obstacles that prevent children from exercising their rights proclaimed in the Convention on the Rights of the Child, and that the economic underdevelopment of the state hits children with one of the worst plagues – poverty. Unfortunately, this results in inequality and negatively affects the position and needs of children and young people. Let us also remind ourselves of the necessity imposed on us by this situation now, when together with the rest of the world we are facing the economic consequences of the COVID-19 pandemic, which are becoming increasingly pronounced. Through measures, programmes and laws we must anticipate higher financial allocations and services that will directly or indirectly affect children and that are crucial for poverty reduction. It is not easy to provide all this in times of crisis, but we must and we will, for the sake of the children of Montenegro, our young fellow citizens, but also for those who are yet to be born.

Structured and comprehensive studies, containing clear guidelines and constructive recommendations for future action, are key instruments that every society should use to formulate and improve the public policies and laws that affect the most vulnerable groups, including children. Thus, this multidimensional study, which represents "research on the different ways in which poverty manifests itself among children in Montenegro – affecting their health, education, security and future human development", provides significant insight into the future development of society and has the potential to serve as a guide to decision makers in defining strategic goals that most effectively improve the social position of this vulnerable population group.

The statistics presented in this study indicate that 23.8% of the population in Montenegro is at risk of poverty, that the relative poverty rate is highest among children aged 0 to 17 (32%), and that "a child who is born today in Montenegro will reach only 62% of its potential productivity in adulthood due to the lack of quality education and health care". Such indicators are a real wake-up call for the whole society and impose the need for strategic directions of action in order to reduce poverty.

Therefore, in the coming period, it is necessary to draw the attention of the public to the issues that are most important in our lives – the life and development of children, the availability of education and health care, and the chance to enjoy basic human rights. One of the priorities is intensified activity to strengthen the legal and political environment in which the necessary reforms would be implemented, aimed at creating opportunities for children to exercise their rights in accordance with the Convention on the Rights of the Child and the EU Charter of Fundamental Rights. In line with the 2030 Agenda for Sustainable Development, adopted by the UN member states, including Montenegro, we will work to focus our national strategic plans on achieving the necessary conditions for prosperous development, in order to protect the rights of children and ensure their bright future.

A more certain future for young people, new jobs, higher salaries, pensions and social benefits remain our goal. We will achieve this goal by "arming ourselves" with knowledge and acting in unison. I call on decision makers to closely monitor and support programmes, measures and legal solutions to improve the conditions of children for education, to ensure freedom and equality, so that everyone equally realizes their potential and participates in strengthening democracy.

As for the Parliament of Montenegro and its competences, especially through the work of the Committee on Human Rights and Freedoms, I assure you that you can count on its readiness to cooperate more closely at any moment, all in the interest of the state and of all of its citizens.

Young people share the fate of all Montenegrin citizens. If the state and its economy are underdeveloped, if life is difficult, it is not logical that our young people will be able to be in a better position. If the parents of our youngest fellow citizens are not employed, we cannot expect the children from such families to be financially provided for. If conditions are created in the state to give adults the opportunity to work as equal members of society, their families will certainly be better off. And when the family lives well, this is beneficial for the wider community and for the entire state. Therefore, it is the obligation of all stakeholders in society, to the extent we can individually and jointly, to create the necessary conditions in which our young people will not lag behind their peers from the region, from EU member states and the developed world in terms of generating human capital. A peaceful, stable and prosperous future for our children and young people must not be called into question.

Aleksa Becic,
President of the Parliament of Montenegro

Photo: Duško Miljanić / UNICEF Montenegro

Executive summary

Overview

Efforts to measure poverty among children have traditionally relied on calculations of household income. But it has become clear that such a one-dimensional approach is inadequate for understanding the depth, breadth and consequences of child poverty. This report takes a different, far more nuanced, approach. As a study of *multidimensional* poverty, it explores the various ways in which poverty is manifested among Montenegro's children – affecting their health, education, safety and future human development. The approach, developed by the United Nations Children's Fund (UNICEF) and its Office of Research - Innocenti, is known as a 'multidimensional overlapping deprivation analysis' (MODA). Using data from surveys designed to measure the situation of children and households, it identifies the different ways that children experience poverty and how these overlap, creating a series of obstacles that prevent children from fulfilling their rights as defined in the United Nations Convention on the Rights of the Child (CRC), acceded to by Montenegro on 23 October 2006.

The reasoning behind this approach is that since children have multiple needs and rights, when deprivation of these rights occurs in several areas at once, children's development is hindered.

This phenomenon requires that policymakers and decision-makers pay increased attention to *all dimensions of deprivation simultaneously*. For example, if a child lives in a dilapidated, unhealthy home it is not enough to undertake home repairs – the child's health is likely to be affected as is his/her self-esteem when, for example, the child cannot bathe or wash clothes, leading to reluctance to attend school due to negative comments from peers. Poor health and school drop-out then create obstacles for upward mobility and increase the chances for inter-generational transmission of poverty.

The purpose of the report is to provide Montenegro's policy- and decision-makers with concrete data upon which to base future efforts to eliminate child poverty. For when segments of the population are unable to escape from poverty, it weighs down and threatens the future prospects of the entire society.

The present study uses data from household surveys conducted in Montenegro in 2018 as the basis for calculating the extent of multidimensional child poverty in the country. In 2018 UNICEF collaborated with the country's national statistics office (MONSTAT) to gather data for two surveys: one for Montenegro's general population and a second for the population of Roma settlements. Based on the results of these multi-indicator cluster surveys (MICS), this report calculates the extent of deprivations experienced by children of different ages and different circumstances (e.g.,living in a rural or urban area, living in a single-parent household) and illustrates how they overlap.

An innovation of the present report is that it goes beyond the strictly quantitative approach to offer information gathered from Montenegrins through focus group discussions and in-depth interviews, augmenting the mathematical calculations with the personal perceptions and experiences of children, parents and professionals working with poor populations. It is thus a 'mixed-method' MODA, relying on both quantitative and qualitative data and information.

The report's methodology is explained in detail in the sections below. In short, it should be noted that all definitions of the various dimensions of child poverty were arrived at in an inclusive manner during workshops convened by UNICEF in 2019, involving representatives from Montenegro's government, the Parliament, and civil society as well as international organizations working in the field. Through this collaboration, deprivation dimensions for households were defined: *housing*, *water and sanitation*, *ability to pay utility bills* and *access to information* (e.g., internet connection). For children specifically, the dimensions measured were: *health*, *nutrition*, *early child development* (ECD) and *education*, *neglect and discipline* (child protection), and *child labour* (domestic chores and other work performed).

The study examined two age groups: children under five years of age (0–5) and children aged 5–17. In some cases, special attention was paid to children under two years of age (0–23 months) since their critical nutritional needs differ from those of older children.

Key findings

Single deprivation analysis:

The report first presents results for single deprivations in all of the categories mentioned above, for both the general population of Montenegro and for its Roma population.

General population:

- For children aged 0–23 months, the greatest deprivation is for *nutrition*.
- Deprivation of *protection* (exposure to violent discipline or neglect) ranks highest for children in the other two age groups.
- Deprivation in the domains of *education, water and sanitation, utilities and information* is generally high among children living in rural areas.
- Deprivations in *health* and *child protection* are more prevalent in urban than rural areas.

The report also analysed some of the background characteristics that increase children's risk of deprivation. With regard to single deprivations in the general population it found that: children aged 0–5 years who live in female-headed households experience greater deprivation in all dimensions except child protection, and children living in households where mothers have only a primary school education are more deprived in all dimensions, especially *water and sanitation*.

Roma population:

- Extremely high levels of deprivation were found for all age groups in most dimensions, especially with regard to *living conditions*.
- Among children aged 0–23 months, deprivations were particularly high for *nutrition and ECD*.
- For older children, difficulties related to access to education and educational achievement were widespread.

With regard to background characteristics, the report found that children living in rural areas generally experience greater deprivation than those in urban areas, irrespective of their parents' education level. Overall for this population, children's risk of deprivation appears to be more linked to where they live than to any specific characteristic of the head of household (e.g., sex, age, education level). Children living in both the northern and southern regions are significantly more likely to be deprived in four out of six dimensions (*ECD, water and sanitation, housing and utilities*) than those living in the central region.

Multiple overlapping deprivation analysis

General population:

- About 20 per cent of Montenegrin children aged 0–5 years do not suffer from any deprivation.
- Children aged 0–23 months experience the highest prevalence of multidimensional child poverty: 77.6 per cent of children in this age group suffer from two or more deprivations. About 64.3 per cent of children aged 0–59 months and 49.1 per cent of those aged 5–17 years are deprived in multiple dimensions.

- For children aged 0–23 months, *nutrition* has the most frequent overlap with other dimensions. *Child protection* is the dimension overlapping the most for children under 5 years and those aged 5–17 years.
- More than 80 per cent of Montenegrin children experience deprivation in at least one domain of well-being, regardless of age; younger children usually experience simultaneous deprivation in more dimensions than older children.

Roma population

- *ECD and housing* show the most significant overlap for children under 5 and those 5–17 years of age, except that for children aged 0–23 months the greatest overlap with other dimensions of deprivation is *nutrition*.
- Almost two-thirds of children under 2 years of age (65 per cent) experience simultaneous overlapping deprivations for *nutrition, ECD* and *housing*.
- Almost all children are deprived in two or more dimensions of wellbeing. Overall, children who experience three or more deprivations simultaneously tend to live in extreme poverty.

Background characteristics that increase the likelihood of multi-dimensional deprivation for Roma children are: living in the northern region and living in a female-headed household. Qualitative interviews with children, families and others also verified that: poor housing is associated with both deprivations in water and sanitation and poor health outcomes for Roma children; hunger resulting from poverty affects children's health and school performance; segregation of the Roma populations in Roma settlements or designated housing leads to social exclusion; and the expectation among many Roma households that children should begin to work at a young age can impede their right to education and child protection.

Beyond MODA

The report addresses two other important issues – the coping strategies used by families affected by poverty to meet household needs and how multidimensional poverty is understood and experienced by adolescents – relying mainly on qualitative research.

Parents in low-income households rely heavily on two main sources of assistance: borrowing from family members or others and assistance from the state. Another main coping strategy, shared by children and adolescents, is to limit spending and consumption on food, clothing and other goods and services. This practice affects several dimensions of deprivation, particularly nutrition. It also affects education when children lack adequate clothing and shoes to attend school, results in stigma and lowers self-esteem among poor children. Another coping mechanism used by low-income households is to involve older children in various forms of child labour.

Among the key concerns reported by adolescents were their inability to access information on sexual and reproductive health (especially in rural areas), the lack of mental health services and the impact of violence, peer pressure and substance abuse.

Key findings

- Shame and stigma significantly affect the lives of poor children and adolescents
- Social exclusion and discrimination constrain poor children's access to quality services and opportunities
- Education and employment are viewed by parents and children as a key avenue for overcoming poverty
- Poor families' reliance on social welfare may entrap children in intergenerational poverty.

Recommendations

The report strongly stresses the importance of tackling multidimensional child poverty through an integrated, cross-sectoral approach that takes into consideration the many ways that poverty manifests itself among and affects children of all ages. Because young children are especially vulnerable to poverty in many dimensions it will be important to focus on their needs, particularly to ensure that they are properly nourished and can attend pre-school. Violence in the home and community also affects all children; programmes are needed to educate parents, communities and children about violence reduction. Problems related to monetary poverty and poor housing conditions result in stigma and discrimination, which in turn contribute to school drop-out. Thus social protection and housing policies should be carefully reviewed in light of their long-term impact on children, especially Roma children. The combination of deprivations affecting the adolescent population has resulted in a variety of factors that put their mental health at risk, calling for more targeted and accessible mental health services. Finally, well-designed social protection programmes that integrate complementary services for child protection, health, education and employment, can empower and enhance the capacity of poor families and their children to lead independent and productive lives.

Photo: Duško Miljanić / UNICEF Montenegro

1. Introduction

Country context

Montenegro is an upper-middle-income country situated in south-eastern Europe, with a territory of 13,812 square kilometres. It has a coast on the Adriatic Sea to the south-west and is bordered by Croatia to the west; Bosnia and Herzegovina to the north-west; Serbia to the north-east; Kosovo (under UNSCR 1244) to the east; and Albania to the south-east. The capital and largest city is Podgorica, although Cetinje is the former royal capital. Montenegro is divided into 24 municipalities and three regions: northern, central and southern.

According to the 2011 census, the total population of Montenegro was 620,029: 306,236 men (49.4 per cent) and 313,793 women (50.6 per cent). Children under the age of 18 numbered 145,126, or 23.4 per cent of the total population: 75,367 boys (51.9 per cent), 69,759 girls (48.1 per cent). The population was increasing at a rate of 1.4 per cent. According to available census data, 6,251 persons (1.01 per cent of the total population) declared Roma nationality[1]. Montenegro also contains a small population of Egyptians (just over 2,000) many of whom face problems similar to those of the Roma. The qualitative portion of this study included adults and children from both minority ethnic populations.

1 The largest number of Roma live in Podgorica (3,988), followed by Berane (531), Niksic (483), Bijelo Polje (334) and Herceg Novi (258), while the majority of Egyptians live in Podgorica (685), Niksic (446), Tivat (335) and Berane (170).

Within Montenegro there is broad national and political consensus in favour of accession to the European Union (EU). The country applied for EU membership in December 2008. After seven years of accession negotiations, 32 out of the 35 negotiating chapters had been opened, of which three have been provisionally closed. It is worth noting that the EU is the largest provider of financial assistance to Montenegro.[2] In 2019 Montenegro was ranked very high for human development for the fourth consecutive year.[3] Yet the data indicates that inequalities remain, due to poverty and regional disparities, as well as exposure to violence and social exclusion that negatively affect Montenegro's most vulnerable groups, including children.

The state has signed and ratified several United Nations conventions that directly or indirectly refer to children, among which are the: Convention on the Rights of the Child (CRC), Convention on the Elimination of All Forms of Discrimination against Women and the Convention on the Rights of Persons with Disabilities. Since then, the Government of Montenegro and national non-governmental organizations (NGOs) have prepared reports on progress toward implementation of these conventions, which were submitted to the relevant UN committees.

2 European Union, *Montenegro on its European path*, accessed in February 2020. URL: <https://ec.europa.eu/neighbourhood-enlargement/sites/near/files/near_factograph_montenegro.pdf>.

3 United Nations Development Programme 2019, *Beyond Income, Beyond Averages, Beyond Today: Inequalities in Human Development in the 21st Century*, UNDP, New York, 2019.

Over the last decade, Montenegro has made progress toward creating a legal and policy environment favourable for the implementation of reforms aimed at ensuring that children can exercise their rights in accordance with the CRC and the EU Charter on Fundamental Rights. In addition to improving national public policies and laws, in 2013 the state ratified the Optional Protocol to the CRC on the Communication Procedure. Montenegro also approved the 2015 global Sustainable Development Agenda. In 2016 the Government of Montenegro adopted the National Strategy on Sustainable Development (NSSD 2030), effectively incorporating the UN Agenda 2030 into Montenegrin policy goals. The NSSD is a comprehensive, long-term national development strategy that addresses not only the environment and the economy, but also building the human resources and social capital that are indispensable for prosperous development.

Montenegro experienced impressive economic growth during the last decade. In 2016 and 2017, real GDP growth was 2.9 and 4.7 per cent, respectively, while in 2018 the reported growth rate was 5.1 per cent.[4] The outlook remains positive, but external uncertainties present a major risk due to the country's high level of public debt, which has led to calls for fiscal consolidation, improved governance, accelerated structural reforms and more efficient public spending. Economic growth has, to some extent, led to improvements in the country's social welfare system. Yet despite these improvements and robust economic growth, the development process has not equally benefited the most vulnerable groups. Nonetheless, expenditures on social protection and social transfers (1.78 per cent of GDP in 2018) contributed to reducing poverty by 7 per cent.[5]

Economic inequality, as measured by the Gini coefficient, has slightly declined (38.5 in 2013 vs. 34.8 in 2018), but children remain disproportionately affected by poverty. A 2018 statistical report on income and living conditions found that 23.8 per cent of the population is at risk of poverty.[6] The relative poverty rate is highest for children aged 0–17 years (32 per cent) and those living in the north (40 per cent) or in rural areas (36 per cent). Montenegrin children lag behind their global peers in human capital formation. The World Bank estimates that a child born in Montenegro today will reach only 62 per cent of her/his potential adult productivity, due to lack of quality education and health care.[7]

Policy/strategy environment

Since regaining independence in 2006 the country has been striving to complete its transition to a functioning democratic society, for which the EU integration process has played a major role. Legal and institutional frameworks have been under continuous development, with support from international development partners. In addition, human rights treaty bodies have helped to shape Montenegro's legal and institutional environment through periodic monitoring of efforts by states parties and the resulting recommendations.

While the country does not have a national poverty-reduction strategy in place, the NSSD affirms Target 1 of the Sustainable Development Goals (SDGs): "By 2030, reduce at least by half the proportion of men, women and children of all ages living in poverty in all its dimensions according to national definitions". Moreover, several national plans and strategies address issues of paramount importance to children's wellbeing and development:

- National Action Plan for EU Acquis Chapter 23 – Judiciary and Fundamental Rights[8]
- National Action Plan for EU Acquis Chapter 24 – Justice, Freedom and Security[9]
- Strategy for exercising the rights of the child 2019–2023[10]

4 World Bank, *World Development Indicators*, World Bank, Washington D.C., 2020.

5 At-risk-of-poverty rate before social transfers: 31.4 per cent; at-risk-of-poverty rate after social transfers: 23.6 per cent. MONSTAT, 'Survey on Income and Living Conditions', MONSTAT, Podgorica, 2018.

6 Statistics on income and living conditions report 2018, MONSTAT, Podgorica, 2019 <http://monstat.org/userfiles/file/anketa%20o%20doahotku/SILC%20saopstenje%202019.pdf>.

7 Montenegro Human Capital Index, 2020 <https://databank.worldbank.org/data/download/hci/HCI_2pager_MNE.pdf>.

8 Government of Montenegro, 'The accession negotiations to the EU', <http://www.eu.me/mn/23>.

9 Ibid. , <http://www.eu.me/mn/24>.

10 Government of Montenegro, Ministry of Labour and Social Welfare, *Strategy on Exercising the Rights of the Child 2013-2023*, Ministry of Labour and Social Welfare, Podgorica, 2019. <http://www.mrs.gov.me/ResourceManager/FileDownload.aspx?rid=364844&rType=2&file=STRATEGIJA per cent20ZA per cent20OSTVARIVANJE per cent20PRAVA per cent20DJETETA per cent202019-2023.docx>.

- Strategy for developing a social and child protection system 2018–2022[11]
- Strategy for social inclusion of Roma and Egyptians in Montenegro 2016–2020[12]
- Strategy on early and preschool education in Montenegro 2016–2020[13]
- Strategy on inclusive education of children in Montenegro 2019–2025[14]
- 2015 Results and education policy recommendations by the Programme on International Student Assessment (PISA)[15]
- Strategy on prevention and protection of children from violence 2017–2021[16]

11 Government of Montenegro, Ministry of Labour and Social Welfare, *Strategy on Social and Child Protection System Development 2018-2022*, Ministry of Labour and Social Welfare, Podgorica, 2017. <http://www.mrs.gov.me/biblioteka/strategije>.

12 Government of Montenegro, Ministry of Human and Minority Rights, *Strategy for Social Inclusion of Roma and Egyptians in Montenegro 2016-2020*. Ministry of Human and Minority Rights, Podgorica, 2016. <https://www.rcc.int/romaintegration2020/docs/6/strategy-for-social-inclusion-of-roma-and-egyptians-in-montenegro-2016–2020–montenegro>.

13 UNICEF Montenegro and Montenegro Ministry of Education, *Strategy on Early and Preschool Education in Montenegro 2016-2020*, UNICEF Montenegro and Ministry of Education, Podgorica ,2016. <https://www.unicef.org/montenegro/en/reports/strategy-early-and-preschool-education-montenegro-2016-2020>.

14 UNICEF Montenegro and Montenegro Ministry of Education, *Strategy on Inclusive Education of Children in Montenegro 2019-2025*, UNICEF Montenegro and Ministry of Education, Podgorica, 2019. <http://www.mpin.gov.me/ResourceManager/FileDownload.aspx?rId=363254&rType=2>.

15 UNICEF Montenegro and Government of Montenegro Ministry of Education, *PISA 2015 Results and Policy Recommendations (including an Action Plan for 2019 –2021).* UNICEF Montenegro/Ministry of Education, Podgorica, 2019. <https://www.unicef.org/montenegro/en/reports/pisa-2015-results>.

16 Government of Montenegro, Ministry of Labour and Social Welfare, *Strategy on Prevention and Protection of Children from Violence 2017-2021*. Podgorica: Ministry of Labour and Social Welfare, 2017. <http://www.mrs.gov.me/ResourceManager/FileDownload.aspx?rId=274449&rType=2>.

- Strategy on protection from domestic violence 2016–2020[17]
- Strategy for integration of persons with disabilities in Montenegro 2016–2020[18]
- Strategy for the protection of persons with disabilities and promotion of equality 2017–2021[19]
- Strategy for judiciary reform in Montenegro 2019–2022[20]
- Strategy for health care development in Montenegro 2003–2020[21]
- Master plan for development of the health system in Montenegro 2015–2020[22]

17 Government of Montenegro, Ministry of Labour and Social Welfare , *Strategija zaštite od nasilja u porodici 2016-2020*, Ministry of Labour and Social Welfare, Podgorica: 2015.< http://www.gov.me/ResourceManager/FileDownload.aspx?rId=223283&rType=2>.

18 Government of Montenegro, Ministry of Labour and Social Welfare, *Strategija za integraciju lica sa invaliditetom u Crnoj Gori 2016-2020*, Ministry of Labour and Social Welfare, Podgorica, 2016. <http://www.mrs.gov.me/ResourceManager/FileDownload.aspx?rId=251538&rType=2>.

19 Government of Montenegro, Ministry of Human and Minority Rights, *Strategija za zaštitu lica sa invaliditetom od diskriminacije i promociju jednakosti 2017-202*, Ministry of Human and Minority Rights, Podgorica, 2016. < http://www.gov.me/ResourceManager/FileDownload.aspx?rId=260554&rType=2 >

20 Government of Montenegro, Ministry of Justice, *Strategija reforme pravosuđa 2019-2022*, Ministry of Justice, Podgorica, 2018. <http://www.gov.me/ResourceManager/FileDownload.aspx?rId=380326&rType=2>.

21 Government of Montenegro, Ministry of Health,, *Strategija razvoja zdravstva Crne Gore 2003-2020*, Ministry of Health, Podgorica, 2003. <http://www.mzdravlja.gov.me/ResourceManager/FileDownload.aspx?rid=217283&rType=2&file=Strategija per cent20razvoja per cent20zdravstva per cent20Crne per cent20Gore per cent20(2003).pdf>.

22 Government of Montenegro, Ministry of Health, *Master plan razvoja zdravstva Crne Gore 2015-2020*, Ministry of Health, Podgorica, 2015.<http://www.mzdravlja.gov.me/ResourceManager/FileDownload.aspx?rid=216169&rType=2&file=MASTER per cent20PLAN per cent20RAZVOJA per cent20ZDRAVSTVA per cent20CRNE per cent20GORE per cent202015-2020.pdf>.

- Youth strategy 2017-2021.[23]

The CRC, the most widely ratified human rights treaty in history, is premised on the notion that children are individuals with rights. Under Article 27, the CRC calls upon all signatory countries "to recognize the right of every child to a standard of living adequate for the child's physical, mental, spiritual, moral and social development". Therefore, the concluding observations made by the UN Committee on the Rights of the Child following a country's submission of its progress reports carry significant weight in all matters related to the realization of children's rights. The process of drafting the national strategy for fulfilling the rights of the child – Montenegro's five-year commitment to children – included data and evidence from:

- Concluding observations by the CRC Committee on Montenegro's combined second and third periodic reports[24]
- Child and adolescent voices, gathered through consultations
- Data on relative child poverty, based on the annual survey of income and living conditions (SILC).

Reducing children's monetary poverty is one of the strategy's 10 objectives, to be measured using three main indicators: the 'at-risk-of-poverty' rate, the material deprivation rate and the 'at-risk-of-poverty-or-social-exclusion' rate.

Monitoring the situation of children, women and men in Montenegro

The multiple indicator cluster survey (MICS) is an international household survey developed by UNICEF. The surveys, carried out in most countries, provide internationally comparable data on living conditions, health, education and other topics relevant for assessing the situation of children and women in all countries. MICS data allows countries to monitor progress towards the SDGs, EU integration and other internationally agreed-upon commitments. Montenegro participated in the 5th and 6th rounds of

23 Government of Montenegro, Ministry of Sports and Youth, *Youth Strategy 2017-2021*, Ministry of Sports and Youth, Podgorica, 2016 < http://www.un.org.me/Library/Youth-Empowerment/1a%20Youth%20Strategy%202017-2021.pdf>.

24 Available at: <https://tbinternet.ohchr.org/_layouts/15/treatybodyexternal/Download.aspx?symbolno=CRC%2fC%2fMNE%2fCO%2f2-3&Lang=en>.


Photo: Duško Miljanić / UNICEF Montenegro

MICS surveys. The latest available report is based on two 2018 surveys, one for Montenegro's general population and the other for its Roma settlements. The surveys were conducted by the Statistical Office of Montenegro (MONSTAT) with technical support from UNICEF and financial support from the government, UNICEF and UNHCR.[25]

Measuring child poverty in Montenegro

Although monetary measures are still widely used to calculate the proportion of children living in poverty, recent evidence and analysis reveal that such approaches do not fully capture what it means for a child to be poor.[26] The argument that monetary indicators offer only a limited perspective on children's welfare and living conditions can be summarized as follows. Monetary resources

25 Statistical Office of Montenegro (MONSTAT) and UNICEF, *2018 Montenegro Multiple Indicator Cluster Survey and 2018 Montenegro Roma Settlements Multiple Indicator Cluster Survey*, Survey Findings Report, MONSTAT and UNICEF, Podgorica, 2019. <https://go.aws/2VCqxla>.

26 See for example, G. Main and J. Bradshaw (2016), 'Child poverty in the UK: Measures, prevalence and intra-household sharing,' pp. 38–61, and A. Carraro and L. Ferrone (2020), 'Measurement of Multidimensional Child Poverty', in W. Leal Filho W.et al., eds., *No Poverty — Encyclopedia of the UN Sustainable Development Goals.*

alone cannot fulfil children's needs because many elements of what is important for their well being and development (such as education, health, access to clean water, safety etc.) may not necessarily be quantified in market terms. Standard monetary measures of poverty also do not account for inequalities in intra-household distribution of, for example, food, education and labour. Moreover, children tend to lack equal decision-making power within the household, and many of the goods and services they need to flourish cannot be expressed in monetary equivalents.

The aim of measuring multidimensional poverty is to assess children's welfare through a broader measure that goes beyond income. Multidimensional child poverty is a complex and multi-layered phenomenon that is revealed multiple key dimensions of well-being deemed essential for children's lives and encompasses the various deprivations experienced by poor people in their daily lives.

It is with this understanding that SDG 1 calls for a commitment to eradicating extreme poverty for all people everywhere (Target 1.1) and "reducing at least by half the proportion of men, women and children of all ages living in poverty in all its dimensions according to national definitions" by 2030 (Target 1.2). SDG Target 1.2 stresses the need to complement monetary with non-monetary measurements of poverty, pointing to the urgent need for a multidimensional response to addressing the underlying causes of child poverty. More recently, the Atkinson Commission on Global Poverty endorsed the measurement of both monetary and multidimensional poverty to track progress towards this important goal[27].

In response to this need, this report provides an update on the status of child deprivation in Montenegro and assists in the identification of particularly vulnerable groups by analysing the deprivations currently hindering children's growth and development, which if not addressed will likely be transmitted to future generations.

Taken together the CRC (UN, 1989), the World Summit for Social Development (UN, 1995), the Millennium Development Goals (UN, 2000) – along with the rationale and guidelines for monitoring them (UNDP, 2003) – and the more recent SDGs (UN, 2015) provided guiding principles for selecting the dimensions of child well-being utilized here. This report consists of an in-depth examination of the range of deprivations experienced by children in Montenegro, using a mixed-method multiple overlapping deprivation analysis (MM-MODA). The approach used in this study is innovative in that it is the first attempt to combine UNICEF's quantitative MODA methodology, developed in 2012, with qualitative research.

Understanding the degrees of child poverty and complementing them with a thorough analysis of poverty profiles are key steps for the design and implementation of effective policies. As part of UNICEF Montenegro's long-terms plans to generate quality evidence on child poverty and disparities and advocate for reducing child poverty, the present study was developed jointly by UNICEF's Office of Research – Innocenti and the UNICEF country office in Montenegro.

The study uses quantitative data collected for the MICS6 in 2018, which was undertaken among both Montenegro's general population and its Roma settlements (total population c. 8,500), as well as qualitative data collected through focus groups and in-depth interviews with key informants. These sources of information permit an in-depth analysis of child deprivation within both populations from two different, but integrated, perspectives.

The goal is to identify the extent to which children in Montenegro are deprived of adequate nutrition, health, education, water and sanitation, housing, utilities, and information, and are exposed to labour and violence. The selection of deprivation dimensions, indicators and thresholds was carried out during a national consultation workshop organized in May 2019 by UNICEF Montenegro with local stakeholders – including representatives from line ministries, international organizations, NGOs and think-tanks – during which the report's methodology was presented to stakeholders. In December 2019 the preliminary results of both quantitative and qualitative analyses were presented to the same stakeholders at a workshop where stakeholders discussed the findings, directions and potential actions that had emerged from the study and offered advice on recommendations and benchmarks for the drafting of policy recommendations. The report presents data on variations in single and multiple deprivations across three age groups: 0–59 months (with a focus on children aged 0–23 months for data availability reasons) and children aged 5–17 years. The analysis presented here goes beyond an estimate of child poverty to provide a profile of poor children and identify which overlapping dimensions have the strongest impact on multidimensional child deprivation. It also highlights the challenges to be addressed to improving child well-being. This quantitative analysis is enriched with qualitative insights about the ways in which children and parents/caregivers experience poverty, including their coping strategies, as well as how adolescents perceive various deprivations and how this affects their transition to adulthood.

27 World Bank (2017), 'Monitoring Global Poverty: Report of the Commission on Global Poverty', Washington, DC: World Bank, 2017.

The report is comprised of eight sections. Section 2 describes the data collection methodology, Section 3 presents the main findings related to a single deprivation analysis among children in the general population and those from Roma settlements, while Section 4 discusses the results of the multiple overlapping deprivation analysis. In Section 5 the authors simulate different scenarios to test the sensitivity of three final estimates of deprivation. Section 6 presents qualitative issues related to children's experiences of poverty that were identified by children and caregivers, but not captured in the original set of MODA indicators. Section 7 describes the main strategies adopted by children and parents to cope with deprivation. Section 8 summarizes the study's main conclusions and offers policy and programming recommendations.

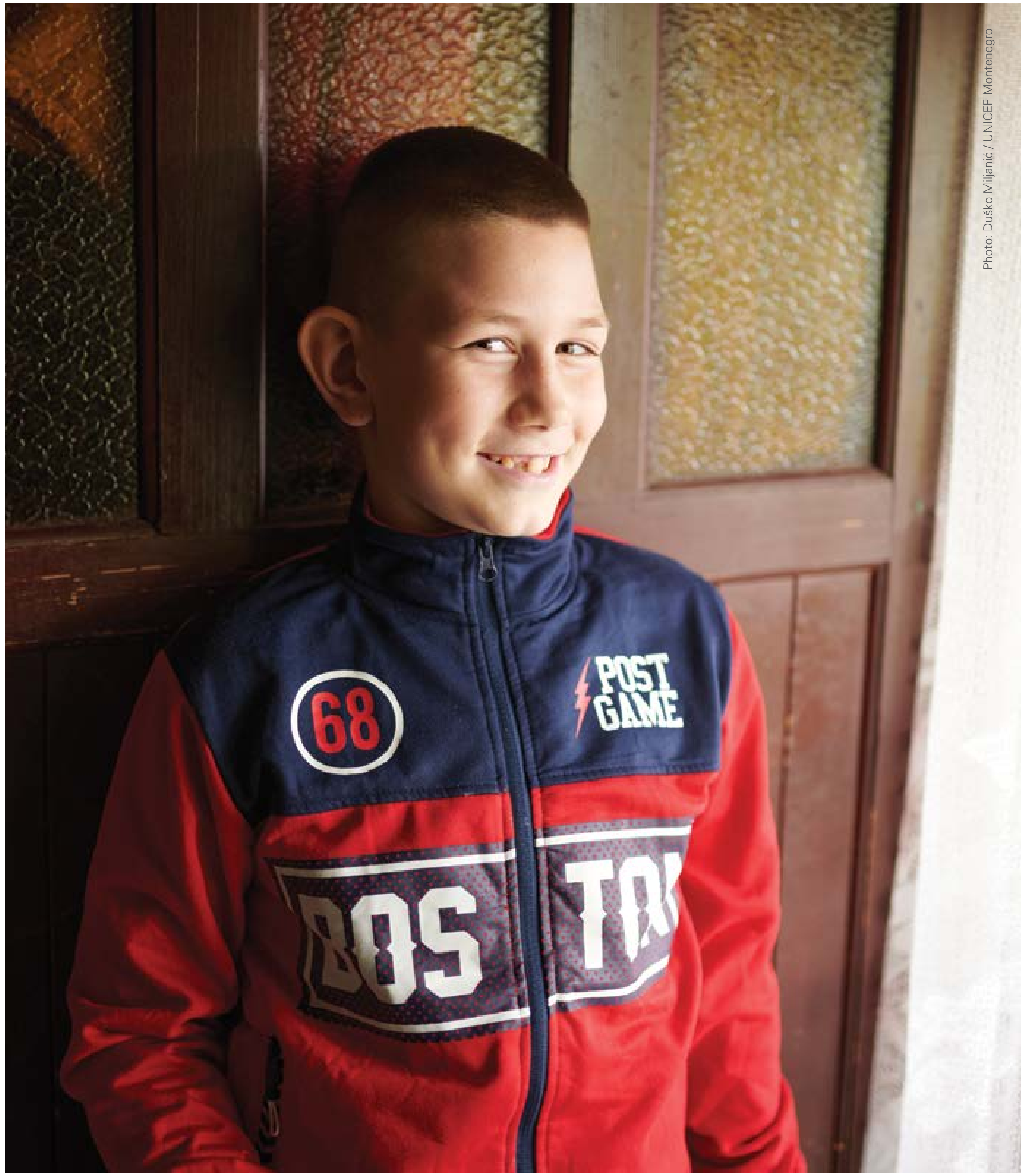


Photo: Duško Mijanić / UNICEF Montenegro

2. Mixed-method MODA

This report is structured as a mixed-method research study. It represents the first attempt since the MODA methodology was developed to combine quantitative and qualitative methods to explore the phenomenon of multidimensional child deprivation. This approach was adopted for two main reasons. First, given that one research approach alone can rarely fully examine complex and multifaceted phenomena (such as child poverty and deprivation), a mixed-method approach makes it possible to obtain answers to the same research question from different angles, thus increasing the scope and validity of the analysis. Second, a mixed-method approach enabled the researchers to generate a more comprehensive and vivid understanding of childhood deprivations in Montenegro and to use qualitative inquiry to clarify and illustrate deprivation trends and patterns identified through quantitative analysis. The authors then identified areas where results from both methods converged and sought to uncover the extent to which they were aligned, contradictory or led to new or emerging perspectives. The mixed methods were integrated through a sequential research design, whereby the quantitative component preceded and informed the qualitative inquiry. The quantitative MODA framework sets out core indicators across nine dimensions of deprivation, to identify those that affect children of different age groups. Qualitative research then serves to confirm, interpret and further explain the quantitative findings. The next two sub-sections provide an explanation of the two methods.

Quantitative MODA

Multiple overlapping deprivation analysis was developed by UNICEF to determine the degree and nature of multiple deprivations experienced by children[28]. The **MODA approach is based on international human rights standards, which inform the choice of dimensions to measure child well-being in the domains of survival, development, protection and social participation.** Children's rights, as defined by the CRC, the World Summit on Social Development and the Millennium Development Goals, guided the construction of "**a core set of dimensions that are essential to any child's development irrespective of their country of residence, socio-economic status, or culture**".[29] These dimensions cover areas that are not directly tied to household income, but are rather subject to influence by context-specific factors (e.g., local infrastructure or source of energy) that cannot be directly controlled by the household. Following a rights-based approach MODA takes children's life-cycle into consideration – acknowledging that dimensions and indicators are age-specific (i.e., differing needs for young children, school-age children and adolescents) – and goes beyond simply counting deprivations to identify overlapping dimensions and profiles. Exploring the profiles of deprived children (showing, for example, differences between those living in rural vs. urban areas or based on their parents' education) helps to understand their situation and thus to develop effective policy recommendations.

Another unique feature of the MODA methodology is the way that it combines other distinctive features in its measurement of multidimensional child poverty. **First,** it applies a "child-centric" approach by selecting the child (rather than the household) as the unit of analysis. Adopting household-level measurements assumes homogeneous resource distribution with children in the same household, which is often not the case. Calculating multidimensional deprivation levels separately for children means recognizing that they have different needs than adults and that poverty is more detrimental for them, due to its life-long effects. **Second**, MODA broadens the scope of compartmentalized, single-dimension analysis by evaluating the presence of *simultaneous* deprivations in each child's life, identifying those aspects that should be addressed concurrently. **Third**, MODA combines indicators into dimensions using the 'union approach' (see Appendix C2), which identifies a child either as deprived in one specific dimension (if he/she is deprived in any of the indicators included in that dimension), or as not deprived. MODA then counts the number of deprivations. Each indicator is considered equally important for the child in a given dimension; thus MODA gives equal weight to all dimensions. **Fourth**, MODA purposely does not define a unique threshold (i.e., cut-off) to count multi-dimensionally poor children. MODA reports multiple deprivation estimates for each possible cut-off point, leaving the definition of multidimensional poverty to a policy discussion in each national context. **Fifth**, MODA uses the 'dual cut-off' method for counting deprivations on the individual or household level and aggregating them into a national measure of multidimensional poverty.[30] The approach produces estimates of headcount, intensity and adjusted headcount of poverty. **Finally**, MODA allows for the generation of profiles based on the geographical and socio-economic characteristics of those who are deprived in multiple dimensions.

As demonstrated by existing studies in middle-income countries, MODA is flexible and largely adaptable to different country settings.[31] Key decisions for the present study – particularly the selection of dimensions and indicators to adapt MODA to the Montenegrin context (for children both in the general population and in Roma settlements) – were agreed during the May 2019 workshop in Podgorica. During the workshop, participants from the public sector, academia and civil society who are mandated to, or active in, realizing child rights reached a consensus on what should be considered relevant for children in Montenegro, defining age groups, discussing the outcomes, grouping them into domains and setting thresholds for defining deprivations for the indicators under analysis. The final selection reflects the opinion of workshop participants, methodological standards and data availability. The quantitative component of the analysis uses data from the two MICS carried out in 2018. These surveys are designed to provide a comprehensive update on the status of children's, women's and men's well-being using a broad set of indicators (including health, nutrition, child development, education, child discipline and others) at the national level.[32] The datasets provide information on the general population of Montenegro and on Roma settlements. Data can be disaggregated and analysed by urban and rural areas and according to region (south, centre and north).

28 de Neubourg et al. (2012), 'Step-by-step guidelines to the multiple overlapping deprivation analysis (MODA)', working paper, UNICEF Office of Research WP-2012-10, Florence.

29 Ibid, p. 6.

30 S. Alkire and J. Foster (2011), 'Counting and Multidimensional Poverty Measurement', *Journal of Public Economics*, 95, 7, pp. 476–487.

31 de Neubourg, et al. (2015), "Wellbeing of Children in Kosovo: Poverty and Deprivation among Children using the Multiple Overlapping Deprivation Analysis", UNSCR 1244, UNICEF; L. Ferrone and Y. Chzhen (2016), 'Child Poverty in Armenia', working paper, UNICEF Office of Research WP-2016-24, Florence; Y. Chzhen and L. Ferrone (2017), 'Multidimensional child deprivation and poverty measurement: Case study of Bosnia and Herzegovina,' *Social Indicators Research*, 131, 3, pp. 999–1014; L. Ferrone and Y. Chzhen (2018), 'How to reach the sustainable development goal 1.2: Simulating different strategies to reduce multidimensional child poverty in two middle-income countries,' *Child Indicators Research*, 11, 3, pp. 711–728.

32 Financial and technical support for data collection was provided by the United Nations Children's Fund (UNICEF), the Government of Montenegro and UNHCR.

Therefore, MICS data lends itself to MODA and geographical profiling. The 2018 Montenegro MICS used a stratified, two-stage cluster sample design.

MODA analysis follows the same framework, covering both the general population and children living in Roma settlements. Participants at the inception workshop identified both household- and child-level dimensions of deprivation. Household-level dimensions **(water and sanitation, housing, utilities and access to information)** are the same for all children regardless of age (except access to information). Child-level dimensions, **(child discipline/neglect, health, nutrition, early childhood development (ECD)/education, child labour/domestic chores)**, are assigned to each age group based on age-appropriate developmental needs and data availability. All chosen dimensions reflect a basic child right. In line with the CRC, all dimensions are weighted equally because no right carries more weight than another. The sample is broken down into two main age groups: 0-59 months and 5-17 years, but also devotes attention to children aged 0-23 months to better capture their specific nutritional needs. All age groups include both individual and household-level dimensions, as detailed in Tables 1 and 2: four dimensions are measured at the household level and five at the child level, for a total of nine dimensions.

Following standard MODA procedure, this report identifies deprivation in a specific dimension as meeting the deprivation threshold for at least one of the indicators selected to construct that dimension (union approach). For example, a child aged 0–59 months is considered to be deprived in the health dimension if he/she lacks either a selected paediatrician or has experienced delayed (or no) vaccinations despite the availability of vaccination services. The thresholds chosen for each indicator are specified in Table 2.

The MODA analysis framework followed the steps noted below for each age group in both the general population and Roma settlements.

1) Single deprivation analysis: Gives a sector-specific perspective on childhood deprivation. It estimates the prevalence of children deprived in each indicator and dimension for the respective age groups. A single deprivation profile, based on children's background characteristics, is also provided. A logistic regression was run to evaluate the association between background characteristics and the probability of experiencing deprivations in any given dimension.

Photo: Duško Miljanić / UNICEF Montenegro

2) Multiple overlapping deprivation analysis: Offers an understanding of ***the depth of multidimensional deprivation children face***, providing: a statistical distribution of the number of children experiencing from zero to seven deprivations simultaneously, the deprivation overlaps between the three most prevalent dimensions and a socioeconomic and geographic profile of multi-dimensionally deprived children.

3) Multidimensional deprivation indices: The MODA methodology allows for the construction of three indices to quantify multidimensional child poverty. The 'headcount ratio' (H) captures the prevalence of poverty, permitting calculation of the number of children experiencing multidimensional poverty according to different cut-off points and percentages, relative to the total number of children in Montenegro (i.e., "how many children are poor?"). This report focuses on the cut-offs of two, three or four dimensions. The 'average intensity of deprivation' (A) captures "how poor are the poor?" by measuring ***the depth or intensity of poverty*** across different age groups among those children classified as multi-dimensionally poor. It is defined as the ratio between the number of deprivations children face divided by the total dimensions. The 'adjusted headcount ratio' (M^0) is the third measurement; it adjusts the headcount in accordance with the intensity of poverty, thus reflecting ***both the incidence and intensity of poverty***. Annex C provides a detailed description of the construction of indices.

Dimension	**0–23 months**	**0–59 months**	**5–17 years**
	Child discipline/neglect Health Nutrition ECD Water and sanitation Housing Utilities	Child discipline/neglect Health ECD Water and sanitation Housing Utilities	Child discipline/neglect Education Child labour (domestic chores) Water and sanitation Housing Utilities Information

Age groups

Table 1 - Age groups and dimensions

Photo: Duško Miljanić / UNICEF Montenegro

Dimension	Indicator	Deprived if..	Level	Age group			Availability
				0–23 months	0–59 months	5–17 years	
Child protection (discipline/ neglect)	Psychological aggression	Child experienced either being shouted, yelled or screamed at, or being called dumb, lazy or another name.	Child				>11 months & <15 years
	Physical punishment	Child experienced any of the following: being shaken; being spanked, hit or slapped on the bottom with a bare hand; being hit on the bottom or elsewhere with belt, brush, stick, etc.; being hit or slapped on the hand, arm; being hit or slapped on the face, head or ears; being severely beaten.	Child				>11 months & <15 years
	Neglect	Child has been left alone for more than one hour or left in the care of another child younger than 10 years old at least once in the last week.	Child				0–11 months
Health	Selected paediatrician	Child lacks a selected paediatrician.	Child				
	Immunization hesitancy	Child experienced parental delay in acceptance or refusal of vaccines despite availability of vaccination services for the following reasons: vaccines not available, occupied with other tasks, doubts about the vaccines, other reasons.	Child				
Nutrition	Exclusive breastfeeding	Child is not exclusively breastfed.	Child				0–5 months
	Feeding	Infant and young child feeding frequency falls short either of minimum dietary diversity (MDD) (meaning the child during the last day or night received foods from fewer than five of the following food groups: 1) grains, roots, and tubers; 2) legumes and nuts; 3) dairy products (milk, yogurt, cheese, infant formula); 4) meat/meat products; 5) fish, fresh or dried; 6) poultry and liver/organ meats); 7) eggs; 8) vitamin-A rich fruits and vegetables; 9) leafy vegetables; 10) cereals, bread and pasta; and 11) other fruits and vegetables) or of the minimum acceptable diet (MAD) (child under the age of six months not exclusively breast-fed; child aged 6–8 months (who is currently breastfed) receiving solid or semi-solid food less than twice a day; child aged 9–23 months (who is currently breast-fed) receiving solid or semi-solid food less than three times a day. Non-breast-fed children aged 6–23 months receiving solid, semi-solid or soft foods and milk fewer than four times a day; or a child aged 24–59 months receiving fewer than five meals a day).	Child				6–23 months
ECD	Plays with toys	Child has no toys to play with, i.e., homemade toys, toys from shops/ manufactured toys, or household items/items found outside.	Child				0–35 months
	Early stimulation	In the past three days the child was not involved in at least three of the following activities with any household member aged 15 or above: book reading, telling stories, singing songs, taken outside, playing together, naming/counting/drawing things).	Child				36–59 months
	Books at home	Child has no books suitable for his/her age at home.	Child				0–59 months

Education	Grade for age	Child is two or more years behind age-appropriate grade.	Child					5–17 years
	School attendance	Child of compulsory school age not attending school.	Child					5–14 years
Child labour (domestic chores)	Domestic chores	Child is engaged in more than five of the following activities: fetching water, collecting firewood, shopping, cooking, washing dishes/ cleaning, washing clothes, caring for children, caring for elderly household member, other HH tasks.	Child					5–17 years
Water and sanitation	Unimproved source of drinking water	Household's main source of drinking water is unimproved: unprotected dug well; unprotected spring; rainwater; tanker truck; bottled water if source of main non-drinking water is unimproved; other. (Definition by World Health Organization/ WHO).	Household					
	Absence of shower unit / bathtub	Household is not equipped with shower unit or bathtub.	Household					
	Unimproved sanitation	Household usually uses unimproved toilet facility: flush to pit latrine/ open drain/somewhere else; pit latrine; composting toilet; no facility/ bush/field; other. (Definition by WHO).	Household					
Housing	Overcrowding	Household has on average more than three people per sleeping room. (UN-HABITAT definition).	Household					
	Multiple housing problems	Household has problems with dwelling: Leaking roof/damp walls, floors or foundation/rot in window frames or floor.	Household					
Utilities	Delay in bills payment	Household was late with payment of utility bills in past year.	Household					
	Energy affordability	Household cannot afford to keep the dwelling adequately warm.	Household					
Information	Internet access	Household is not equipped with internet connection.	Household					
	Mobile	No person at the home has a mobile phone.	Household					

Table 2. Dimensions, indicators and definitions: General population and Roma settlements

[2] I.e. the number of deprivations experienced by multi-dimensionally deprived children.

[3] See de Neubourg et al. (2012). The union approach applies if the dimension is constructed with more than one indicator.

Qualitative MODA

Purpose and design

The main goal of the qualitative component of the study was to enrich the findings of the quantitative study, permitting better understanding of how children perceive and experience different dimensions of poverty. Qualitative methods were designed to give voice to children and their parents/guardians to identify, describe and explain in their own terms the way they live and cope with different deprivations and to gather their views on priority actions for overcoming poverty. In line with the MODA methodology, the qualitative research adopted a lifecycle approach to assessing deprivations across two age groups of children (0–5 years and 5–17 years), and examined how different circumstances (e.g., location, ethnicity, household composition, gender and disability) shape children's experience of multiple deprivations.

The following research questions guided the qualitative data collection and analysis:

- How do children perceive and define poverty (in each specific dimension of deprivation)?
- How do the deprivations overlap?
- How does poverty manifest itself with specific groups of children in terms of their material, psychological and social wellbeing?
- How do children and parents/caregivers cope with poverty?
- What kind of future can children and parents/caregivers that live in poverty expect?
- What strategies are required to reduce multi-dimensional poverty in Montenegro?

The qualitative research design complements the quantitative MODA framework in several ways. First, data collection was structured across the nine core dimensions of deprivation used in the MODA to identify the most prevalent types of deprivations affecting different age groups of children. Qualitative data was collected to confirm (or dispute), as well as to contextualize and interpret, the quantitative MODA findings. Second, qualitative data was used to illustrate how different deprivations overlap and reinforce each other. Finally, an exploratory approach to data collection was adopted to identify deprivations identified by children and caregivers but not captured in the original set of MODA indicators.

Sampling and research methods

Qualitative data was collected via semi-structured focus groups, exploratory focus groups and individual interviews, including in-depth and key informant interviews. Overall, the approach to sampling locations and participants in qualitative data collection was purposive[33] and included a relatively complex sample design to compare the experiences of children living in different settings (e.g. rural and urban), different household composition and socio-economic background. The sample was designed to include those at risk of poverty, based on the latest available SILC data. Data was collected in Podgorica, Niksic, Rozaje and Bijelo Polje to capture regional variations in poverty dynamics, consistent with the quantitative approach.

Research methods used to collect the data included:[34]

1) Twenty-six *focus group discussions* (FGDs) with parents/caregivers of children aged 0–17, as well as with boys and girls aged 15–17. The sample included households with unemployed adults, beneficiaries of social assistance, children living in large and single-parent families and children of Roma origin. Each FGD consisted of eight parents/caregivers or eight adolescents (aged 15-17 years). An estimated 208 respondents, recruited by staff from the Centres for Social Work, were interviewed.

2) Eight *in-depth interviews* sought to understand the experiences of very vulnerable children, such as those living in foster families and children with a disability.

3) Twelve *key informant interviews* were conducted with decision-makers and social service providers, (e.g., social workers, youth workers, NGOs). These interviews were held to offer a complementary perspective on child poverty in Montenegro and discuss the type of support available from the government and service providers to help children and their families cope with and overcome poverty.

Thematic analysis was undertaken to interpret and analyse the data. This approach allowed the authors to organize data into themes and use overarching research questions to unpack the stories and important emerging topics.

33 Purposive sampling means that informants who can contribute information related to the evaluation questions are intentionally sought out, rather than randomly selected from the population (Skoval and Cornish, 2015).

34 Detailed descriptions of each focus group and individual interview conducted (category of respondent, area of residence, data collection method and other details) are presented in Annex D.

Ethical review

The research team adhered to the 2015 UNICEF guidelines on *Ethical standards in research, analysis and data collection* to ensure that the highest ethical standards were applied in each phase of the research. Participants were informed of the context and purpose of the research and voluntary nature of their participation as well as the guaranteed privacy and confidentiality of discussions. Informed consent was sought from all individuals aged 18 and above participating in the research. Prior written consent of both participating adolescents and their parents/caregivers was obtained to allow them to participate in research. Finally, participants were assured that information obtained during the discussion would be treated with care and their identity would remain anonymous at all stages of data analysis and reporting. All team members signed a statement of confidentiality, according to which they are legally bound not to share any information obtained during the research. The study protocol was submitted to UNICEF's Independent Ethical Review Board, which granted permission for this phase of the research.

Photo: Duško Miljanić / UNICEF Montenegro

3. Single Deprivation Analysis

3.1. Analysis for the general population of Montenegro

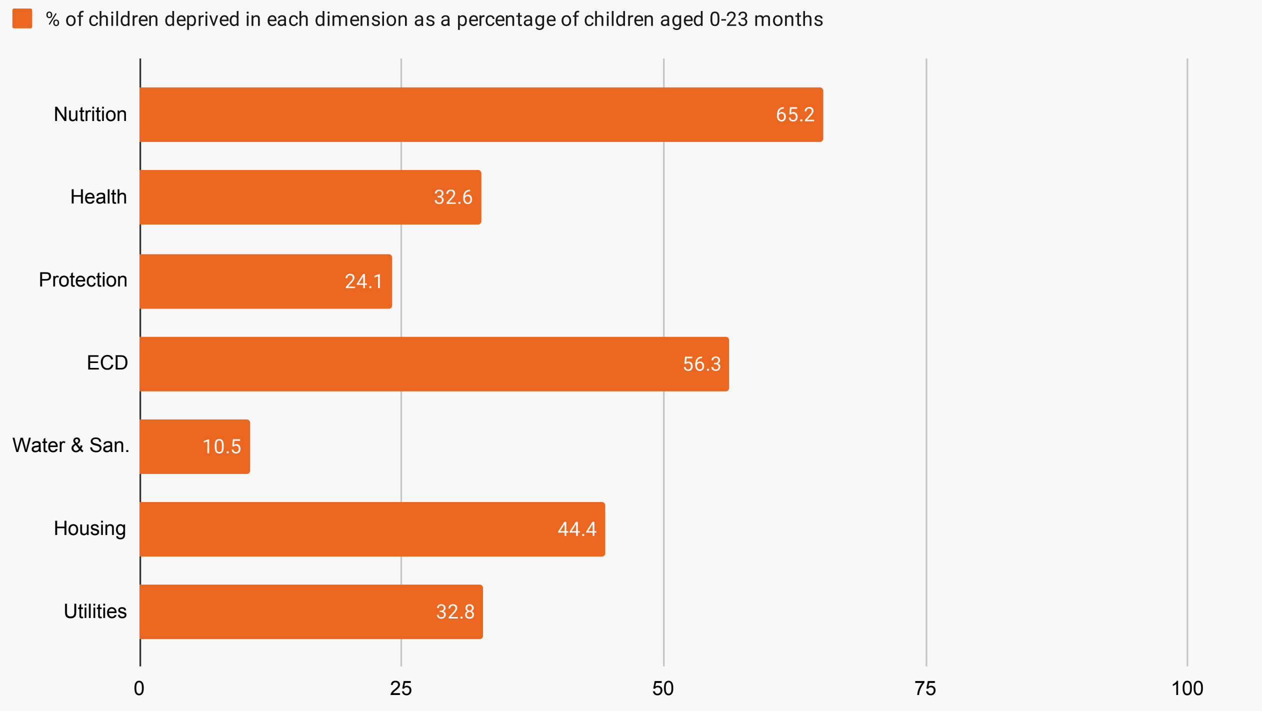
Summary of key findings:

- Levels of deprivation in a single dimension differ across age groups. For children aged 0–23 months, the highest scoring deprivation dimension is nutrition. Deprivation of protection (exposure to violent discipline or neglect) ranks highest for children in the other two age groups, who suffer psychological aggression or physically punishment by a caregiver.
- Deprivation in the domains of education, water and sanitation, utilities and information is generally high among children living in rural areas. Health deprivations, particularly immunization hesitancy, and deprivations in child protection are significantly higher in urban areas.
- Children under 5 years of age living in female-headed households are more deprived in all dimensions (except child protection) than their peers living in male-headed households. Fifty-two per cent of children living in male headed households are deprived in child protection vs. 36 per cent living in female headed households, respectively.
- Parents' education matters. In general, children living in households where mothers lack secondary education are more deprived in all dimensions; especially water and sanitation (by a factor of 6.5) and ECD (threefold difference).
- Children's risk of deprivation across most dimensions is linked to specific background characteristics: **mother's education, household head's education and, for some dimensions, the gender of the head-of-household.**

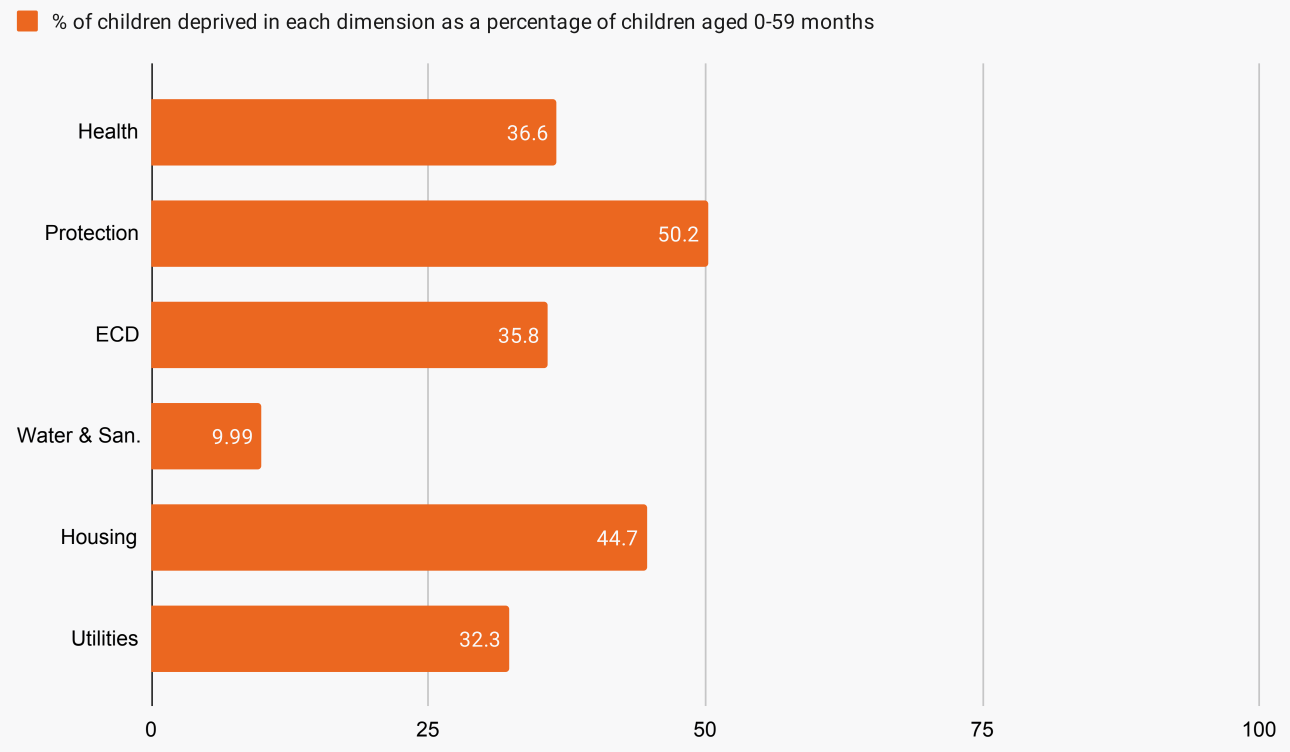
This section examines the prevalence of child deprivation for each indicator and dimension by age group, using the definitions of deprivation described in Table 2. Figures 1 (a), (b) and (c) report deprivation rates for the three age groups (0–23 months, 0–59 months and 5–17 years), while Table 3 presents the rate of deprivation by indicator and age group in more detail. The ranking of deprivations varies according to age group and the set of dimensions considered. These quantitative results are complemented by qualitative findings to present a multifaceted and more detailed account of how children experience each deprivation.

Figure 1. Deprivation rates by dimension: General population (%)

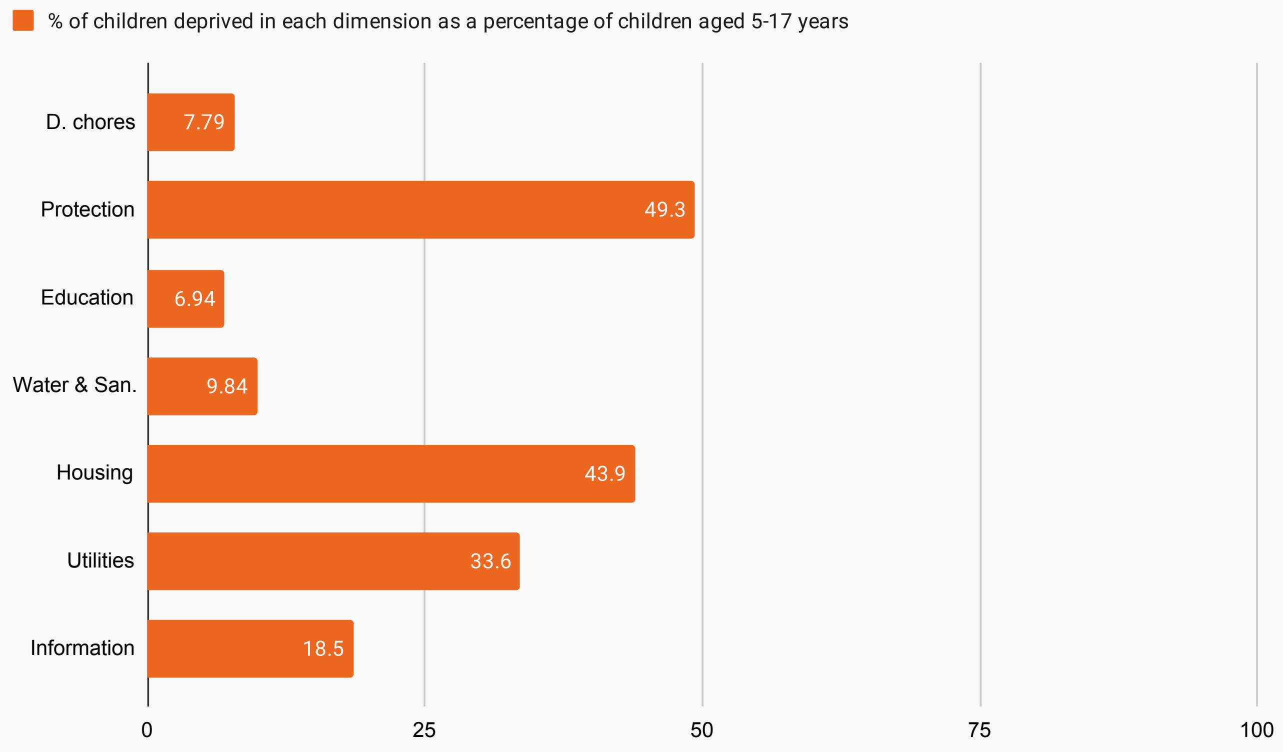
a) 0-23 months



b) 0-59 months



c) 5-17 years



Dimension	Indicator	Age group		
		0–23 months	0–59 months	5–17 years
ECD/Education	Has no books suitable for the age or doesn't play with any toys	56.68	46.82	-
	Has no books suitable for the age or is not involved in early stimulation activities	-	21.64	-
	Grade for age	-	-	3.35
	School attendance	-	-	1.74
Health	No selected paediatrician	13.11	9.24	-
	Delayed vaccination	23.11	29.68	-
	Refused vaccination	8.96	13.15	-
Nutrition	No exclusive breastfeeding	64.48	-	-
	Minimum dietary diversity	38.72	-	-
	Minimum meal frequency	48.96	-	-
Child protection	Psychological aggression*	41.91	59.21	46.92
	Physical punishment*	25.20	33.67	21.70
	Neglect**	1.84	4.59	-
Child labour	Domestic chores	-	-	7.78
Water and sanitation	Unimproved drinking water	0.95	1.02	0.74
	Absence of shower unit / bathtub	5.23	5.86	5.42
	Access to unimproved sanitation	7.87	6.96	6.33
Housing	Multiple housing problems	37.05	35.99	36.92
	Overcrowding	17.42	19.28	17.98
Utilities	Delay in bill payment	23.95	23.90	28.85
	Energy affordability	14.47	12.21	12.54
Information	Internet access	-	-	18.29
	Mobile	-	-	0.81

Table 3. Deprivation rates by indicator and age group as a percentage of total children in the age group: General population (%)

Source: MICS6 2018.

* The reported means refer to children aged 11–23 months and those aged 0–59 months.
Overall psychological aggression and physical punishment data are available for 11–59 month-olds.

** The reported means refer to children 0–11 months old

Dimensions

Nutrition

As shown in Figure 1a, children in the 0–23 months age group experience a high incidence of deprivation, particularly for **nutrition** (65.2 per cent) and **ECD** (56.3 per cent). According to Table 3, which displays in detail deprivation rates by indicator and age group, the high deprivation level registered for **nutrition** results from the large proportion of children who are not exclusively breastfed (64 per cent of children 0–6 months), combined with shortfalls in infant feeding patterns, including limited dietary diversity and inadequate meal frequency for children aged 6–23 months.

Qualitative evidence on nutrition largely focused on children 5–17 years of age, to compensate for the lack of data on nutrition for this age group in MICS6 datasets. Overall, nutrition deprivation for this age group manifests itself more in the poor quality of food consumed and limited dietary diversity than in food insecurity and hunger. Although neither parents nor children mentioned difficulty accessing food, poor families found healthy food to be largely unaffordable, and thus typically opted for less healthy choices. As a result, most respondents felt that their diets were limited in diversity and quality:

> I think [poor diets] are a problem everywhere. For example, fruit salads in a healthy food grocery shop are pricey. They cost about 3 or 4 euros, and here at fast-food shops, you can buy a pancake or a slice of pizza for one euro. It is more affordable and cheaper than this healthy food. And it should actually be the opposite.
>
> – FGD with children from single-parent families, Podgorica

Individual dietary preferences also play an important role in shaping children's diets and nutrition behaviour, as the majority of child respondents expressed choices for "fast food" rather than healthy food choices. In addition, older children rely on school meals and snacks, which they have to purchase in school in the absence of government-supported school feeding programmes. Many parents reported a struggle to pay the €1-2 required for this every day. However, the practice has taken on social significance, with homemade snacks viewed among children as socially unacceptable and an indication of poverty. Several children who were unable to afford school snacks and/or supermarket treats reportedly felt ashamed and refused to eat home-prepared food:

> My daughter one day came out of school crying. She says this friend told me... You gave me a grilled sandwich, "concentrated" syrup, a bottle of water and some crisps. She told me that I couldn't hang out with other kids since I was of a lower class than she because she can afford to buy stuff at the supermarket.
>
> – FGD with unemployed parents from Podgorica

Early child development

Turning to ECD, the relatively high levels of deprivation reflected in the quantitative data are **mostly driven by the absence of suitable books at home.**[35] As shown in Table 2, this indicator is used to construct the ECD dimension for children by combining it with the indicators for "plays with any toys at home" or "involvement in stimulating activities", respectively. Slightly more than half of children aged 0–23 months lacked either suitable books or toys to play with; similar incidences are found for the 0–59 months group (*see Table 3*).

Early cognitive development begins at home during children's interactions with parents, but school environments – including children's engagement with teachers and peers – are also critical for their emotional and intellectual development. Most respondents had a positive attitude toward preschool education, particularly its contribution to children's socialization. However, early education is reportedly not uniformly accessible, and some children are unable to attend kindergarten due to the distance from their homes or the unaffordable indirect costs of enrolment, such as transport, clothing etc.

Education

Deprivation levels for school attendance and achievement are relatively low for children aged 5–17 (see Figure 1 (c). Qualitative research confirms that most children in the general population attend primary and secondary school, and both parents and children generally express very positive attitudes towards education. However poorer families – particularly those without access to social assistance – face financial barriers to education and reported struggling with additional indirect expenses such as transportation, books, school supplies and presentable clothes and shoes for their children.

Children with special educational needs face constraints in accessing education. The majority of parents feel that children continue to suffer discrimination and stigma from their peers and teachers. Many respondents believe that the government and relevant service providers working in inclusive education need to intensify their efforts to improve access to and quality of inclusive education to improve schooling outcomes among children with special educational needs.

Child protection

The results show that children also experience high levels of deprivation in child protection. As shown in Table 3, a large proportion of children are subject to **psychological aggression from a caregiver** at home, including both the under-5 (59 per cent) and 5–17 age groups (47 per cent). Qualitative findings confirmed that children are also subjected to violent discipline and physical punishment at home.

Perceptions and attitudes about child discipline are multifaceted and complex. Children reported that methods used by parents to discipline children generally change as children grow older, when 'reward and punishment' is replaced by talking. However, both children and parents reported corporal punishment as a dominant means of discipline used at various ages – although parents are divided in their opinion of its usefulness. Some believe that 'only talking works', while others openly admit to using physical punishment to discipline children. Likewise, many parents feel that new child protection laws restrict their autonomy over child discipline, as children are protected by 'too many [child] rights… nowadays':

> **Moderator:** Is it ok to spank a child when they cause trouble? **Respondent 1:** No, never. **Respondent 2:** I think it is. Spare the rod, spoil the child. A little; not too much. **Respondent 1:** Pain doesn't teach us anything… **Respondent 3:** I don't think they should be spanked all the time, but it should be known that he who pays the piper calls the tune, as the old saying goes.
>
> – FGD with unemployed parents from Bijelo Polje

The study found some evidence of neglect. Working parents, particularly single parents, are often unable to be at home with their children as much as they consider necessary or desirable. Several parents reported leaving their younger children at home in the care of older siblings while they were at work. The inability to care adequately for their children and concerns for their safety cause psychological distress for parents (and siblings).

> **Moderator:** Are you afraid when you leave them home alone? **Respondent:** Well, yes, of course. One never knows what one of them might do. For example, when my child needs to warm up a baby bottle, I worry if they would forget to turn off the stove or not? These are all at risks, but we need to take them, because we need to provide for them.
>
> – FGD with parents from Nikšić

Another important qualitative finding was that children at various ages are exposed to violence outside of the home, mainly in school environments. **Peer-based verbal and psychological abuse** in school settings was a widespread problem reported by adolescents. Children also reported limited access to safe mechanisms at schools for reporting maltreatment, along with inadequate response and support from school authorities to address risks. Many respondents believe that school experts, such as psychologists and teachers, must become more involved and work preventively with children who find themselves in vulnerable situations, rather than focusing only on addressing the most extreme cases of peer-to-peer violence. In some cases, however, recourse is seen as impossible due to the danger of reporting abuse:

> [...] a dorm is a disaster for boys. It sometimes happens that they set someone's T-shirt on fire for five euros while this person is asleep... It's much harder on boys… They break into rooms at night. Everyone locks their door, but some come kicking in, breaking down doors... And no one tells on them, since they're threatened that - if you tell anyone, we'll beat you up and that's it.
>
> – FGD with adolescent boys from Nikšić

Health

The health dimension, according to data on immunization hesitancy and access to a chosen paediatrician, shows relatively large rates of deprivation among children younger than 5 years of age; about one out of three of children experiences problems in this dimension (*see Figures 1.a and 1.b*). In line with the MICS findings, qualitative reports reflect a high degree of hesitation about immunization due to rumours of links to autism or other harmful effects on child health. Although required by law, immunization

Photo: Duško Mijanić / UNICEF Montenegro

is an uncomfortable topic for many parents, who reported receiving information about potential risks of vaccinations mainly from media sources rather than directly from healthcare providers, resulting in general confusion and doubts about vaccine safety. Heavy work burdens were also reported to constrain parents from seeking healthcare for their sick children, while the cumbersome administrative procedures and paperwork needed to access health services may further restrict child visits to paediatricians.

Housing

Study results show that about 44 per cent of children in each age group live in a household reporting **multiple housing problems** or **overcrowding** (*see Figure 1*). In the qualitative sample, respondents often described their housing as poor and inadequate. Many children reported living in run-down, overcrowded apartments in which boys, girls and parents typically share bedrooms, resulting in the spread of illness and discord over a lack of privacy. Some parents explained that the houses they live in lack bathrooms and a source of drinking water, while others described them as being in disrepair and leaking, pointing to further potential health and safety hazards such as damp, rot and structural instability.

> **Parent:** There are seven of us, and the house is in bad condition. Sometimes, when the winters are cold there are five of us in one room and we all get [sick]. **Moderator:** How does it affect the children? Parent: It does – the colds start, coughing, bronchitis.
>
> – FGD with unemployed parents from Niksic

35 About half of the sampled children aged 0–59 months had no textbooks or picture books suitable for their age at home. Data available on request.

Information

Information-related indicators show, on average, relatively lower levels of deprivation (18 per cent), thanks to reasonable internet accessibility at home (*see Table 5*) and widespread availability of mobile phones (almost 100 per cent coverage). However, according to the qualitative sample, access to information is far from uniform, and there is a general perception that poverty negatively influences children's access to information. Many children reportedly do not have computers at home or lack regular internet access, particularly in rural areas. Interestingly, school reforms that appear to be progressive, such as e-learning, were perceived to widen the gap between children with different socio-economic status or areas of residence.

> **Moderator:** Do you think poverty has an impact on how informed children are? (Everyone confirms). **Respondent:** They may not have access to internet, they don't have computers and such…In the city, [the] power supply comes back right away. If there is something that we should copy from the internet and someone does not have internet or a computer at home, they cannot do it and they will get a bad grade because of it.
>
> – FGD with rural children from Bijelo Polje

Utility costs

About one out of three children is living in a household where parents experience problems in being able to afford the cost of utilities. (*See Figure 1.*) Most parents spoke extensively about their difficulties paying bills for electricity, fuel and other living costs. Several parents were often unable to pay for electricity. This can have a significant impact on children, as heating, electricity and cooking fuel are critical to children's health and ability to study, while regular access to electricity is essential for home use of computers and the internet.

Child labour

Finally, the study found some evidence of child labour in both the domestic and productive domains. According to quantitative data, around 7 per cent of children aged 5–17 report being engaged in at least five domestic chores (*see Figure 1.c*). Qualitative data confirms child labour among poor children, who tend to engage in productive activities. Children (primarily boys) reported earning money mainly by selling goods or performing physical work, such as helping at construction sites or carrying furniture. While child labour is common mostly among adolescent children aged 15 years and older, some children reportedly begin working as young as 10. The study was unable to establish the frequency and average working hours of children or the spill-over effects of child labour on education outcomes. Nevertheless, most children expressed an appreciation for their parents' efforts to provide money for them, and believe it is their responsibility to start contributing financially as soon as possible, earning at least enough to buy books, clothes and other personal items.

> **Respondent:** After these three months [of working during summer break] I can plan what to do with this money by myself and that helps my parents a lot. I think that we need to help ourselves first of all, and then, in that way we are helping our parents.
>
> – FGD with children of unemployed parents from Nikšić

Which children face a higher risk of deprivation?

Evaluating the prevalence of single deprivations within the general population is crucial to understanding the most urgent issues facing children, which must be prioritized in the policy agenda. However, national averages tend to mask the heterogeneity of children's deprivations experienced in different social, geographical and economic contexts. To fully grasp how single deprivations are connected to characteristics of households and children, the quantitative analysis identified 11 background characteristics (factors) common to all age groups, which the authors considered as the most likely determinants of deprivation. These include both demographic and geographical factors: child's sex and age, head of household's gender, mother and household head's education level, household head's age and household size, number of children between 5 and 18 years old and the area and region of residence.

Table 1 of Annex A provides a summary of statistics on background variables for the general population. The table shows that two out of every three children under 5 years of age live in urban areas, 34 per cent live in Podgorica, 23 per cent in the north, 20 per cent in the south and 23 per cent in the centre (excluding Podgorica). Similar proportions apply for the other two age groups. The average household is composed of more than five members. Among 5–17-year-old

Photo: Duško Miljanić / UNICEF Montenegro

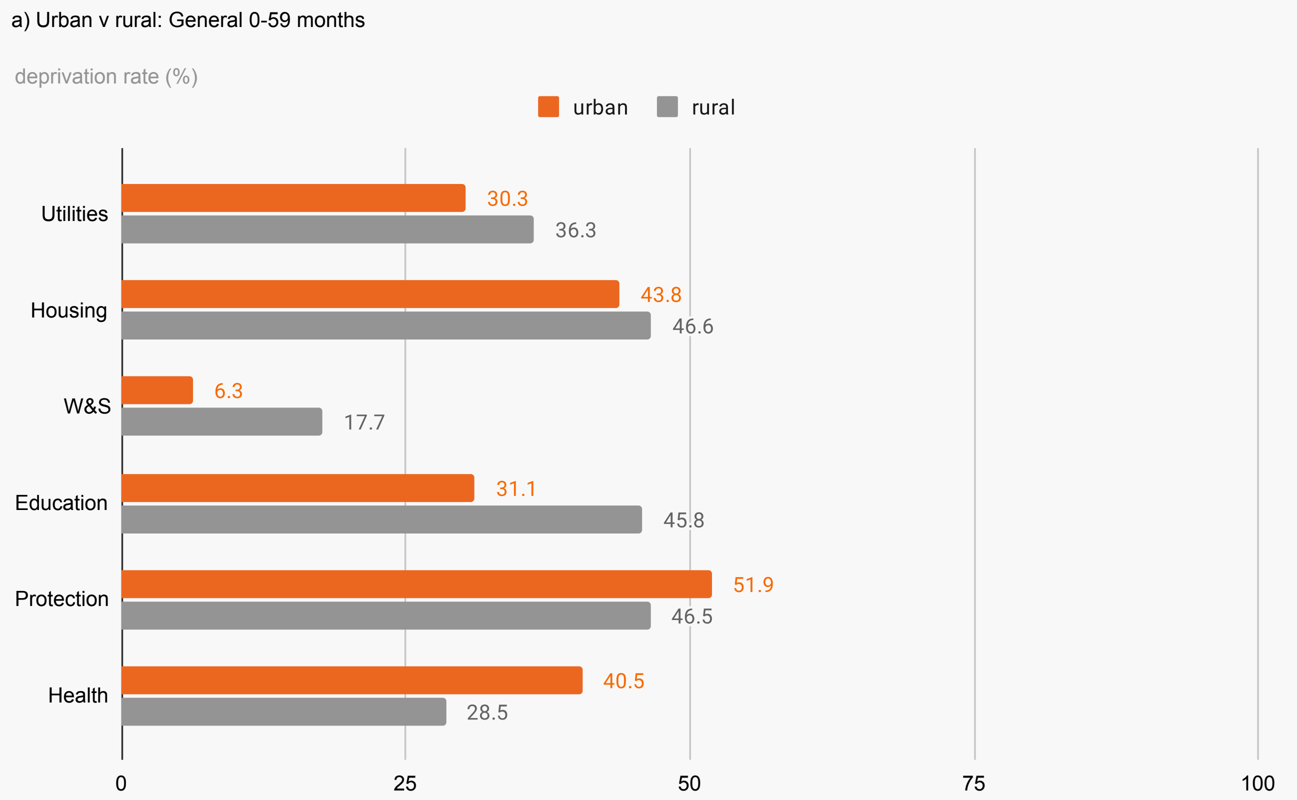
children the sample is almost equally divided by gender, while slightly more than half (55 per cent) of children aged 0–23 and 0–59 months are boys. The majority of children (89 per cent) live in male-headed households. Most heads of household and mothers have completed secondary or higher education: 79 per cent of all children live in a household whose head is highly educated, with minimum distinctions across age groups.

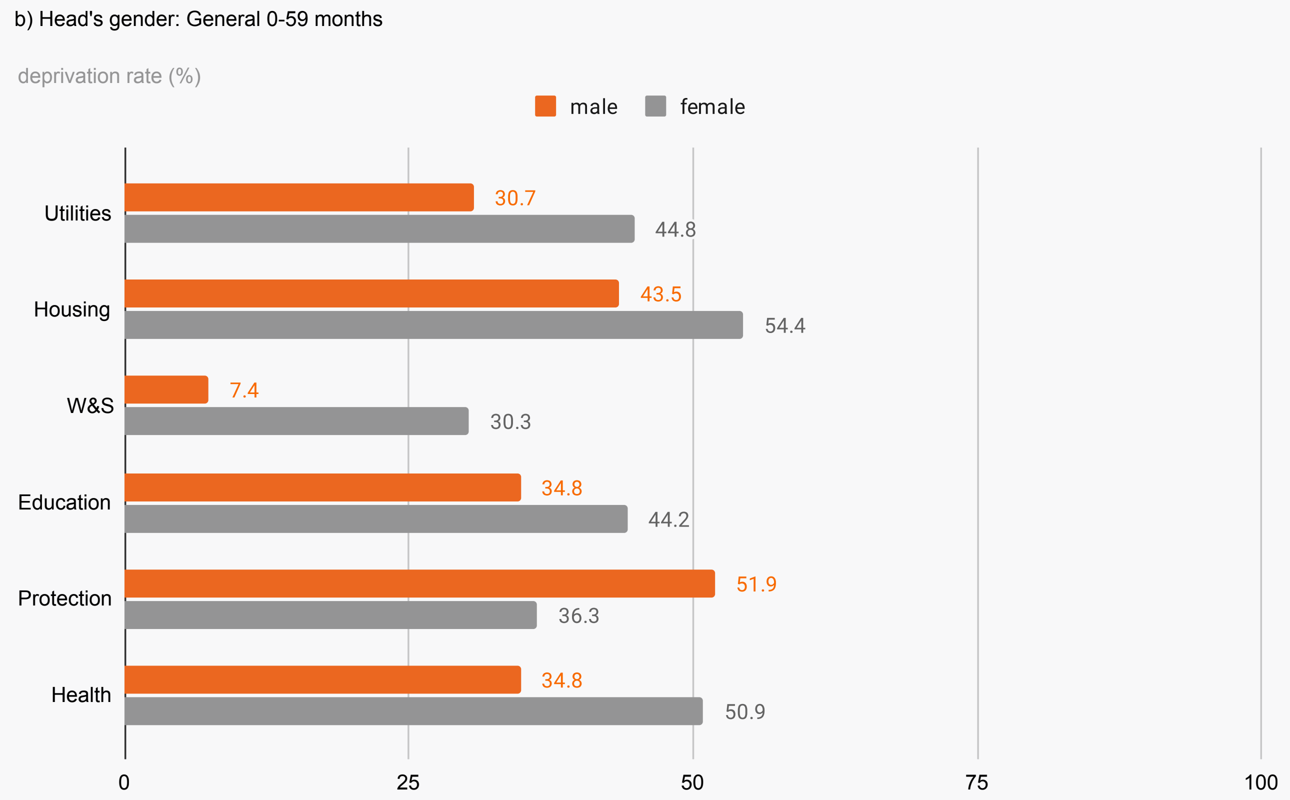
The deprivation rates calculated across different background characteristics help to understand which children are subject to some degree of deprivation, as well as to target those at higher risk of being deprived. For this analysis the study profiles children by analysing the incidence of deprivation for the dimensions under review for three of the 11 background characteristics (*see Annex A for the complete distribution of deprivations, and statistical tests of mean differences*). Figures 2 and 3 show the differences between children aged 0–59 months and those aged 5–17 years in terms of: rural vs. urban areas, male vs. female household heads, low vs. high education level of the mother (*see Annex A Table 1 for full results, including for the 0–23 months subgroup*).

The qualitative analysis also examined certain socio-economic variables to determine whether some groups of children are more vulnerable to specific deprivations than others. Specifically, the analysis examined deprivations experienced by children living in rural and urban areas, children living with single parents and/or in large families and children living in families where primary caregivers received social welfare due to unemployment.

Figure 2. Dimension headcounts by area of residence, sex of household head, mother's education: General population (children 0–59 months)

Source: Author's calculations based on 2018 MICS.





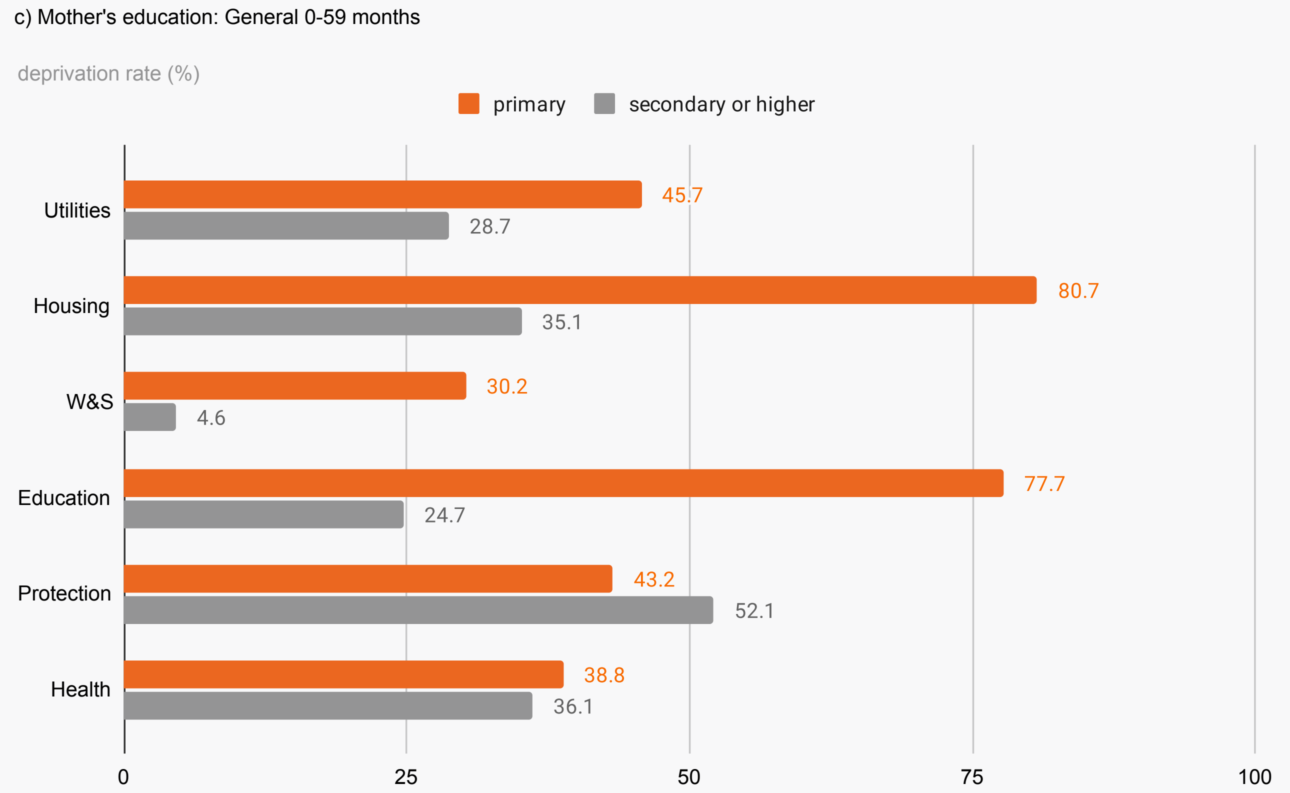
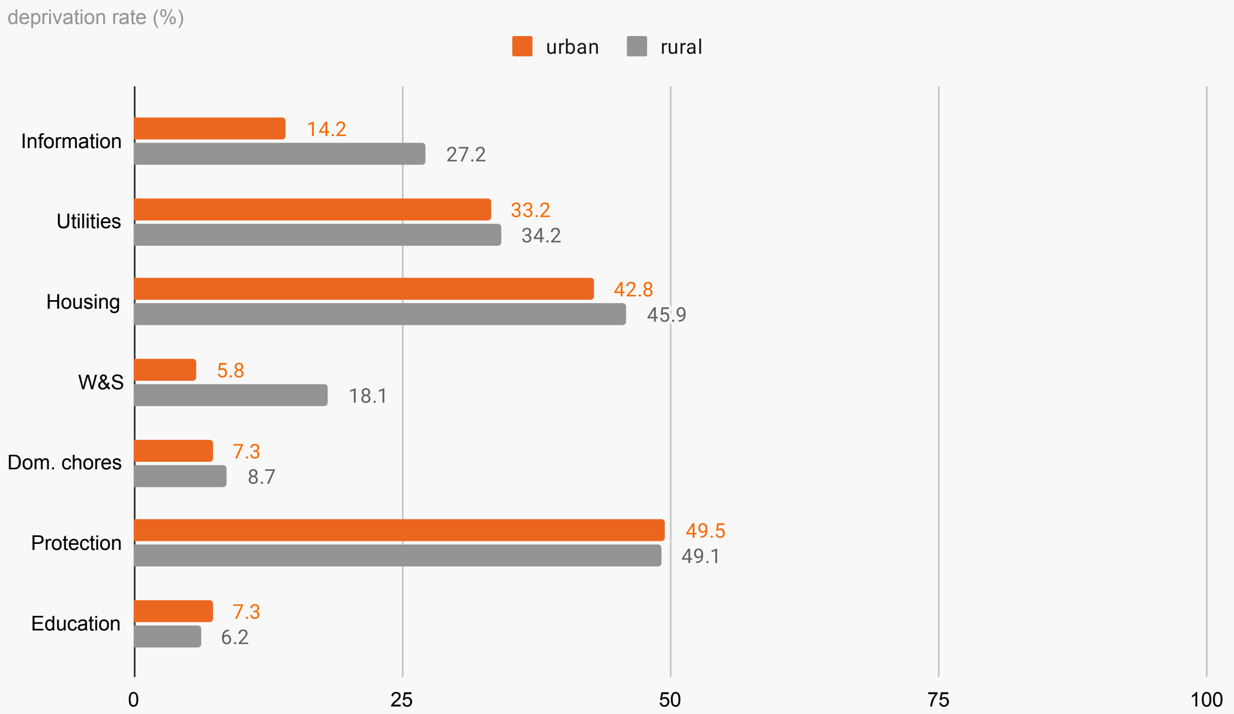


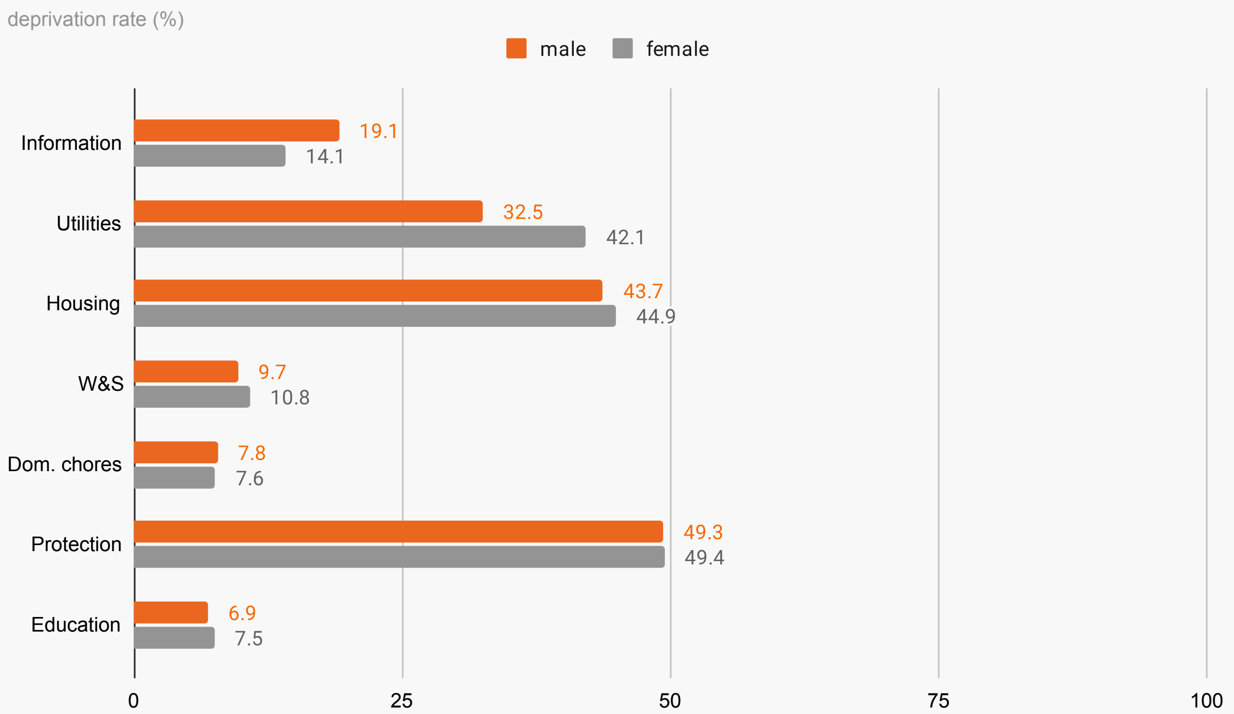
Figure 3. Dimension headcounts by area of residence, sex of household head and mother's education: General population (children 5–17 years)

Source: Authors' calculations based on MICS6 2018.

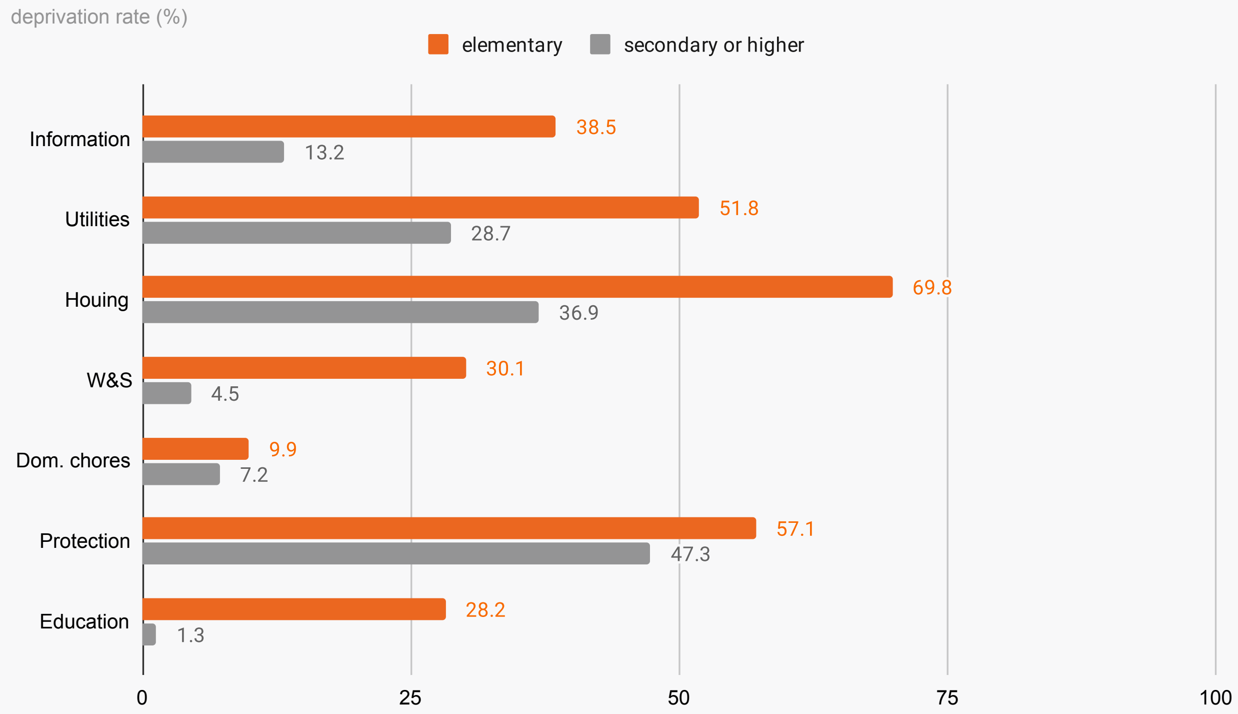
a) Urban v rural: General 5-17 years



b) Head's gender: General 5-17 years



c) Mother's education: General 5-17 years



Urban/rural

When looking at urban-rural inequalities, the data shows that the **incidence of deprivation is generally higher in rural areas, particularly for water and sanitation, education, utilities and information**. Qualitative data broadly confirms this result. Rural respondents, both children and parents, cited concerns about the quality and availability of safe drinking water, which is lacking in many homes. Thus children must collect unimproved, unsafe water from nearby sources, underscoring concerns for children's health.

> We have no water at home; they [the children] go a bit further down and carry it over here. This water has not been tested as safe, they will get sick!
> – FGD with unemployed parents from rural parts of Rožaje

Education: Distance and the remoteness of many rural towns were reported to negatively impact children's access to educational facilities, particularly primary school. Access to public transport is often unaffordable or unavailable, forcing many children to walk to school. Walking long distances presents safety hazards for children, including heavily trafficked roads without sidewalks, bad weather conditions and dangerous animals such as wild dogs that have been known to bite children. These concerns often lead to school absence, particularly for young children.

> I know of a lot of my friends who hike 10-15 kilometres every day to go to school.
> – FGD with adolescent boys from Rožaje

> I pay for transport for two [of my children], because it's dark, there are puddles, dirt road. The other two go by foot.
> – FGD with unemployed parents from Podgorica

Access to information: Rural-urban imbalances are also manifested in unequal access to information among adolescent children living in rural areas. Rural children are less likely to have internet access at home, which affects their educational attainment. An association between families living in rural settings and lack of access to information leaves many parents feeling concerned that their children may be falling behind children living in urban settings. Children, meanwhile, equate internet access and ownership of computers and smartphones with social status, noting a clear distinction between urban and rural families.

> Poor children live in remote villages and have no phone or internet access.
> – FGD with rural adolescent girls from Bijelo Polje

Health: Evidence of differences in health deprivations between rural and urban areas is somewhat mixed. When looking at immunization hesitancy, data suggests that some children living in urban areas are more deprived than those in rural areas. This result is not surprising, since caregivers in urban areas may have more exposure to deceptive information about vaccination safety and thus restrict their children's access to immunization.

Yet the **evidence also shows that children living in rural regions outside of Podgorica face wider gaps in access to health services and paediatricians**, as a result of spatial inequalities. For example, participants outside Podgorica mentioned significant differences in health protection between the capital and other municipalities, particularly in the northern region. They reported limited numbers of paediatricians in the north, leading to long wait times or the need to seek healthcare elsewhere, which is financially problematic for poor families.

> **Respondent 1:** [In Rožaje] we don't have healthcare workers and doctors. It's chaotic, given our town's population, [there are] 3,000 patients for a single doctor. This doctor cannot handle so many patients even if he is working with two computers and five assistants. **Moderator:** So what do people do? **Respondent 1:** They barely manage. **Respondent 2:** Those who can afford it, go to Berane; those who can't, they keep waiting their turn to receive treatment here.
> – FGD with parents from Rožaje

Housing/child protection: Qualitative data shows that **deprivations in housing and child protection are more prevalent in urban areas**. Urban life for the poor is often characterized by overcrowded housing involving multiple generations and inadequate or unsafe building conditions. Many poor families reported either renting homes, moving in with **large, multi-generational extended families** or building illegal houses. Access to credit to buy or build housing is limited, particularly for those informally employed, social welfare clients and single-parent families. Qualitative research found that some urban settings can serve as a predictor of child protection risks tied to criminal activity and substance abuse.

Head of household

The assessment of the distribution of deprivations by gender of the household head yielded interesting quantitative findings. The back-to-back plot reported in Figure 2 clearly shows that **children under 5 years old living in female-headed households are generally more deprived with respect to their peers living in male-headed households** in all significant dimensions except protection. These differences apply to the 0–23 month subgroup in all dimensions except education and nutrition (the latter is not available for the 0–59 month group), but disappear in six out of seven dimensions for children between 5 and 17 years (*see Figure 3 and Annex A Table 2*).

Mothers' education

Deprivations related to mothers' education levels are significant. In general, **children living in households where the mother's education is low are more deprived in all dimensions**. The differences are all statistically significant except for the health dimension (among the 0–59 age group) and domestic chores (for the 5–17 age group). The greatest differences were identified in the dimensions of water and sanitation (6.5 times larger) and education (three times larger). Deprivation in education reflects an even wider gap when comparing children aged 5–17 in families with more educated and less-educated mothers (1 per cent vs. 28 per cent).

What drives the risk for single deprivations?

The next step is to make use of the background characteristics to identify and examine the underlying determinants of child deprivation in each individual dimension. To accomplish this task, the study evaluated the extent to which each demographic or geographical predictor can influence the likelihood (probability) of deprivation, all else being equal.

A complete list of results for single deprivation logit-regressions can be seen in Annex B, which also includes further details on methods and analysis. Among children aged 0–23 months (*see Annex B, Table 1*), **mother's education, household head's education level and to some extent the head of household's gender** were found to be crucial in explaining the risk of deprivation for most dimensions. The average child of a mother with low education levels has a 39.3 per cent chance of being deprived in ECD, 28.2 per cent in housing, 23 per cent in nutrition and 10.8 per cent in water and sanitation, in comparison to children with more highly educated mothers. Approximately 22 per cent of children aged 0–23 months live in a household in which the mother has low education levels, and the analysis for this age group shows that the mother's education level is the most relevant determinant affecting their early development. Head-of-household education levels have a similar impact on four out of seven deprivations (health, protection, housing and utilities), while the household head's gender matters primarily for indicators related to water and sanitation and the ability to pay utility bills.

Among the **geographical variables**, children aged 0–23 months living in rural areas are more likely to experience deprivation in water and sanitation, while those living in the centre (excluding Podgorica) and north are less likely to be suffering from deprivation in the utilities dimension than those living in Podgorica (which we selected as the reference region). But children living in the north face lower deprivation risks for health and protection dimensions than those living in Podgorica.

Similar patterns emerge from analysis of the 0–59 months age group (see Annex B, Table 2). Children in **female-headed households** are again less likely to experience any form of physical or psychological aggression but are more likely to suffer from household-related deprivations, such as water, sanitation and utilities. The analysis shows the same associations registered for the 0–23 months group in relation to the education level of heads of household and mothers. The **ECD** dimension can also be explained by household size and area of residence. Children living in **larger households** are more likely to be subject to household resource constraints, which in turn will affect ECD access and outcomes and their likelihood of experiencing a deprivation for early child development. The chance of being deprived in ECD is also greater for those living in rural areas than for urban children.

Analysis of the results for children aged 5–17 (*see Annex B, Table 3*) again point to the importance of parental education in determining deprivation status in each domain. Less education is significantly associated with a higher probability of deprivation in household-level domains, including access to information. It is interesting to note that children living in a household where the head has less education are 8.9 per cent less likely to be deprived in terms of the number of domestic activities performed at home; girls are 10 per cent more likely than boys to be involved in such activities. The gender-differentiated impacts registered for children do not seem to apply for adults. The gender of the household head had no significant impact on any of the dimensions under review. Region of residence influences deprivation levels in education, and to some extent access to information: children living in the central, south and northern regions are 9 per cent, 5 per cent and 4 per cent more likely to be deprived in access to education and schooling outcomes, respectively.

UNICEF Montenegro / Zoran Jovanović Maccak

Finally, the study found qualitative evidence that household-level factors (such as family composition and size) affect children's exposure to and experience of deprivations, as described in Box 1.

Box 1: Are children from single-parent and large families at a greater risk of disadvantage?

Qualitative research gathered important insights about a range of deprivations affecting **children from single-parent families**. They were found to be more vulnerable than their peers living in nuclear families to specific deprivations, including for parental care, housing and access to services, as well as heightened stigma. Due to limited availability of time, single parents (particularly those who work and lack additional family support) also reported difficulties in caring for and protecting their children with regard to school pickup, ensuring healthy eating, spending adequate time together and taking children to social or cultural activities.

Financial constraints associated with single-parent households can be compounded by a lack of awareness (especially among women) about their social welfare rights and entitlements; many did not know they qualify for state-provided financial assistance. Very low income makes it difficult to afford toys or books for young children, and many also struggled to pay the supplemental costs of school, including supplies, books, transportation and additional classes for older children. This can negatively impact educational achievement and expose children to stigma and bullying. Having a single-parent household is a source of embarrassment in some cases, and teachers may not be equipped to protect children from the cruelty of their peers.

Children living in larger households also face a high risk of overcrowding, which can affect their schoolwork (due to lack of space and quiet to complete assignments) and lead to frustration and friction among family members. Respondents also mentioned the difficulty of asserting parental authority when extended families live together. As noted earlier, health issues can be exacerbated in cramped living conditions and basic expenditures stretched thin for large families relying on limited income.

3.2. Analysis of Roma settlements in Montenegro

Summary of key findings

- Extremely high levels of deprivation were registered for almost all dimensions. Inferior living conditions (housing) are a common denominator for all age groups. Deprivations in the domains of ECD (95.8 per cent) and nutrition (77 per cent) rank high among the 0–23 months age group. Difficulties related to access to education and poor school performance, although not ranking at the top of deprivations, are widespread.
- Children living in rural areas are generally worse off than their urban peers. Interestingly, the parent's level of education does not make any difference in any of the deprivation domains.
- The risk of deprivation in each domain is more likely to be related to the region of residence than to the demographic characteristics of Roma households.

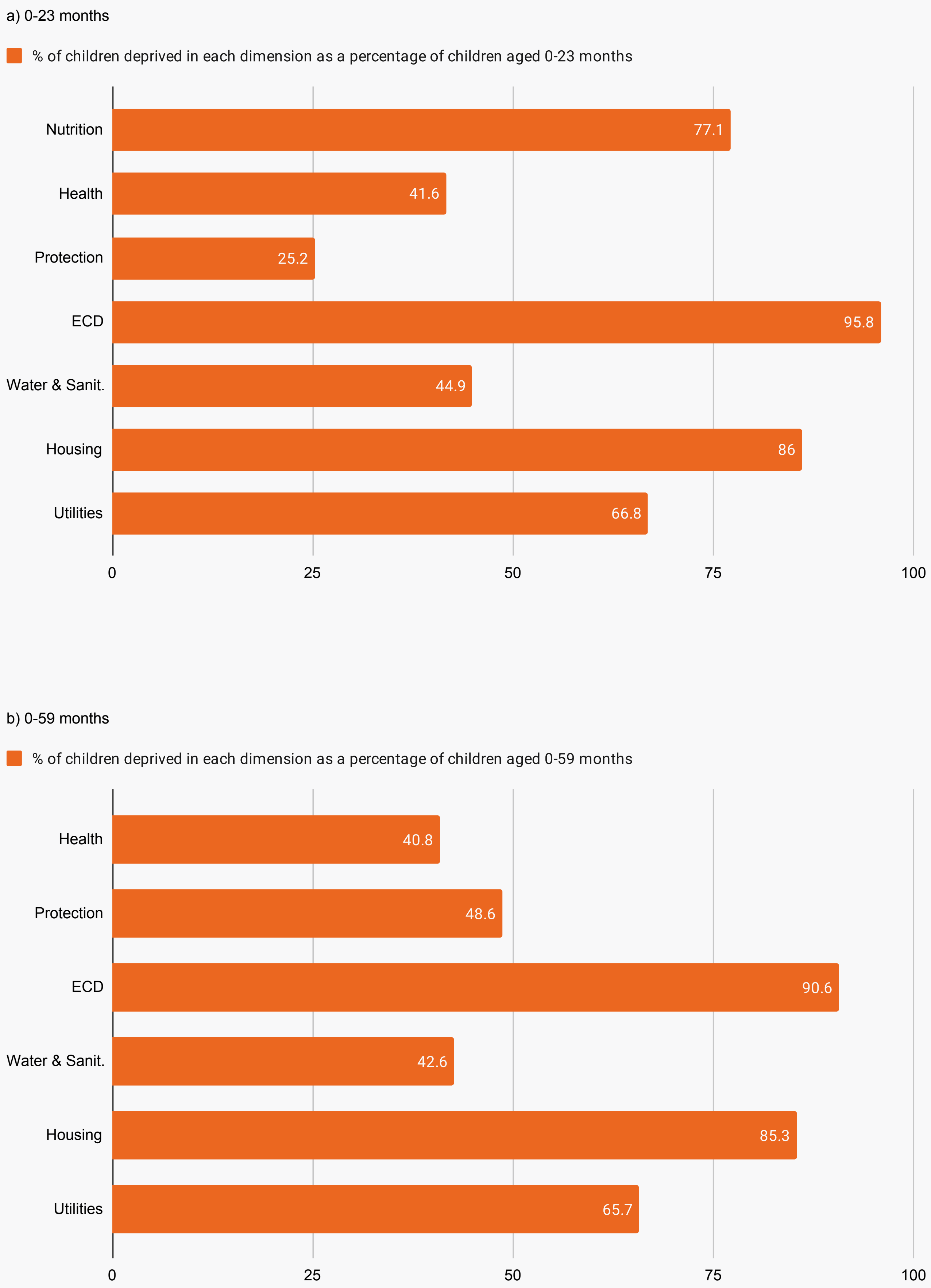
To further inform policies and interventions, the mixed-method MODA analysis was extended to children living in Roma settlements using data from the 2018 MICS. Analysis of the Roma sub-set replicates the procedures adopted for the general population, using the same dimensions, indicators, age groups and profiling variables (*see Table 2*).

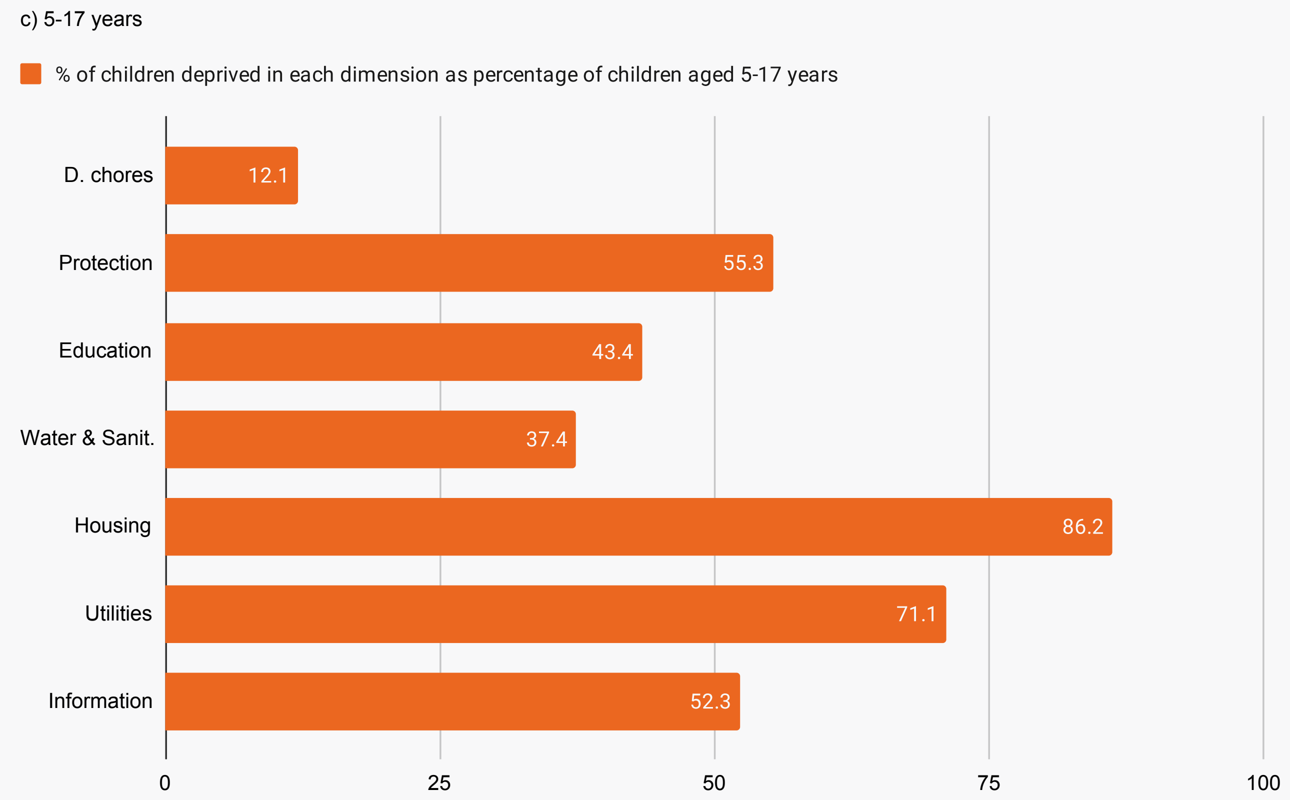
The single deprivation analysis for Roma settlements presents the findings for each of the dimensions and indicators previously identified for the general population. For each age group the analysis calculates the proportion of children living in Roma communities deprived in each dimension and indicator, providing relevant evidence about which aspect of children's life requires greater attention. As in the previous chapters, quantitative results are complemented and contextualized through qualitative inputs, to provide a more nuanced understanding of deprivations faced by children from Roma communities.

Deprivation analysis for children aged 0–23 months (*see Figure 4. a*) shows that of seven dimensions, Roma **children are most likely to be deprived in ECD (95.8 per cent), housing (86 per cent) and nutrition (77 per cent).** The high deprivation incidence registered for ECD is mostly driven by the very large number of children with no books suitable for their age at home (about 90 per cent).

Figure 4. Deprivation rates by dimension: Roma settlements (%)

Source: Author's calculations based on MICS in Roma Settlements, 2018.





Qualitative analysis confirmed that high levels of monetary poverty among minority ethnic populations result in fewer books and toys for young children. Preschool education emerged as an important topic during focus group discussions with mothers of Roma origin. In addition to being very eager to see their children complete their education, mothers placed great value on kindergarten and preschool due to their role in socialization and integration of Roma children in mainstream society. Most parents agreed that children who attend preschool 'fit' more easily into the wider society and have fewer difficulties in school.

Deprivations in the **housing and nutrition dimensions are mainly driven, respectively, by 'multiple housing problems'** (75 per cent incidence among children aged 0–23 months) **and lack of dietary diversity** (80 per cent). **Children of Roma origin appear to experience housing problems more frequently than their peers**. This was confirmed by qualitative research: during focus groups Roma participants reported that the buildings where they live are often of poor quality or illegally built and many children live in houses without bathrooms or safe drinking water.

> The local charity helped me out with the bathroom. I used to give my children baths outdoors.
>
> – FGD with unemployed mothers of Roma origin from Podgorica

For children 0–59 months (*see Figure 4.b*) deprivation levels for all dimensions are similar to those affecting the 0–23 month age group: education, housing and utilities are the three main deprivations experienced. The sole exception is for child protection, which at 48.6 per cent is almost twice as high as the deprivation level for the younger age group.

The results for children aged 5–17 years (*see Figure 4.c*) show that again **housing** (86.2 per cent) and **utilities** (71.1 per cent) are the two domains in which most Roma children are deprived. Exposure to violent discipline is high (over half of all children) and is in line with figures for the 0–59 months group.

Deprivation in education, although not ranking highest, is tied to poverty among children of Roma origin. Indicators in Table 4 show that only 80 per cent of Roma children attend school, and three out of every 10 lag behind in academic achievement. Financial constraints were identified as a key factor hindering school attendance, as many households cannot cover the costs of supplies, transportation and food, and some children said they were ashamed of going to school in the same clothes every day. Financial deprivation can also cause stigma and psychological aggression at school for children unable to afford socially acceptable clothing or shoes. Some Roma children attending secondary school reported receiving scholarships of €70 but said this was not nearly enough to cover school expenses, and many drop out.

> But someone perhaps doesn't have anything to wear, and goes without clothes or shoes, and gets teased at school; then they become too embarrassed to go.
>
> – FGD with adolescent boys of Roma origin, Podgorica

Another important school-related problem faced by Roma children is their parents' inability to assist them in their studies. Helping children to learn at home is quite common in Montenegro, where parents are expected to either aid their children with homework or pay for someone to do so. However, this requires local language skills and/or financial resources that many Roma families simply do not have. Children raised in segregated communities (e.g., Roma settlements) may also have only limited skills in the local language, and therefore need more educational aid than their peers. But few schools offer this type of assistance. This creates a significant problem; some students reportedly leave school without mastering basic reading or writing skills.

> What I believe to be the biggest problem is that, once they start school, they can't immediately get involved because of the language barrier. Then, no marks are given until the 3rd and the 4th grade, and there is also the issue of teaching staff. There are 30 children in a class and they cannot keep track of everything and also pay attention to children from the Roma and Egyptian communities who can't speak the language. These kids can't keep up in class in the early grades, and this continues into the 4th grade, when the grading starts, and then it's most often the case that teachers give them passing marks so they won't repeat the grade.
>
> – Key informant interview with an NGO representative

UNICEF Montenegro / Zoran Jovanović Maccak

Informants in Podgorica spoke of poor living conditions and dangerous social norms in Roma settlements, including alcohol and drug abuse. Such precarious conditions create an unsafe and unhealthy environment for children, affecting their development and overall wellbeing. Roma parents also viewed the purposeful concentration of minority ethnic populations in settlements designed exclusively for them as reinforcing their social segregation and limiting their access to social services, as well as exposing them to the potential risk of discrimination and intolerance from the general population. This can lead to negative outcomes for children's schooling and future prospects:

> This country doesn't know how to handle Roma people. You know why? When we come over here, they've put all of us in one place. And everything remains as it was. If the state had built 10 houses in Berane, 10 in Rožaje, mix the refugees with Montenegrins, then our children would start assuming their mentality.
>
> – FGD with parents of Roma origin, Podgorica

Finally, with regard to nutrition, adolescent children of Roma origin were less hesitant than others to report malnutrition, and drew attention to the many hungry children in their communities who may be falling short of both minimum dietary diversity and minimum acceptable diet:

> Go get a soda, go to our neighbourhood and just watch how many children will gather around you.
>
> – FGD with adolescent boys of Roma and Egyptian origin, Podgorica

Dimension	Indicator	Age group		
		0–23 months	0–59 months	5–17 years
ECD/education	Has no books suitable for the age or doesn't play with any toys	95.93	92.67	-
	Has no books suitable for the age or is not involved in early stimulation activities	-	87.87	-
	Behind in grade for age	-	-	31.01
	Not attending school	-	-	19.41
Health	No selected paediatrician	22.62	19.83	-
	Delayed vaccination	20.36	24.48	-
	Refused vaccination	9.05	12.07	-
Nutrition	No exclusive breastfeeding	49.15	-	-
	Minimum dietary diversity	80.25	-	-
	Minimum meal frequency	52.47	-	-
Child protection	Psychological aggression*	41.12	54.94	52.95
	Physical punishment*	32.71	38.41	29.11
	Neglect**	4.98	7.93	-
Child labour/ domestic chores	Domestic chores	-	-	12.24
Water and sanitation	Unimproved drinking water	0.45	0.34	0.21
	Absence of shower unit / bathtub	32.58	31.72	28.69
	Access to unimproved sanitation	29.77	26.10	23.19
Housing	Multiple housing problems	75.11	72.76	74.89
	Overcrowding	50.23	55.52	50.00
Utilities	Delayed bill payment	45.25	46.21	49.37
	Energy affordability	41.18	41.03	45.36
Information	Internet access	-	-	50.63
	Mobile	-	-	10.55

Table 4. Deprivation rates by indicator and age group, as a percentage of total children in the age group: Roma settlements

Source: Authors' calculations based on 2018 MICS in Roma settlements.

* The reported means refer to children 11–23 months in the age group 0–23 months and to children 11–59 months in the age group 0–59 months. Overall data on psychological aggression and physical punishment is available for children 11 months and older.

** The reported means refers to children aged 0–11 months.

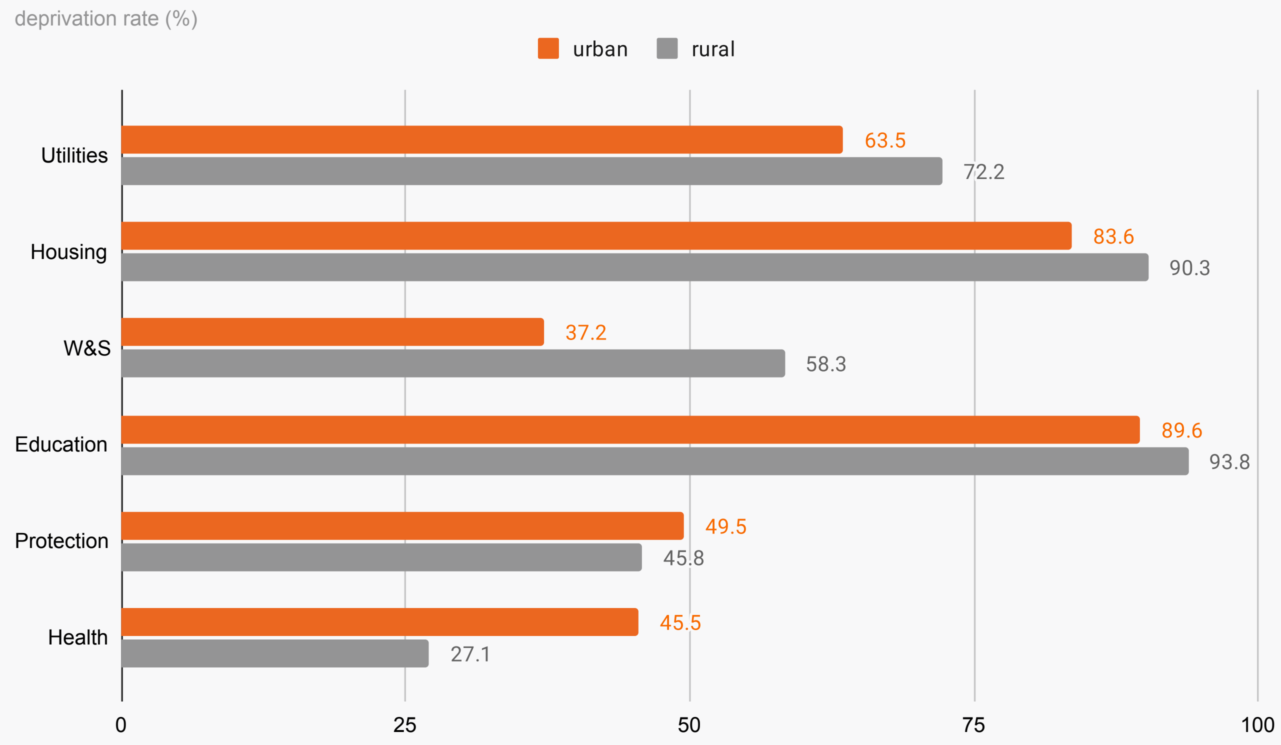
Which children of Roma origin face a higher risk of deprivation?

To analyse the profile of deprived children, the same set of background characteristics previously introduced for the general population was used to assess how these characteristics are correlated with each dimension. Annex A, Table 2 shows the sample composition of these background variables. Roma settlements are mostly concentrated in urban areas and in the central region, which included Podgorica in the MICS6 for Roma settlements. About 10 per cent of Roma children live in the south while 20 per cent live in the north. The sample is equally split between girls and boys. Most children live in families with more than seven household members; 60 per cent live with male household heads around 35-to-40 years old with secondary or higher education. The mothers of about half of the children have low educational levels. Very few children (less than 5 per cent) live with a caregiver reporting a disability. Annex A reports deprivation headcounts by dimension and background characteristic, along with the t-test for the difference in means between subgroups. Figures 5 and Figure 6. show deprivation headcounts for three selected background characteristics: urban vs. rural; low vs. high education level of mother; male vs. female head of household. Full results are available for all age groups, dimensions and background characteristics (*see Annex A, Table 4*).

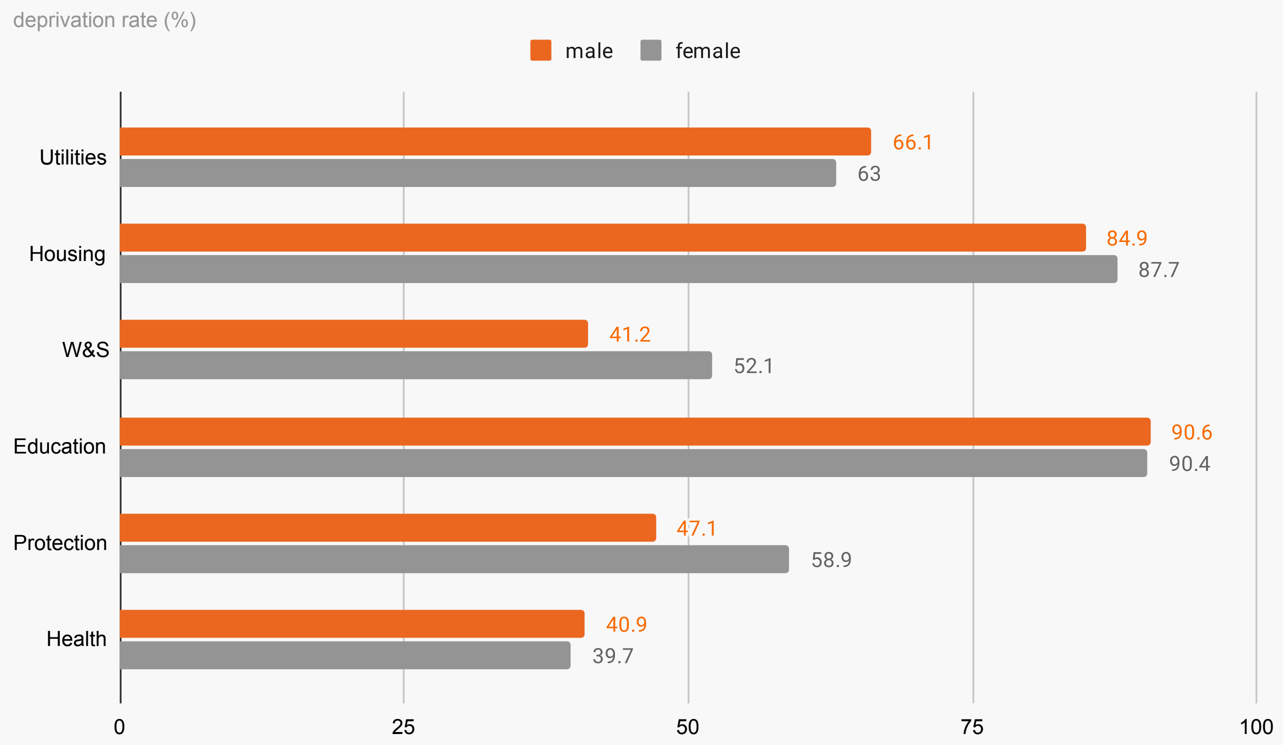
Figure 5. Dimensions headcount by area of residence, sex of household head and mother's education: Roma settlements (children 0–59 months)

Source: Authors' calculations based on MICS6 Roma Settlements, 2018.

a) Urban v rural: Roma 0-59 months



b) Head's gender: Roma 0-59 months



c) Mother's education: Roma 0-59 months

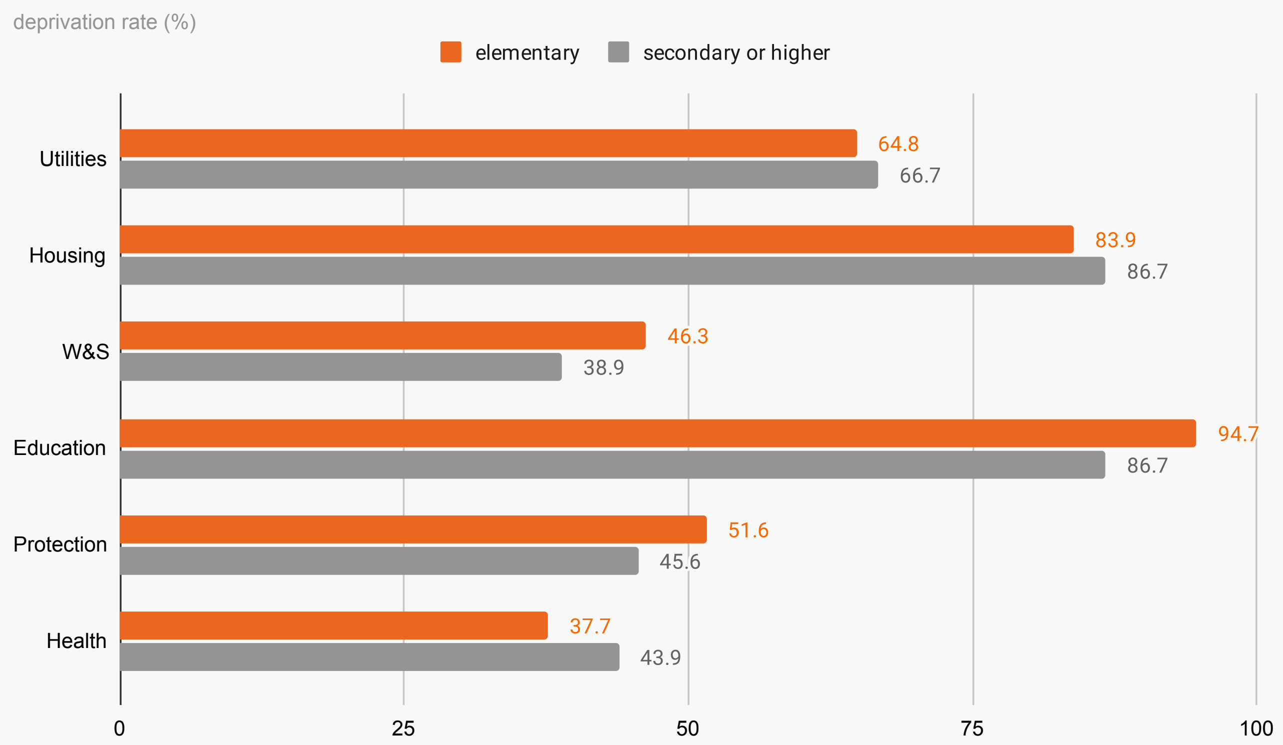
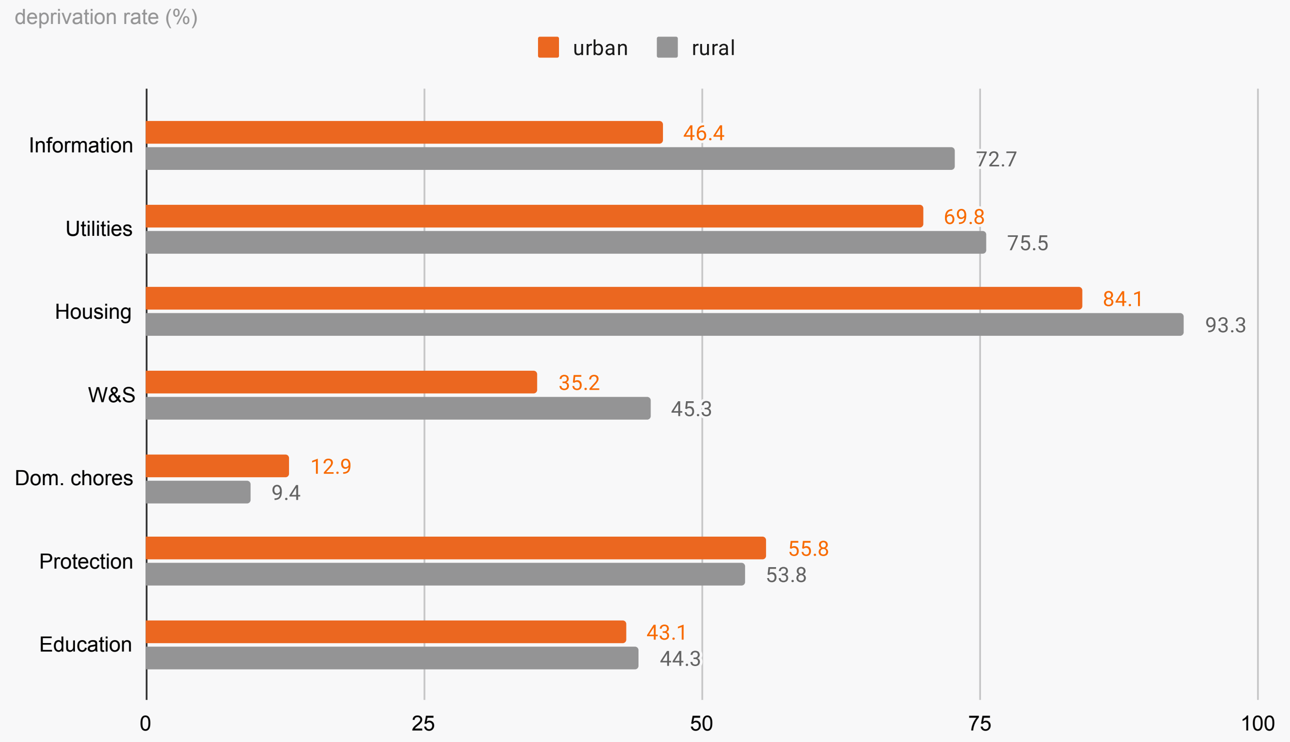


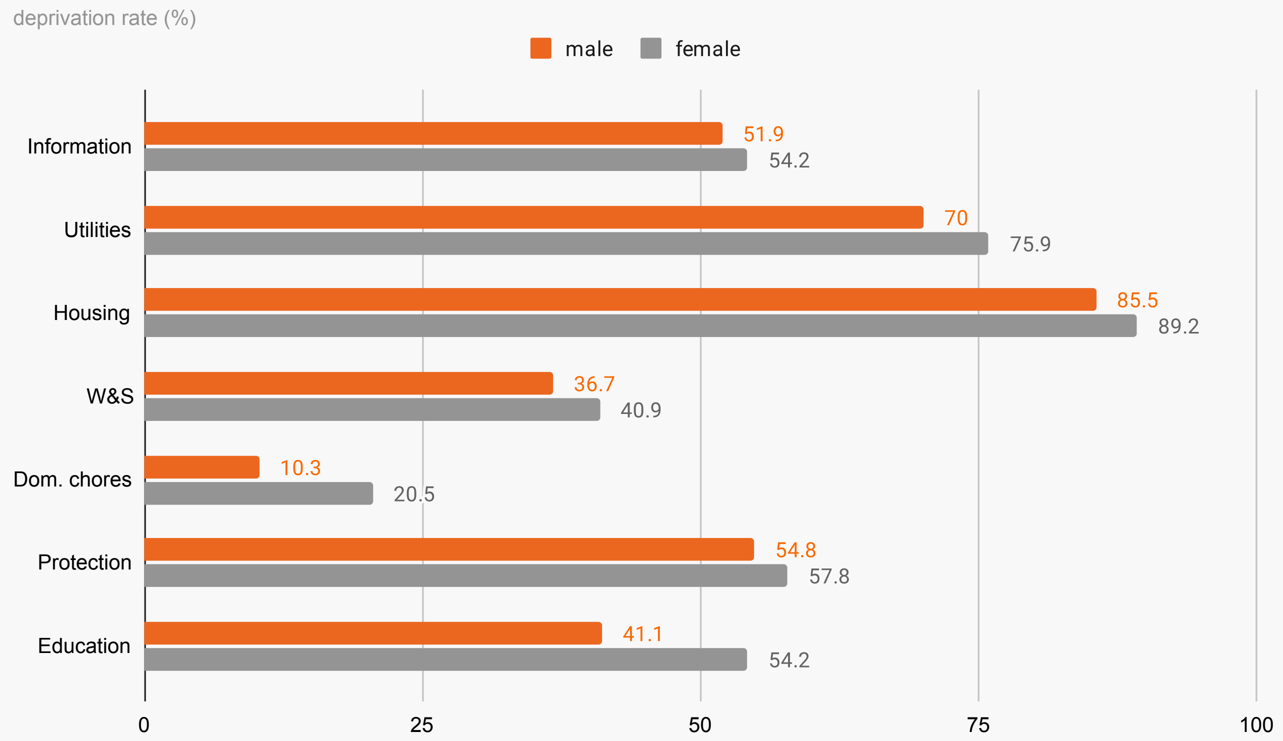
Figure 6. Dimensions headcount by area of residence, sex of household head and mother's education: Roma settlements (children 5-17 years)

Source: Authors' calculations based on 2018 MICS in Roma Settlements.

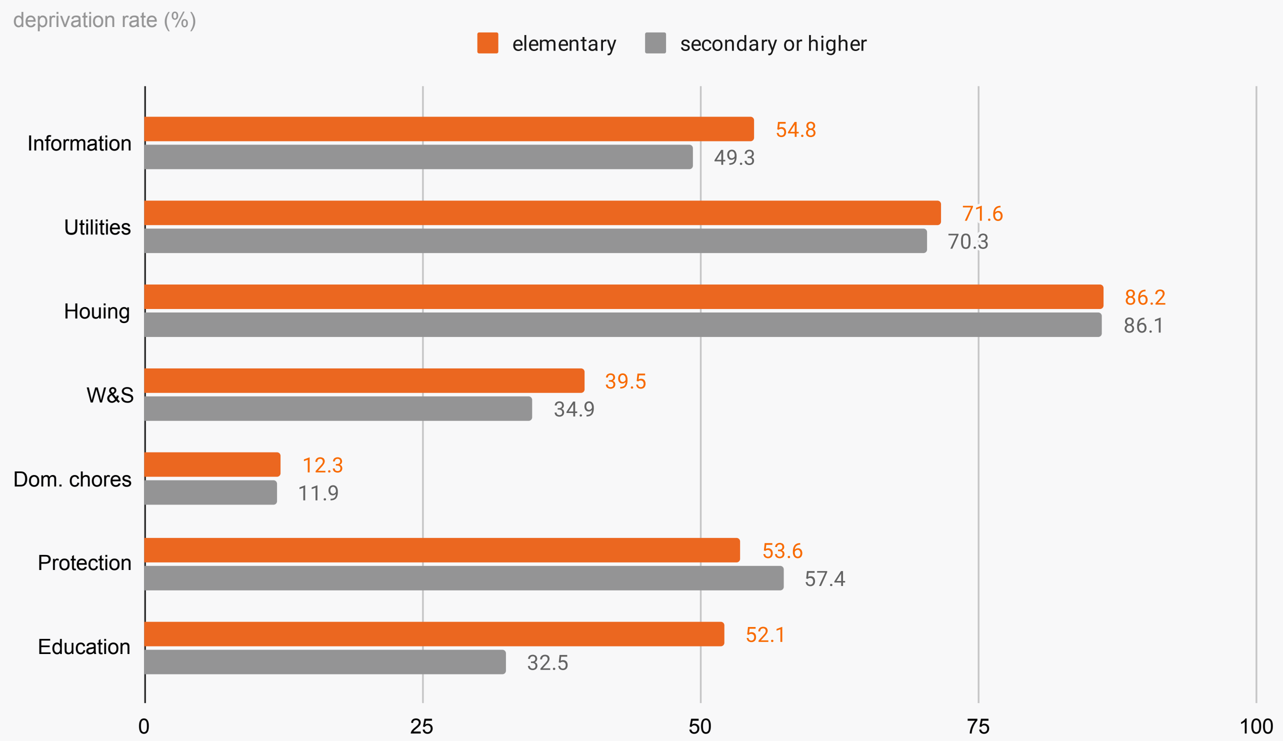
a) Urban v rural: Roma 5-17 years



b) Head's gender: Roma 5-17 years



c) Mother's education: Roma 5-17 years



On average, **Roma children aged 0–59 months living in rural areas experience greater deprivation in the areas of child protection, ECD and water and sanitation** than those living in urban areas. **Older children (5–17 years age group) appear to be worse off in other domains** (**access to information, housing and water and sanitation**). Qualitative findings indicate that housing-related problems disproportionately affect children in rural areas, and children of Roma origin seemed to experience housing problems more often than their peers in the general population. As noted previously, Roma respondents frequently live in informal Roma settlements or poor-quality and/or illegally built houses, lacking bathrooms and clean drinking water.

Roma children aged 0–59 months living in urban areas are highly deprived in the health dimension. Interestingly, there appears to be no statistically significant difference in the incidence of deprivations for the two education-related background characteristics. The share of children deprived in each dimension is approximately the same regardless of the mother's education level. This applies to all dimensions for children age 0–59 months (except nutrition for children aged 0–23 months) and to all except education for 5–17-year-olds. Children aged 5–17 years living in households headed by a more highly educated person are less deprived only in terms of access to information and water and sanitation; no differences in any other domain were registered for the two younger age groups. The **head of household's gender matters only for the education dimension**; high deprivation levels in ECD/education were found for all age groups.

What drives the risk for single deprivations in the Roma population?

To answer this question, the report assesses the extent to which the selected background characteristics are associated with an increased probability of experiencing deprivation in each domain (See Annex B). The resulting picture is mixed; only a few demographic factors appear to be significant, but not for all dimensions. This suggests that no specific demographic factor unequivocally influences the various deprivations simultaneously. Rather there is a complex combination of contributing factors that must be carefully evaluated on a case-by-case basis. For example, holding everything constant, children age 0–59 months living with mothers with low levels of education are significantly more likely to be deprived in ECD, but less likely to experience housing problems. The head of household's education levels have a positive correlation only with deprivation in the water and sanitation dimension, while their age is directly related to deprivation in the areas of health

and utilities. **Geographical factors are found to be better predictors of deprivation than demographic variables, especially for the 0–59 months age group**. Children living in both the northern and southern regions are significantly more likely to be deprived in four out of six dimensions (ECD, water and sanitation, housing and utilities) than those living in the central region.

Finally there is an inverse relationship between the head of household's age and the likelihood that a child will be deprived of protection and access to information, and a positive association between his or her age and the extent of domestic chores required of children. Living in a female-headed household increases a child's chances of being deprived in education by 14.3 per cent and of being assigned domestic chores by 8.3 per cent. The data shows that **the education level of mothers and heads of households has a direct impact on their children's access to education and information**.

Photo: Dusko Miljanić / UNICEF Montenegro

4. Multiple Overlapping Deprivation Analysis

4.1. General population

Summary of key findings:

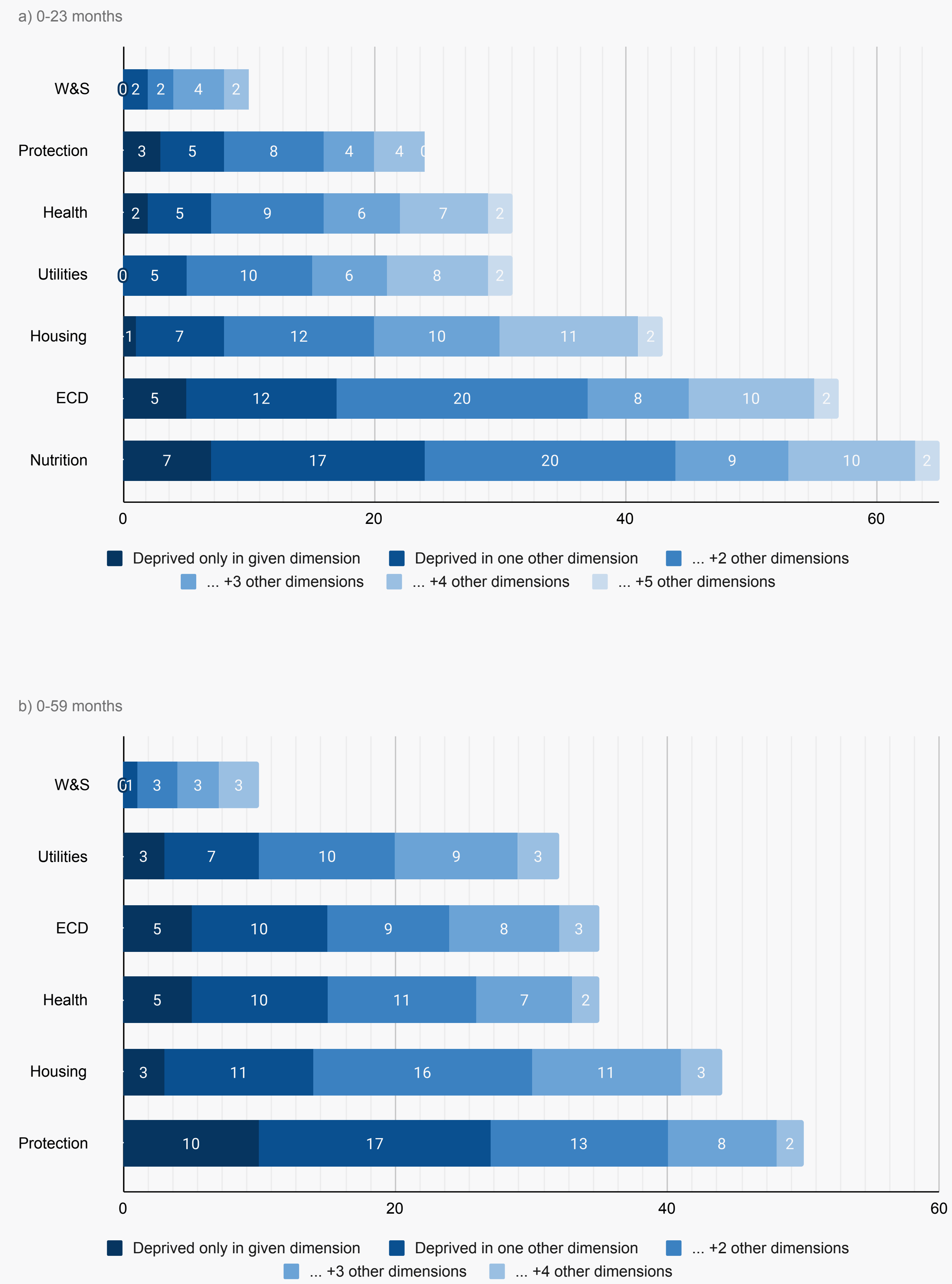
- Around one-fifth of Montenegrin children under the age of 5 do not suffer from any deprivation. This share declines dramatically to one tenth for children 0–23 months and is highest (25 per cent) for children aged 5–17 years.
- Child protection has the greatest overlap with deprivations in other dimensions of wellbeing for children 0-59 months and the 5–17 years age groups. Nutrition overlaps with other dimensions primarily for children aged 0–23 months.
- More than 80 per cent of children experience deprivation in at least one domain of their well-being, regardless of age group; younger children tend to be simultaneously deprived in more dimensions than older children.
- Children aged 0–23 months experience the highest prevalence of multidimensional child poverty: 77.6 per cent of children in this age group suffer from two or more deprivations, with an average intensity of 45.5 per cent of all possible deprivations. Some 64.3 per cent of children aged 0–59 months and 49.1 per cent of those aged 5–17 years are deprived in multiple dimensions.
- The education level of the mother and the head of household has a significant influence over the whether or not a child will be deprived at many levels.

Quantitative Overlapping Deprivation Analysis

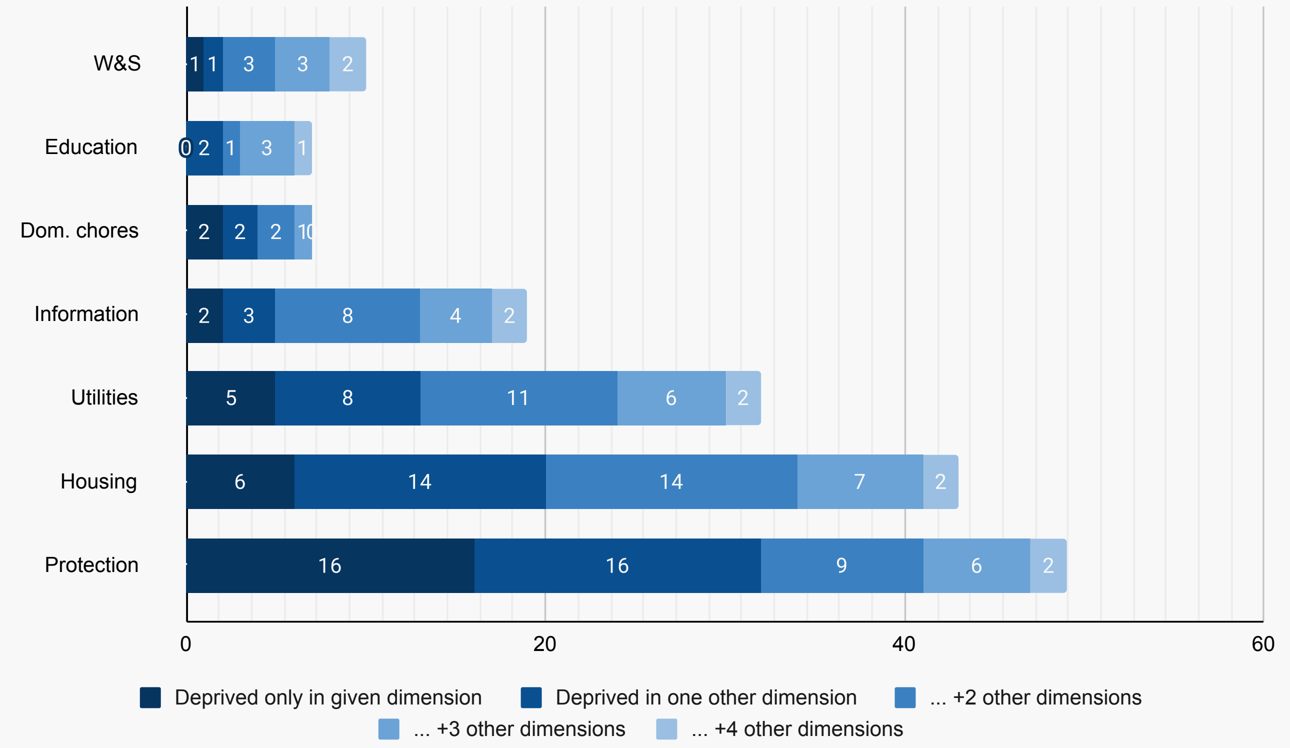
To better gauge the severity of deprivations faced by children it is fundamental to examine the extent to which they experience multiple deprivations simultaneously and to investigate which children suffer from several deprivations at once. What are the dimensions with the greatest overlap? Figure 7. depicts the overlap for each dimension for the 0–23 months age group, showing the proportion of children deprived in the given dimension and in one or more additional dimensions. **Results suggest that most children deprived in any single dimension are also deprived in two, three or four other dimensions** (*see Figure 7*). Large overlaps are particularly evident for ECD, nutrition and housing. Taking ECD as an example, 7 per cent of children are deprived only for the ECD dimension, while 53 per cent are deprived for ECD *and* one to five additional dimensions. Overlaps between dimensions for the 0–59 months and 5-17 years age groups are depicted in Figures 7.b and 7.c. **For children under five, deprivation in child protection has the highest overlap** with other dimensions; 17 per cent of children are deprived of protection and one other dimension; 13 per cent are deprived in two additional dimensions**. For children aged 5–17 years, child protection is the domain that overlaps the most with other dimensions**; housing and utilities come in second and third. The prevalence of deprivation in education and the extent of overlap are lower than for children under five years of age.

Figure 7. Overlap between dimensions: general population (%)

Source: Authors' elaboration on MICS6 2018.



c) 5-17 years



The Venn diagrams included in Figure 8. show (proportionally) the deprivation overlaps for different dimensions for each age group. A Venn diagram of any combination of three dimensions shows the deprivation levels for: (1) each dimension separately, (2) the overlap between two dimensions and (3) the overlap between three dimensions. For simplicity, included here are the three dimensions with the highest prevalence. Deprivations in nutrition, ECD and housing are plotted for children aged 0–23 months; protection, housing and health for children 0–59 months; and protection, housing and utilities for children 5–17 years.

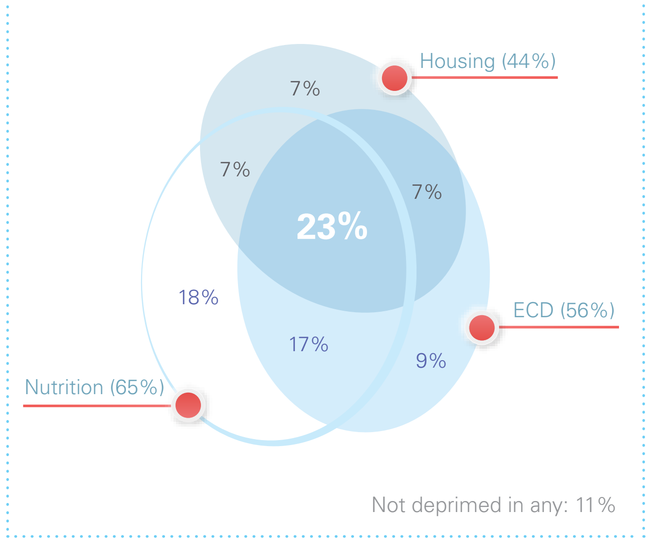
Deprivation in housing is common to all three age groups and in protection for two out of three, highlighting their relevance for multi-dimensional deprivation. Figure 8.a highlights key findings to inform programming interventions intended to tackle multidimensional child deprivation among children in the 0–23 month age group, suggesting the presence of a substantial degree of overlap: 23 per cent of these children are deprived in all three dimensions simultaneously, while just one out of 10 is not deprived in any of the three dimensions analysed. Seventeen per cent are deprived in ECD and nutrition (but not housing), 7 per cent are deprived both in ECD and housing (but not nutrition) and nutrition and housing (but not ECD). Eighteen per cent of young children are deprived in nutrition, but not ECD or housing.

Figure 8.b illustrates a similar analysis for children aged 0–59 months, showing that the degree of overlap between the three main dimensions (protection, housing and health) is low (11 per cent) compared to the overlapping deprivations for the 0–23 months group. At the same time children suffering from two or three of these deprivations simultaneously represent 42 per cent of all children in the 0–59 months age group, highlighting the strong influence of these three dimensions on much of the population. Sixteen per cent of the children in this age group suffer from deprivations in the child protection domain (but not in the other two); nearly one out of five is not deprived in any of these domains. Figure 8.c, depicts the deprivation overlap analysis for children in the 5–17 years age group. Again, the overlap of multiple deprivations is somewhat lower than was the case for the 0–23 months age group. Figure 8.c shows that 13 per cent of children are simultaneously deprived in protection, housing and utilities, while 25 per cent are not deprived in any of these domains. Among the three dimensions, deprivation in protection (19 per cent) is less frequently accompanied by deprivation in the other two, while 11 per cent of children are deprived only in housing and 6 per cent only in relation to utilities.

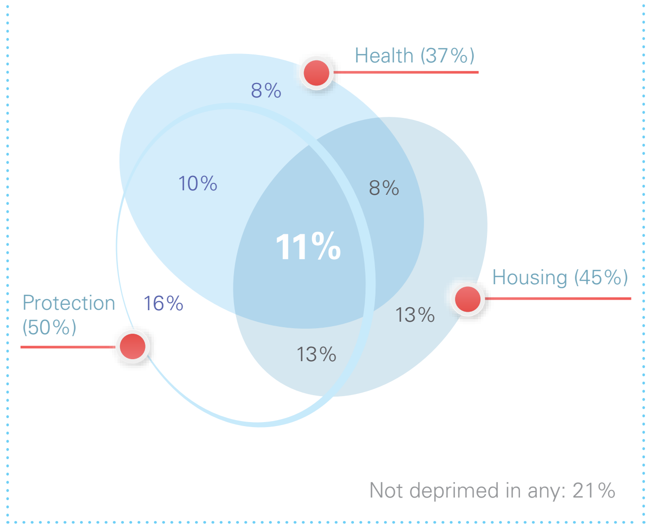
Figure 8. Deprivation overlaps between the three dimensions with highest prevalence, by age group

Source: Authors' elaboration on MICS6 2018.
Note: The area-proportional 3-Venn diagrams were drawn using the EulerAPE tool (Micallef and Rodgers, 2014).

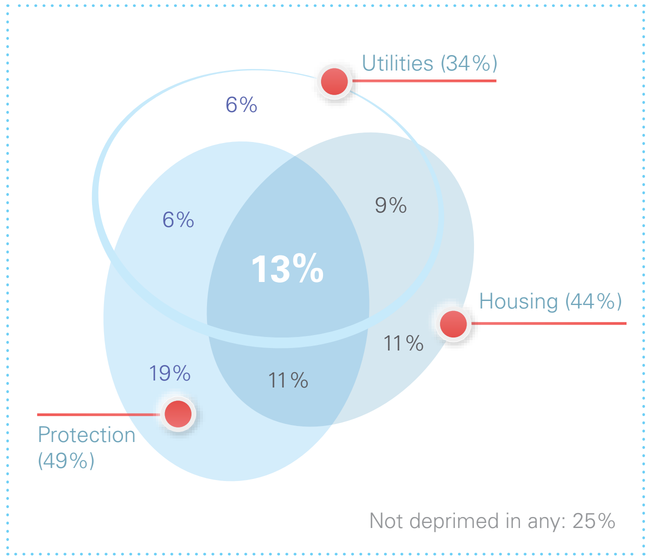
a) general population (0–23 months)



b) general population (0–59 months)



c) general population (5–17 years)



Multiple Deprivation Analysis

The adoption of a single-sector analysis as a tool to measure child survival and development is not sufficient in its scope because children have multiple needs and rights that must be satisfied at the same time. In other words, to evaluate the severity of multidimensional deprivation, increased attention should be devoted to *all dimensions simultaneously*. Multidimensional deprivation analysis is an effective instrument to improve understanding of both the extent and depth of deprivation, as it moves from sector-specific analysis towards a child-centric perspective by counting the number of deprivations children experience simultaneously.

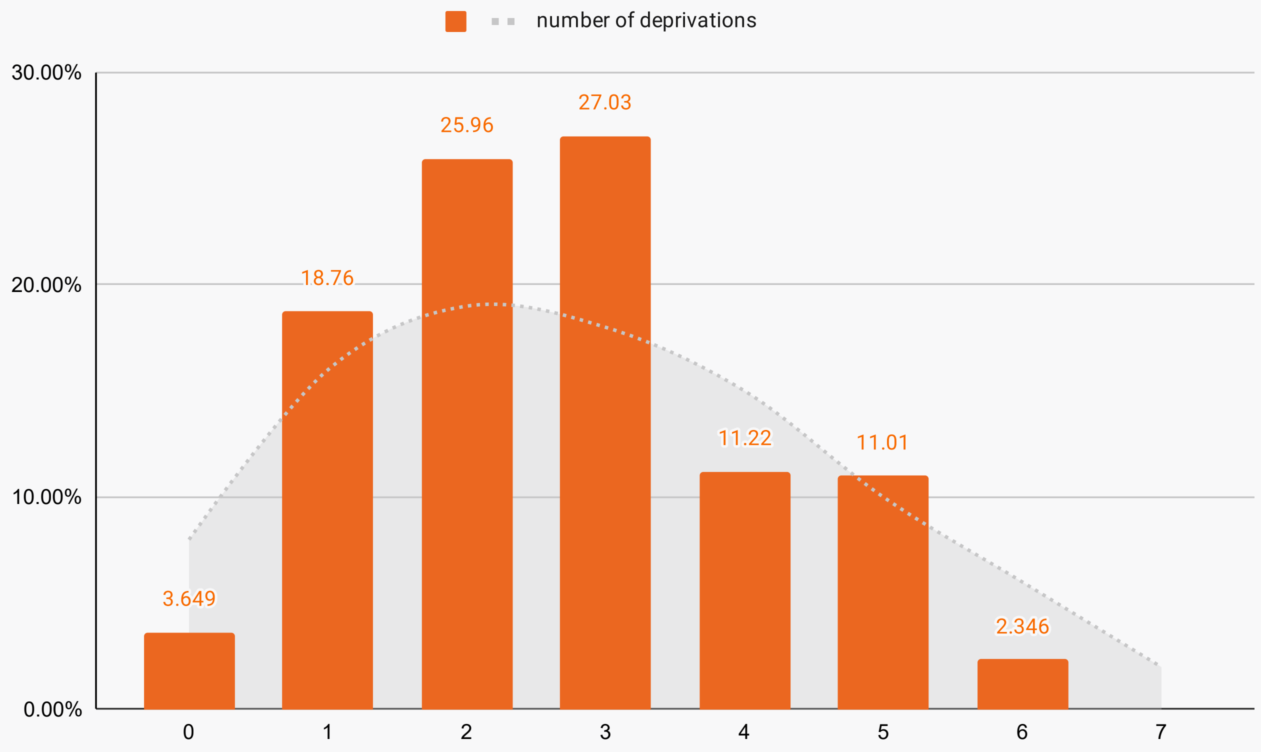
The next step of the present analysis involves calculating the cumulative deprivations suffered by each child and their distribution by age group (*see Figure 9*). Distributions skewed to the left indicate a low prevalence of multidimensional deprivation, while those skewed to the right indicate a higher incidence of multiple deprivations.

In Montenegro more than 80 per cent of children experience deprivations in at least one domain of their well-being, regardless of age group. All distributions are left-skewed with peaks at one dimension, except for children aged 0–23 months, whose distribution peaks at three deprivations (27 per cent). **Overall, quantitative analysis demonstrates that younger children tend to be simultaneously deprived in more dimensions than older children**. At the same time, qualitative analysis revealed that older adolescents (15–17) also face many deprivations simultaneously, threatening

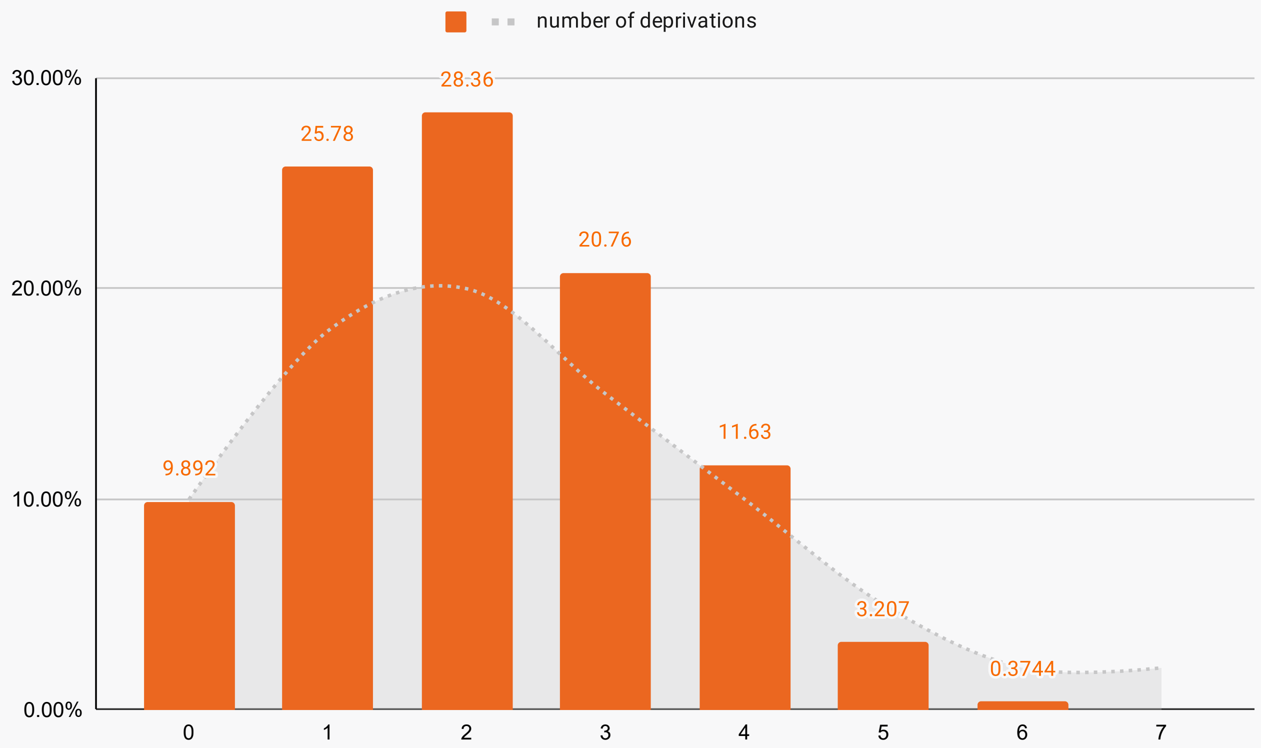
Figure 9. Distribution of number of deprivations, by age group: general population (%)

Source: Authors' depiction based on 2018 MICS data.

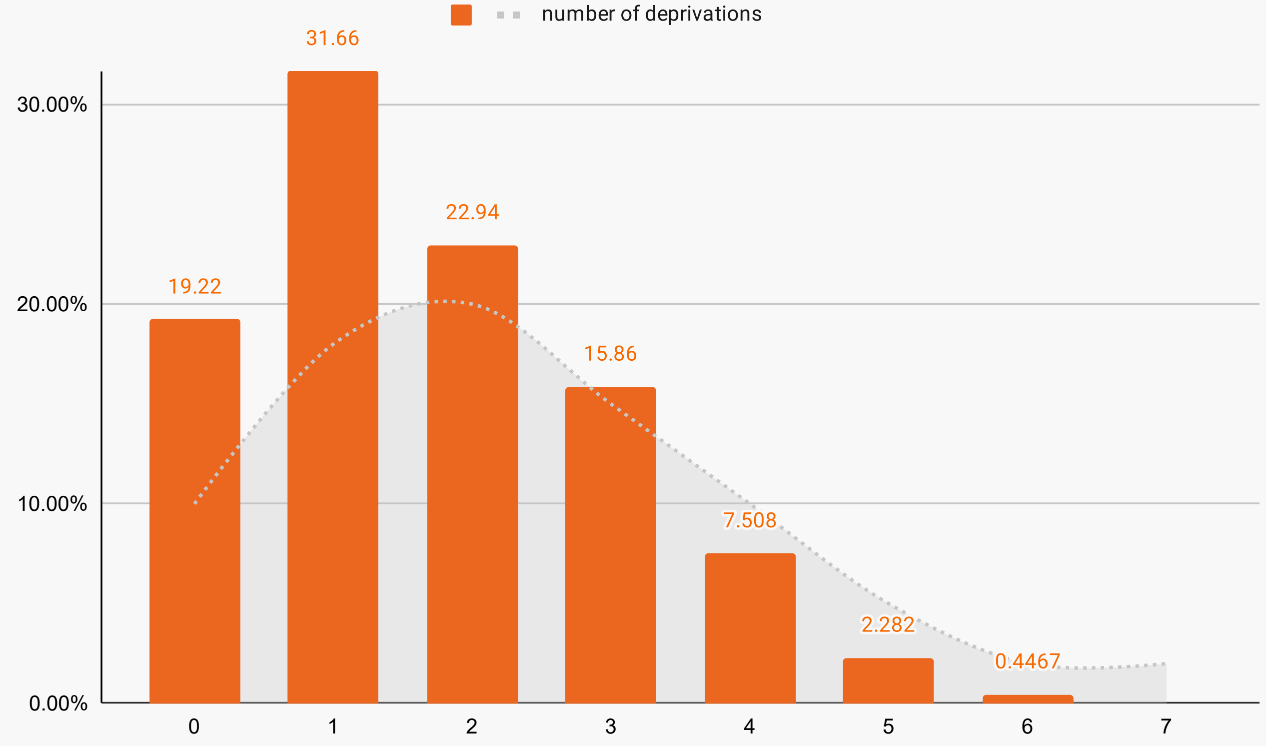
a) General population: 0-23 months



b) General population: 0-59 months



c) General population: 5-17 years



their safe and healthy transition to adulthood, as discussed in Section 6.2.

The above headcount distributions can also be reported using multidimensional deprivation *indices*, which are calculated at different cut-offs from multi-dimensional deprivation: if children face two or more, three or more or four or more deprivations, they are considered to be multi-dimensionally deprived. As shown in Table 7, multidimensional deprivation indices are calculated for each age group. In the case of Montenegro, for each age group the deprivation threshold (k=2) is the minimum threshold to ascertain whether or not children are multi-dimensionally deprived. This threshold identifies children with two or more deprivations as being multi-dimensionally deprived and delivers an accurate reference point to inform policymakers about actual levels of multi-dimensional poverty. The multi-dimensional deprivation headcount (H) represents the proportion of children considered to be deprived based on a given multi-dimensional deprivation cut-off (*k*). The average intensity of deprivation (A) returns the breadth of deprivation for those children identified as multi-dimensionally deprived. Finally, the adjusted multi-dimensional deprivation headcount (M_0) reflects at the same time the incidence and intensity of multi-dimensional deprivation (see Annex C for further details on the construction of multidimensional deprivation indices). The M_0 index cannot be directly interpreted but can serve to compare deprivations among different subgroups of children (i.e., at equal H levels, M_0 is higher on average for those experiencing a higher number of deprivations).

The data shows that 77.6 per cent of all Montenegrin children aged 0–23 months suffer from two or more deprivations, with an average intensity of 45.5 per cent of all possible deprivations. Meaning that children identified as deprived in two or more dimensions are deprived on average in 45.5 per cent of possible dimensions; that is, 3.2 dimensions. By multiplying H by A it is easy to derive the adjusted headcount ratio of 35.3 per cent. Using the same cut-off, 64.3 per cent and 49.1 per cent of children aged 0–59 months and 5–17 years, respectively, are multi-dimensionally deprived. The average intensities of deprivation show approximately the same values across age groups, but as the cut-off increases the incidence of poverty *decreases*, while the average intensity of multidimensional deprivation *increases*.

Age group	Cut-offs	HC (%)	A (%)	M_0 (%)
0–23	2+ out of 7 dimensions	**77.6**	45.5	35.3
	3+ out of 7 dimensions	51.6	54.0	27.9
	4+ out of 7 dimensions	24.6	66.3	16.3
0–59	2+ out of 6 dimensions	**64.3**	47.6	30.6
	3+ out of 6 dimensions	36.0	58.9	21.2
	4+ out of 6 dimensions	15.2	71.0	10.8
5–17	2+ out of 7 dimensions	**49.1**	46.9	23.0
	3+ out of 7 dimensions	26.2	58.7	15.4
	4+ out of 7 dimensions	10.3	72.2	7.4

Table 5. Multidimensional poverty headcounts, intensity of poverty and adjusted headcount ratios at different cut-offs, by age group

Source: Authors' elaboration on MICS6 2018.

As was the case for the single deprivation analysis, the study investigated the push and pull factors among background characteristics most likely to be associated with an increase or decrease in multidimensional child deprivation. Even though the analysis can be performed for each possible cut-off, the authors used the 3 or more and 4 or more cutoffs as dependent variables for demonstrative purposes. Figure 10 presents the marginal effects obtained from the 3 or more deprivations cut-off (10.a) and 4 or more deprivations cut-off models (10.b). Each panel incorporates the coefficients for all background characteristics for children aged 0–23 months (green), children aged 0–59 months (blue) and children age 5–17 years (orange). It is apparent from both panels that, if all else remains constant, **the level of education of the mother and head of household are again strongly associated with the likelihood of experiencing multiple deprivations**. This is evident and systematic for children belonging to each age group at any cut-off. This result is consistent with the results obtained from the single deprivation analysis, suggesting that parents' education plays a central role in determining the risk for both single and multiple deprivation.

Children aged 0–23 and 0–59 months are more likely to be deprived in four or more dimensions if they live in a rural area. Children 0–59 months living in either the southern or northern regions are less likely to be suffering from three or more deprivations than children living in Podgorica.[36]

These results corroborate evidence from the single deprivation analysis, where geospatial predictors were found to influence sector-specific deprivations. The evidence also points to an increased risk of deprivation in four or more dimensions for children living in households managed by females. Children aged 0–59 months living in households where the caregiver has functional difficulties are more likely to be deprived in three or more or four or more dimensions, although the estimates are not precise, as the coeffient refers to less than 2 per cent of the children sampled.

36 Podgorica is taken as reference category.

Figure 10. Marginal effects from logistical regressions, by age groups and cut-offs

Note: To improve readability, the household size variable was excluded from this figure; full estimates are available on request.

Qualitative Deprivation Overlap Analysis

Qualitative evidence confirmed that children face multiple overlapping deprivations, particularly in housing, water and sanitation and health, but also in education and information. Analysis yielded several important insights into the types of deprivations that frequently overlap and how they mutually reinforce each other, with diverse consequences for child wellbeing.

First**, several deprivations are linked primarily to financial constraints and monetary deprivation.**

> Everything comes from finances. It means lack of proper nutrition, lack of education, even lack of health.
>
> – FGD with parents from Rožaje

Children whose parents receive social welfare assistance expressed concern about the connection between financial means and health. They explained that some ill children are not properly treated due to a lack of money.

> … Nowadays, we often hear about four- or five-year-old children that are very ill. They don't have the financial resources to get treatment, and it's really horrible what's happening today.
>
> – FGD with adolescent boys from Bijelo Polje

Similarly, a lack of money for computers and internet connectivity, which are required for many school assignments, affects children's educational outcomes when homework is left undone. Poor utilities also affect education, due to rural electrical outages or inability to pay utility bills, leaving children without light to study. Additional costs associated with school attendance are often unaffordable for poorer families, which can also result in shame and ridicule for their children:

> At school, they always ask for something that costs €1 or €5. It is never cheaper for us who are on welfare. When there's an excursion, we pay the same as everyone, the same goes for a birthday gift, or a gift for the teacher after the fourth grade, we all pay equal amounts. If you don't pay, your children will be ridiculed right away.
>
> – FGD with unemployed parents from Nikšić

Second**, substandard housing is closely linked with poor water quality and ill health**, **as well as quality of education and access to services and information**. Substandard or overcrowded housing is also problematic with regard to education, as children must study in noisy common living rooms or unhealthy environments where they cannot concentrate on their tasks.

Third**, the distance and remoteness of rural locations also negatively impacts access to healthcare and educational facilities, including pre-schools, and is compounded by unaffordable transportation costs.** In addition, walking presents safety concerns, especially for young children, leading to school absence and/or restricted access to healthcare:

> There is an outpatient clinic down there, but it's not working. Only the one here in town works. We go down there by foot, it takes an hour. I have five children, it will be six soon, I do need a car [to take them to the clinic].
>
> – FGD with parents from Rožaje

Finally, **poor nutrition is also closely linked with adverse education and health outcomes for children**, as a poor diet results in ill health, low energy and sub-optimal school performance.

> I believe that housing, nutrition and health are the most important, because if a person is sick or hungry then they cannot get an education.
>
> – FGD with unemployed parents from Nikšić

4.2. Roma settlements

Summary of the key findings:

- Deprivation overlap reaches extremely high levels for almost all dimensions for children in Roma settlements. ECD and housing show the greatest overlap with other deprivations for children under 5 years and those aged 5–17. For children aged 0–23 months, nutrition has the highest overlap with other dimensions.
- Two-thirds (65 per cent) of children aged 0–23 months experience simultaneous overlapping deprivations for nutrition, ECD and housing.

- Almost all children are deprived at the cut-off of two or more dimensions. The majority of children reach high levels of deprivation intensity. Overall, children experiencing three or more deprivations at once tend to suffer from extreme poverty.
- Living in a female-headed household or in the northern region are two key factors that appear to exacerbate a child's probability of being multi-dimensionally deprived.

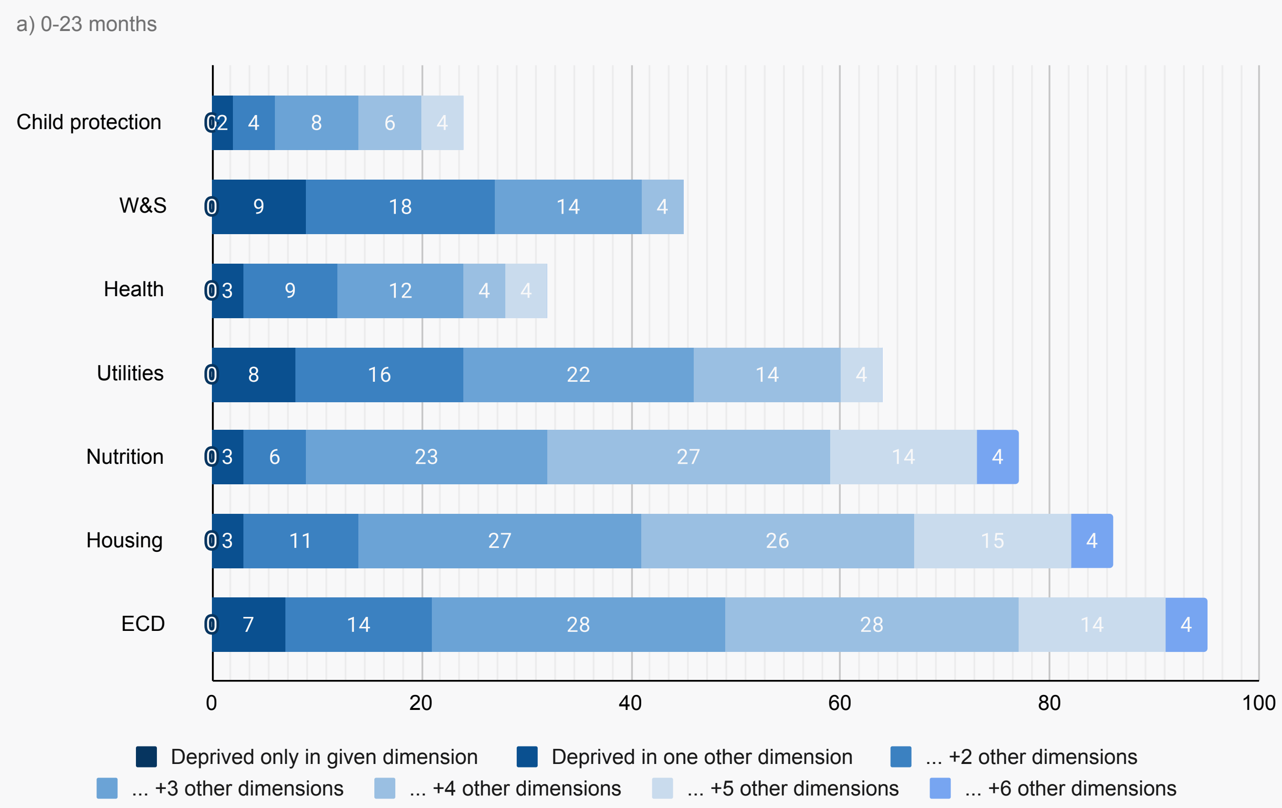
Quantitative Deprivation Overlap Analysis

Figure 11 depicts deprivation overlaps, by dimension, for children aged 0–23 months, 0–59 months and 5–17 years for the Roma subsample. Among the under-2 age group, the extent of overlap is large for ECD, housing, nutrition and utilities. About one out of every four children is deprived in ECD, housing or nutrition *and* three or four additional dimensions. At the same time *all* children are deprived in at least one dimension and 4 per cent are deprived in all dimensions simultaneously (the specified dimension plus six others). The same is true for the 0–59 months age group, but with fewer children deprived in four or more additional dimensions compared to the younger sub-set. Also, it is rare to be deprived in only one dimension (green bars): just 2 per cent of children are deprived only in ECD. For children aged 5–17 years, deprivation in housing overlaps most frequently with deprivation in other dimensions. About one child out of five deprived in housing is also deprived in two, three or four other dimensions. Concerning utilities, which is the second most overlapping dimension, 17 per cent report a deprivation in two other additional dimensions, 21 per cent in three additional dimensions and 19 per cent in four additional dimensions.

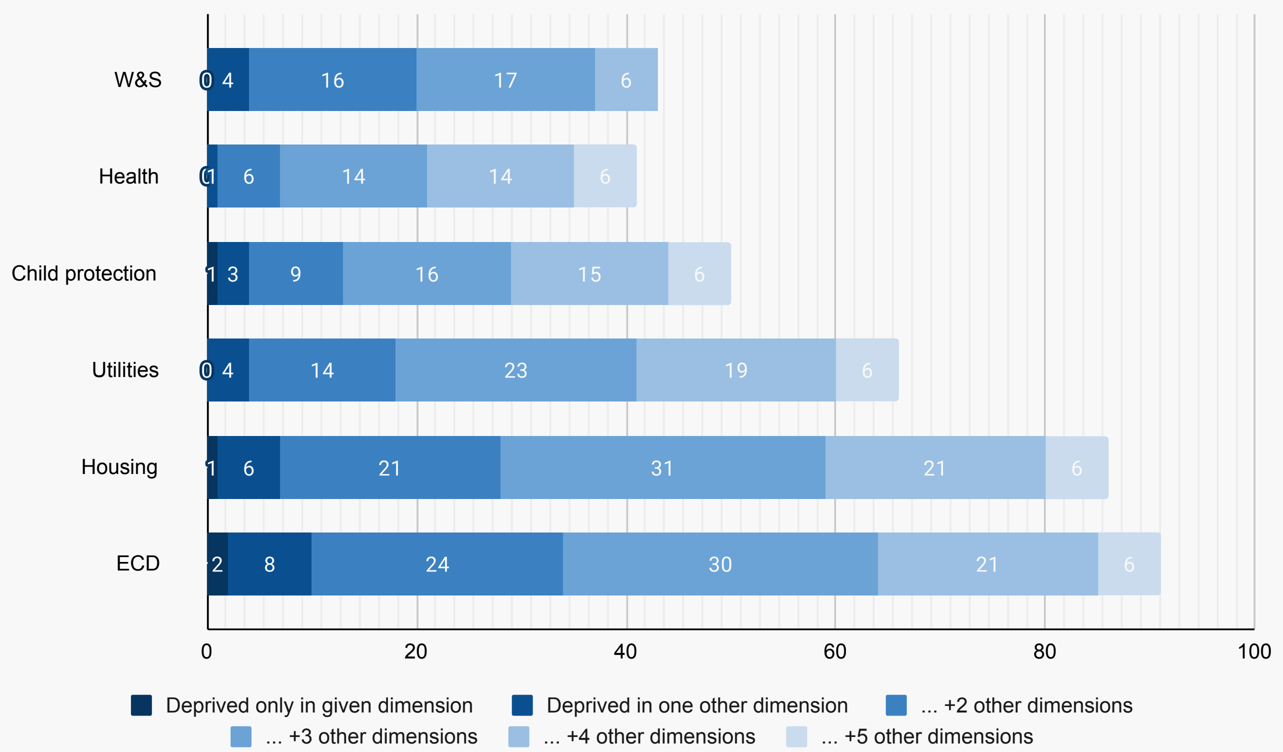
The combinations of deprivations experienced simultaneously are best presented using Venn diagrams. This section depicts only the overlaps among the three most prominent deprivations for each age group. Figure 12 shows a very high degree of overlap in deprivations for nutrition, ECD and housing. **Two-thirds (65 per cent) of children aged 0–23 months are deprived in all three**

Figure 11. Overlap between dimensions: Roma settlements (%)

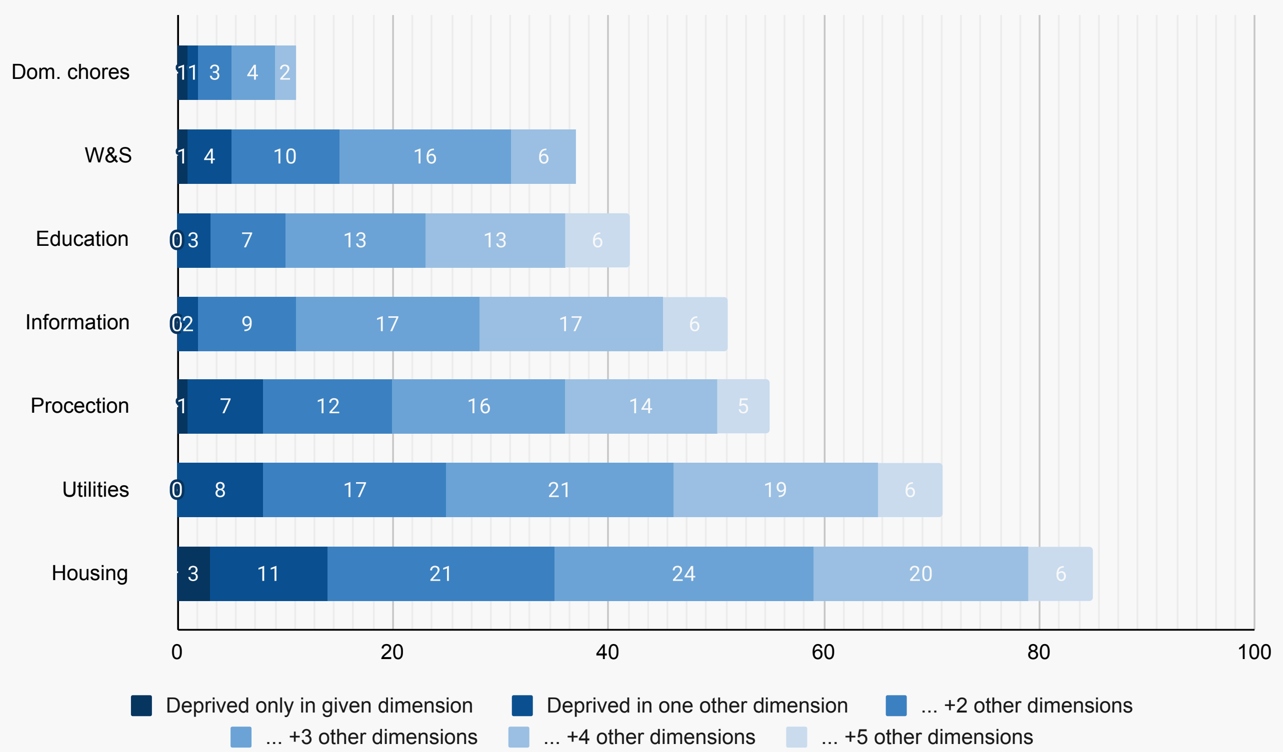
Source: Authors' elaboration based on MICS6 2018 – Roma Settlements.



b) 0-59 months



c) 5-17 years



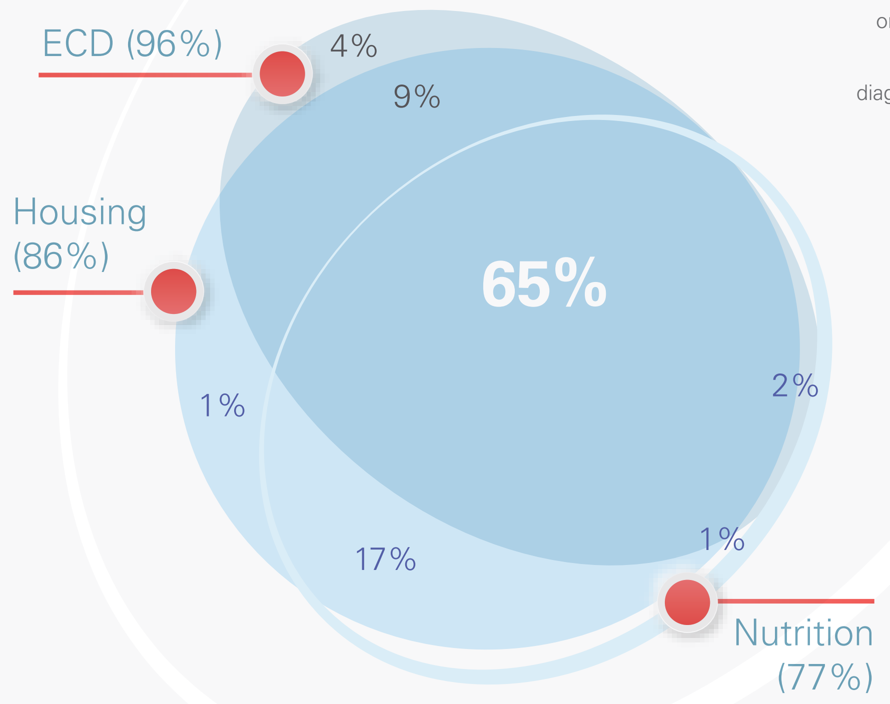
dimensions simultaneously. Nutrition and housing overlap at around 82 per cent. While 77 per cent of children are deprived in nutrition, only 1 per cent are not also deprived in ECD and/or housing. Graphs 12.b and 12.c show a similar, although slightly less pronounced, triple overlap. Fifty-three per cent of children aged 0–59 months and 35 per cent of children aged 5–17 years are simultaneously deprived in all three dimensions.

Figure 12. Deprivation overlaps between the three dimensions with highest prevalence, by age group

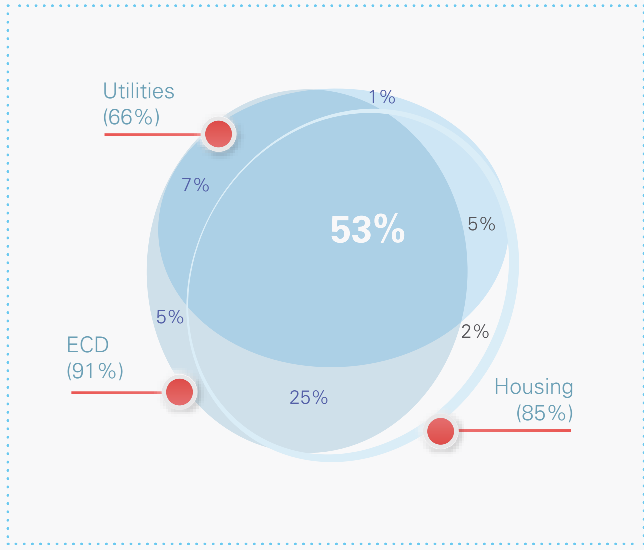
Source: Authors' elaboration based on MICS6 2018 – Roma Settlements.

Note: The area-proportional 3-Venn diagrams are drawn with EulerAPE tool (Micallef and Rodgers, 2014)

a) Roma settlemets (0–23 months)

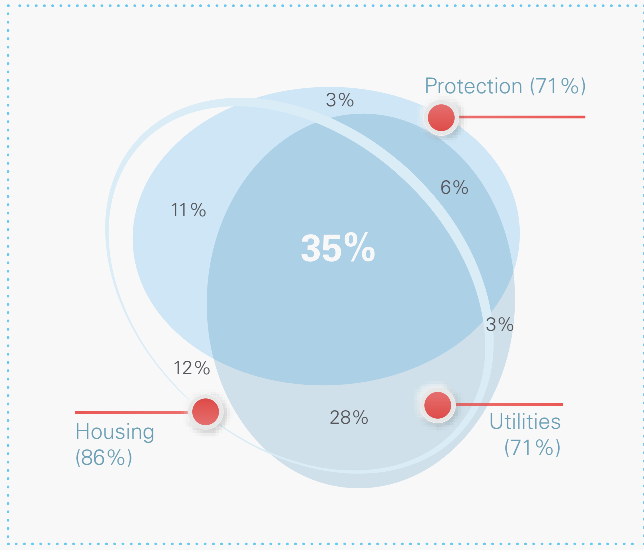


b) Roma settlements (0–59 months)



Not deprived in any: 1%

c) Roma settlements (5–17 years)



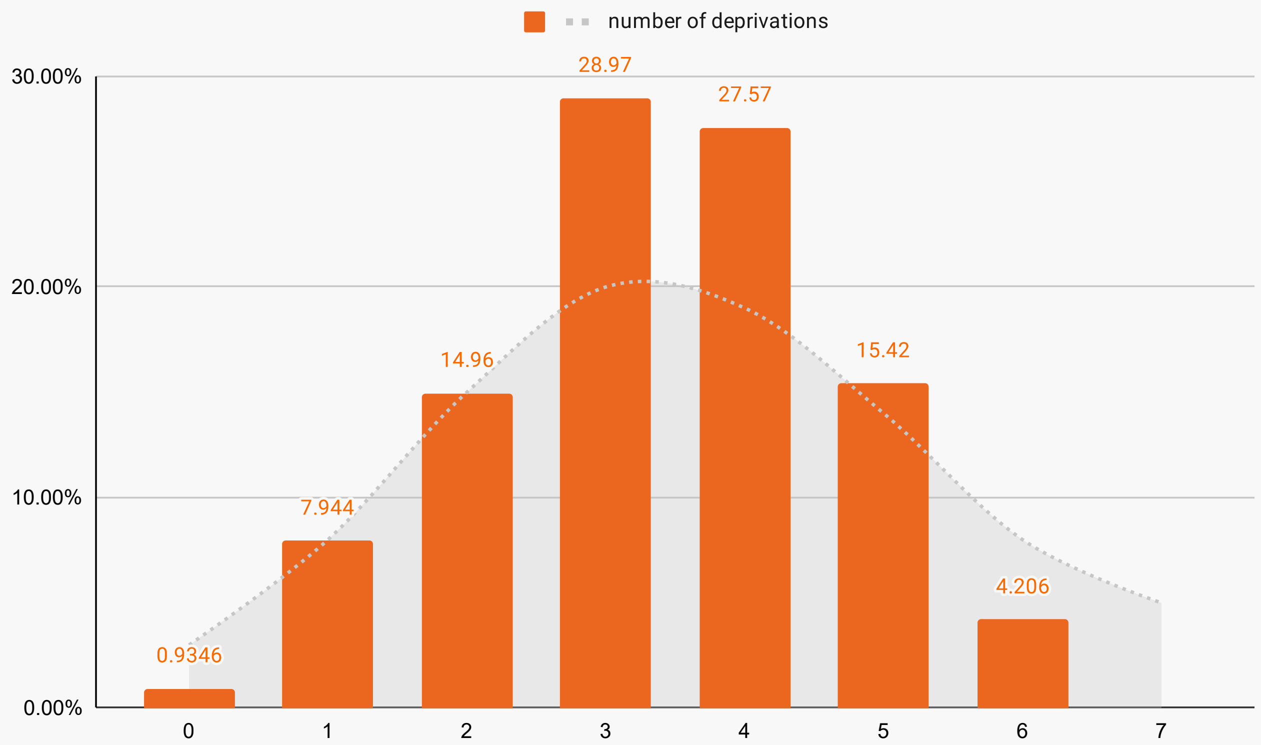
Not deprived in any: 2%

The distribution of the number of deprivations for children living in Roma settlements indicates that **the breadth of deprivation is very high for each age group** (*see Figure 13.*). The distributions are right-skewed**: more than 96 per cent of children suffer deprivations in at least one dimension, irrespective of age group**. Very few children are not deprived in at least one dimension: none in the 0–23 age group and around 1 per cent or less of older children.

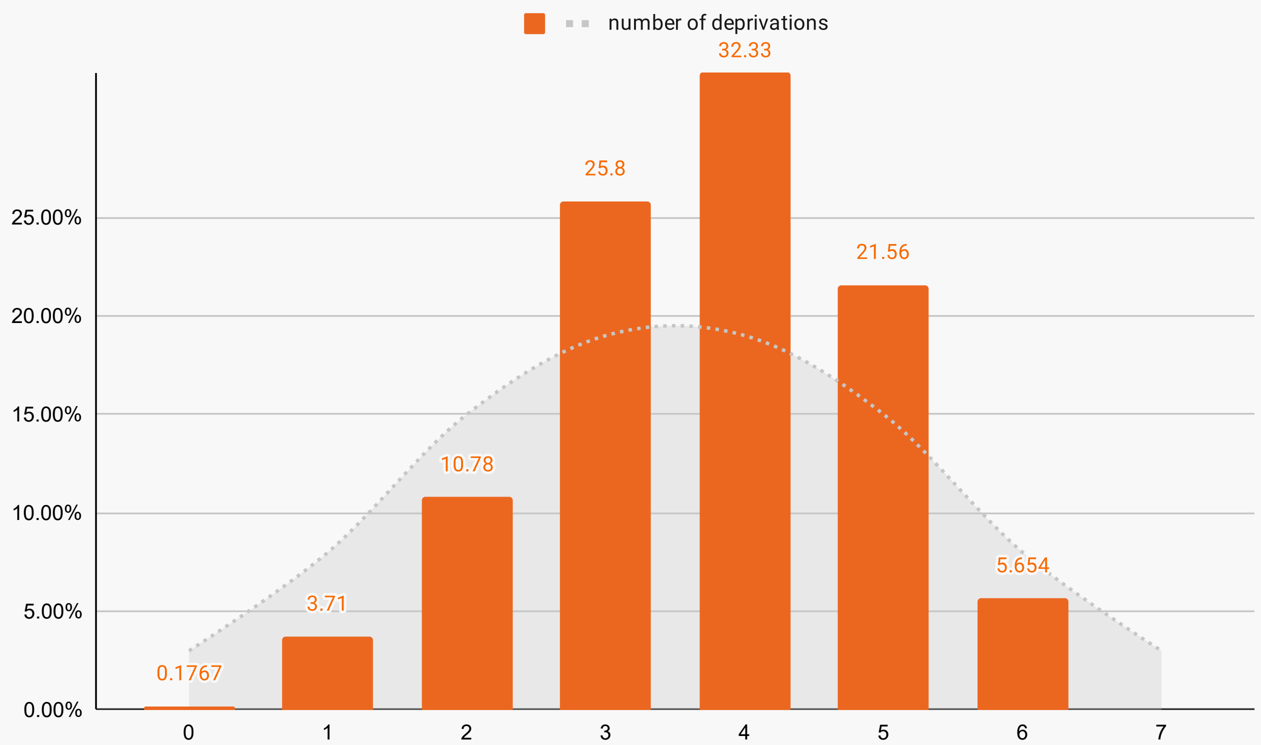
Figure 13. Distribution of deprivations, by age group (%)

Source: Authors' depiction based on 2018 MICS in Roma settlements.

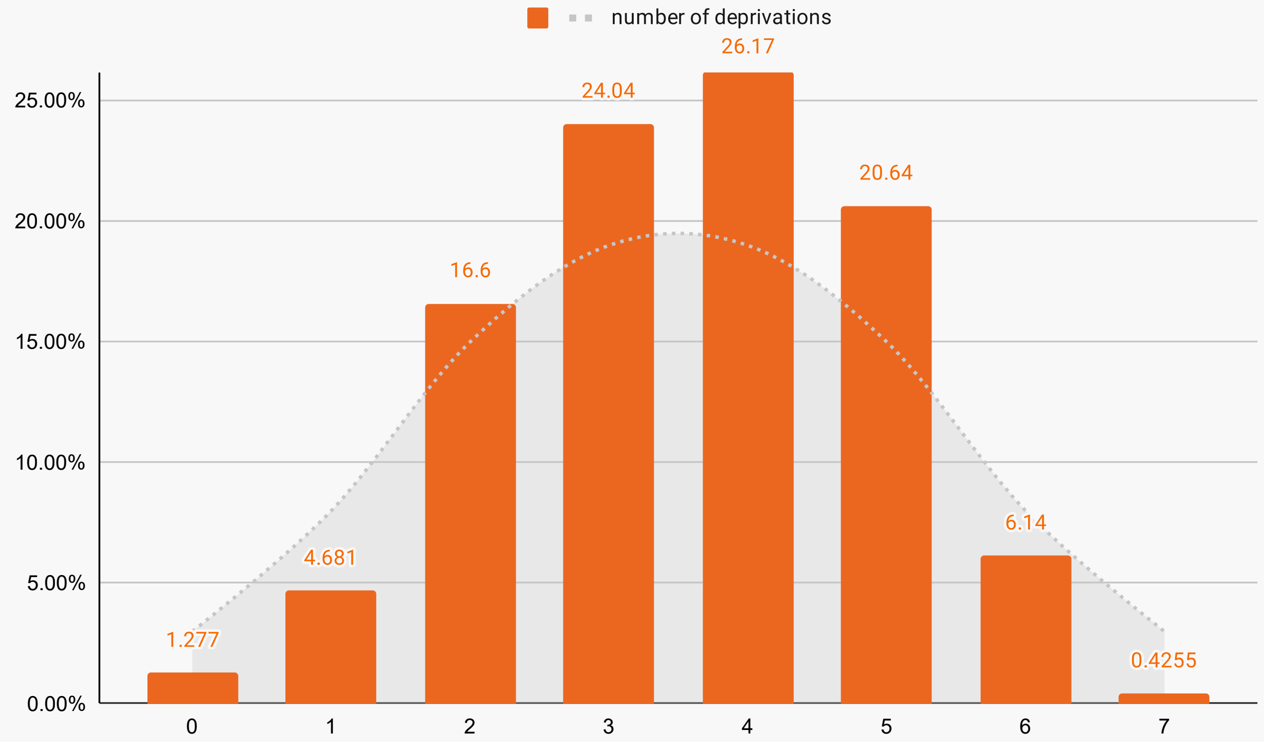
a) Roma settlements: 0-23 months



b) Roma settlements: 0-59 months



c) Roma settlements: 5-17 years



Multiple Deprivation Analysis

Table 6 details the incidence of deprivation, its intensity and the adjusted headcount ratio for Roma settlements. Headcount ratios at different cut-offs reiterate the distributions reported above. The picture for Roma children is quite extreme, since **almost all children are deprived at the cut-off of two or more dimensions**. The prevalence of multiple deprivations decreases at higher thresholds, but the figures remain dramatically high: 76 per cent of children aged 0–23 months, 59.5 per cent of children 0–59 months and 53.4 per cent of children 5–17 years are deprived in four or more dimensions.

Taking into consideration the intensity of deprivation (A), children experiencing three or more simultaneous deprivations tend to suffer from **66–68 per cent of all deprivations** – averaging four simultaneous deprivations. This suggests that for the majority of children **the intensity of deprivations is quite severe**.

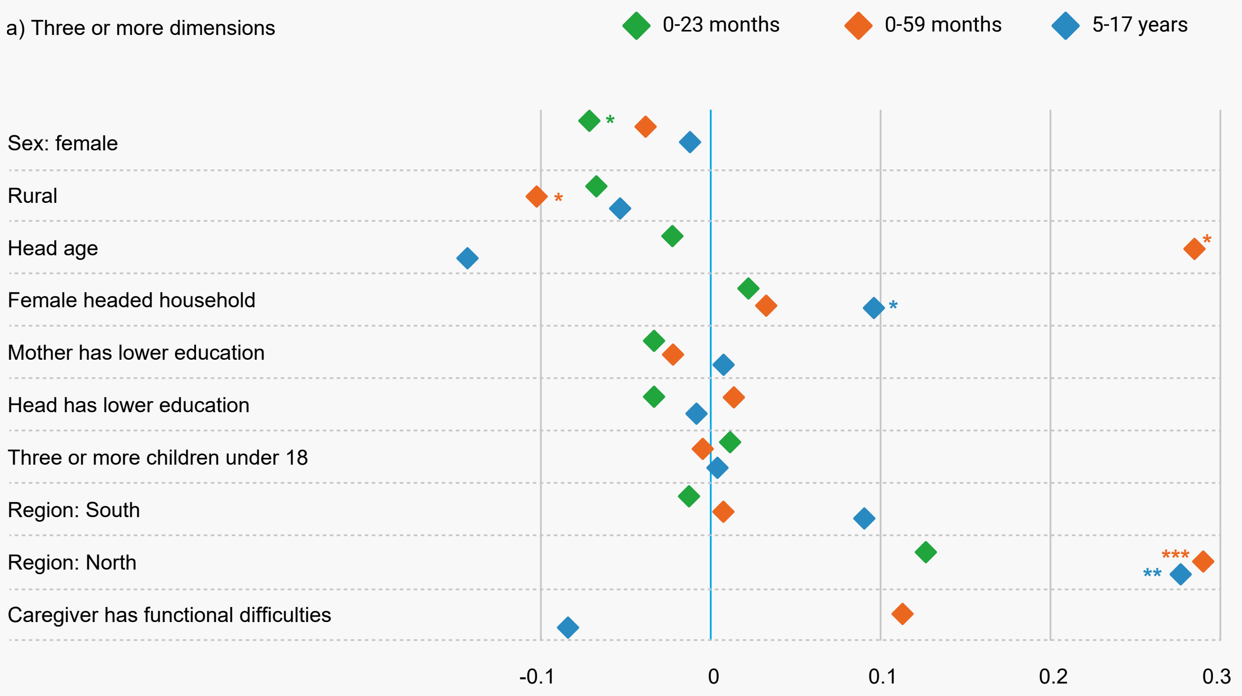
Age group	Cut-offs	HC (%)	A (%)	M_0 (%)
0-23 months	2+ out of 7 dimensions	99.1	62.9	62.3
	3+ out of 7 dimensions	91.1	65.9	60.1
	4+ out of 7 dimensions	76.2	70.5	53.7
0-59 months	2+ out of 6 dimensions	96.1	64.2	61.7
	3+ out of 6 dimensions	85.3	68.0	58.1
	4+ out of 6 dimensions	59.5	75.9	45.2
5-17 years	2+ out of 7 dimensions	94.0	62.6	58.9
	3+ out of 7 dimensions	77.4	68.9	53.3
	4+ out of 7 dimensions	53.4	77.4	41.3

Table 6. Multidimensional poverty headcounts, intensity of poverty and adjusted headcount ratios at different cut-offs, by age group

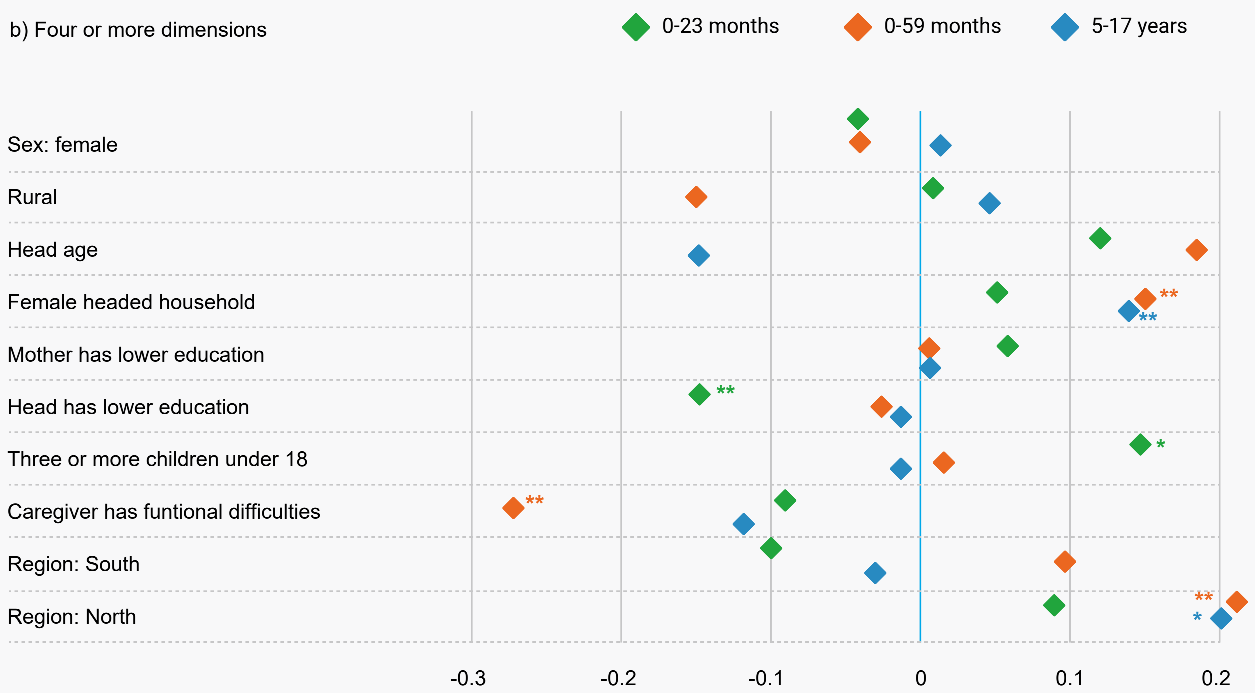
Source: Authors' elaboration based on MICS6 2018 – Roma Settlements.

Figure 14. Marginal effects from logistic regressions, by age groups and cut-offs

Source: Authors' elaboration based on MICS6 2018 – Roma Settlements.



* p<0.1, **p<.05, ***p<.01



* p<0.1, **p<.05, ***p<.01

Examining the most influential variables among the general population (mothers' education and location of residence) it becomes apparent that there is no similar association for the Roma community (*see Figure 14.*). For instance, whether or not the head of household is well educated or whether Roma children live in a rural or urban area has no significant impact on their likelihood of being deprived. The picture is quite complex, the only characteristics likely to exacerbate the incidence of multidimensional poverty are related to the *gender* of the household head (the coefficient is positive and significant at 5 per cent for children 0–59 months and for the 5–17 years age group living in female-headed households) and to the *region of residence.* Living in the north increases the probability of being deprived, all other factors being constant.

Qualitative Overlapping Deprivation Analysis

Qualitative findings also indicate that Roma children generally face greater severity and depth of multidimensional deprivation due to heightened poverty, segregation and discrimination by the larger society. The following types of deprivations most commonly overlap, resulting in a diverse set of adverse consequences for children.

First, housing is associated with water and sanitation deprivation and poor health outcomes among Roma children. Financial hardship and segregation compel many families to live in informal settlements or sub-standard or illegal housing without running water or adequate sanitation, with consequent impacts on child health and wellbeing. Lack of access to potable water or safe sanitation increases exposure to waterborne disease and other safety risks when using outside toilets or fetching water from other locations. Lack of running water is also a cause of stigma when children are unable to wear clean clothes to school, contributing to school drop-out.

Second, the concentration of Roma populations in Roma settlements or designated housing and the associated risk of segregation and social exclusion have multidimensional adverse effects on children's social, educational and professional development. Owing to overall lack of integration into the host population, Roma children may be subjected to stigma and discrimination from their schoolmates, leading to discomfort with attending school. Moreover, school-based discrimination combined with limited parental capacity to support children with schoolwork can lead to poorer learning outcomes for Roma children. Likewise, isolation from mainstream Montenegrin society allows certain harmful social norms and practices to affect Roma children, particularly adolescents (such as hazardous forms of child labour, child marriage, etc.), exposing them to the risk of inter-generational poverty.

Third, education and child protection issues are also closely linked and negatively impacted by heightened deprivation and traditional social expectations among the Roma population, putting children at particular risk of child labour. Roma children are typically expected to begin working at a young age. This early debut into work can lead to missing classes and/or dropping out of school; some families encourage their children to work instead of continuing their education. Some adolescents report engaging in dangerous or taxing physical labour, such as collecting items from dumpsters and carrying heavy items, which can negatively impact their physical health. Parental neglect is a compounding factor in this dimension, as some Roma children were said to be forced into begging as young as 5. Although this was mentioned only once during focus group discussions, organizations that work with the Roma population noted that begging is a significant problem, particularly in urban settings. This presents considerable safety concerns, as well as potentially cyclical poverty effects:

> There are examples where a mother observes her kids while they are begging on the street. […] Maybe it [begging] is not bad for parents but it is bad for the kid. He or she will grow and when he or she gets married the children will beg too.
>
> – FGD with parents of Roma and Egyptian origin, from Nikšić

Finally, hunger reportedly affects many Roma children due to poverty, which in turn affects their educational performance and long-term health.

5. Sensitivity checks

In multidimensional poverty analysis the selection of dimensions is often under debate, either during the inception phase, when brainstorming with the local stakeholders about their selection or at the end of the analysis process, when policy makers have to develop the most suitable approach to tackle multiple deprivations at the same time. The literature on multidimensional poverty measurement has made huge advances in analysing the statistical performance of different approaches to poverty measurement, emphasizing among other things the importance of the sensitivity of multidimensional poverty measurements to variation in the composition of the selected dimensions.[37]

In this section we test how the multiple deprivation index family (H, A, M^0) responds to a variation in the number and type of dimensions used for their estimates. Or – from a different perspective – how the complete eradication of either a child- or household-level deprivation dimension would influence the final headcount and depth of deprivation. Although the latter is an unrealistic scenario, it helps to identify the dimensions that contribute most to the multidimensional measurement, and will yield a picture of how the phenomenon would look if one of the most relevant dimensions were not taken into consideration during the initial selection. To test how sensitive H, A and M^0 are to a variation in the composition of the set of dimensions, the study re-calculated the three indices, excluding one dimension at a time but keeping all other factors fixed.

37 J.Y. Duclos and L. Tiberti (2018), "How to reach Sustainable Development Goal 1.2: Simulating different strategies to reduce multidimensional child poverty in two middle-income countries," *Child Indicators Research*, 11, 3, pp. 711–728; Ferroni and Chzhen (2017).

Now, which criteria were used to select the dimensions to be excluded? The first selection criterion was to choose three dimensions for each age group, namely those showing the highest headcounts in the single deprivation analysis. The second selection criterion was to ensure that at least one child-level dimension and one household-level dimension would be excluded. Household-level deprivations affect simultaneously all children living in the same household, while child-level deprivations may differ within the same household. Thus their exclusion would potentially have different implications for the final estimates.

The exclusion of a dimension was expected to affect the final headcount ratio differently depending on the age group, the reference cut-off and the population under analysis. The dimensions used in this simulation exercise are those showing the highest incidence in the single deprivation analysis, namely protection, health and ECD at the child level, and housing and utilities at the household level. For the sake of simplicity, we restrict the analysis to two age groups (0–59 months and 5–17 years) for both populations at the three different cut-offs (k=2, 3, 4). In what follows we will comment exclusively on the headcount ratio results, reported in Table 9. A full table including the result for all three measures is available in Annex B Table 2.

The multiple overlap analysis (Figure 7.) has shown that each dimension has a different degree of overlap with the other dimensions. In the general population few dimensions dominate over the others, while in contrast Roma children are highly deprived in almost all the dimensions. By looking for example at the 0-59 months group within the general population (Figure 7.b), we find that 11 per cent of children deprived in health are also deprived in two other dimensions and 8 percent of children deprived in protection are also deprived in three other dimensions. As a general consequence we expect that excluding one dimension will reduce the final headcount ratio, and that such a reduction would vary depending on the selected dimension, the age group, and the population. Results in Table 7 confirm our expectations, **for each dimension removed there was an overall drop in the final headcount estimates for both the general and Roma populations[38]. However, the reduction in the poverty rate (k=2) is lower for the Roma than for the general population. This is not surprising given that no deprivation clearly dominates others within Roma settlements, which show higher and more evenly distributed headcounts than in the general population.**[39] Unsurprisingly, **whatever the dimension excluded, the final estimates converge to approximately the same value**: for example, in the 0–59 month age group, about half of the children are multiply deprived at cut-off (k=2) and one-quarter at (k=3).

Given that there are no systematic differences between child-level or household-level dimensions, the study concludes that the simulated headcounts reflect the degree of overlap of the excluded dimension with the rest and the distribution of deprivations, returning different results for the two populations.

Also it is evident that in settings like those under analysis there is no clear domination by any one specific deprivation over the others, but rather **multidimensional poverty is driven by a small number of key dimensions that are highly correlated with each other**. This is true for both populations under analysis but is particularly evident in Roma settlements, where large numbers of children are simultaneously deprived in more than four dimensions. This makes the eradication of multidimensional poverty a pressing challenge. In both cases, a multi-sector approach to poverty eradication would be preferable, although the situation for the Roma population is by far more complex and requires additional investigation. Reducing deprivation in two or three key single dimensions through a multi-sectoral intervention could represent a good strategy for reducing multidimensional poverty.

38 The drop in headcount rates follows the skew of the deprivation distribution. In the general population, the reduction in headcount rates is larger at the cut-off (k=2) for both age groups analysed, because the excluded dimensions mostly overlap with only one other dimension. (*See Figure 7.*)

39 As expected, at each cut-off the larger the simultaneous overlap of the excluded dimension with the other dimensions, the larger the final headcount reduction. If the selected dimension overlaps with a large number of dimensions, a larger variation in final headcount rates at high cut-offs (k=4) is expected; but if the dimension under analysis overlaps with fewer dimensions, a larger variation of the headcount rates is expected to happen at smaller cut-offs (k=2).

General population (%)							
		0–59 months			5–17 years		
Cut-offs		Scenario 1	Scenario 2	Scenario 3	Scenario 1	Scenario 2	Scenario 3
	Excluded dimension	Protection	Housing	Health	Protection	Housing	Utilities
	2+	47	53	54	33	35	41
	3+	23	20	25	17	12	15
	4+	7	4	8	4	3	4
Roma population (%)							
		0–59 months			5–17 years		
Cut-offs		Scenario 1	Scenario 2	Scenario 3	Scenario 1	Scenario 2	Scenario 3
	Excluded dimension	ECD	Housing	Utilities	Protection	Housing	Utilities
	2+	93	90	93	87	83	86
	3+	77	64	84	66	56	61
	4+	44	29	58	37	29	32

Table 7. Multidimensional poverty headcount at different cut-offs, by age group

Photo: Duško Miljanić / UNICEF Montenegro

6. Beyond MODA: Emerging and cross-cutting issues

This section addresses the issues related to children's experiences of poverty identified by children and caregivers, but not captured in the original set of MODA indicators/dimensions. By drawing on these qualitative insights the authors seek to further unpack the complexities and multi-dimensionality of poverty for boys and girls in both the general population and Roma communities, with a greater focus on the psycho-social, rather than monetary, aspects of deprivation. Section 6 also provides more in-depth insights into how adolescents feel, perceive and experience various deprivations in their lives and how this affects their transition to adulthood.

6.1. Perspectives on child poverty and deprivation in Montenegro

- ***Shame and stigma significantly affect the lives of poor children***

For children, material poverty is often associated with shame, stigma and risk of social exclusion from networks that they view as important to their identity and broader wellbeing. The stigma of poverty was a common theme in discussions with both parents and children. Poor children often feel pressed to hide their lack of material goods from peers, such as homemade food, non-branded clothing items and inadequate housing. Visible signs of material poverty – such as inadequate clothing, shoes and mobile phones – can expose poor children to bullying, psychological distress and shaming from better-off children. To avoid humiliation and abuse, poor children use various tactics to hide their poverty, such as demanding money from their parents for popular snacks at school, going hungry while paying for expensive clothing on instalment or borrowing money to afford goods that are considered important status symbols, such as name-brand clothing and shoes, phones or computers. In more extreme cases, stigma and discrimination may compel children to leave school.

> A certain attitude is immediately adopted towards the one who has less.
> – FGD with unemployed parents from Podgorica

> [My son] is now in the second grade of basic school, and as he does not have a telephone, he refuses to go to school.
> – FGD with parents from Rožaje

Children perceive the difference between rich and poor to be of great importance, and a determining factor of their social identity, including popularity and respect among peers. Some used terms such as 'lower class' to depict those with less money, and some poor children expressed feeling humiliated and having a sense of low worth to society, based on their economic status.

> People look at that [economic status]. If you are rich and have money then everyone will hang out with you and if you are not rich then you are somehow rejected and pushed to the sidelines. Kids from the middle class would make fun of those kids who don't have money.
> – FGD with adolescent boys from Rožaje

Interestingly, social exclusion and perceived inequalities seem to be less prevalent in rural communities. The fact that many children live in small communities has its advantages. For instance, since everyone knows everyone, people help and have greater trust in one another. Nonetheless, social pressure is also stronger, and people are sometimes forced to fake their living conditions to avoid embarrassment. There is also some evidence that rural children may be subjected to discriminatory attitudes and expectations from service providers, including teachers.

> "In my elementary school, a teacher once said: 'Find a way to get online – it's the 21st century!."
> – FGD with adolescent boys from rural parts of Bijelo Polje

Finally, shame and stigma related to poverty can also cause social isolation for poor children. For example, hosting or attending birthday parties is an important opportunity for children to socialize with their peers and form social networks. Inability to organize "expensive" parties makes children feel excluded, isolated and ashamed. Parents also said they suffered from feelings of shame and were forced to find a way to pay for things they couldn't afford (expensive parties, birthday presents for children's friends), typically by borrowing money to save face. Respondents explained that children in inadequate housing often avoid inviting friends to their homes due to embarrassment or fear that others would refuse to visit, which can lead to social isolation.

- ***Social exclusion and discrimination constrain poor children's access to quality services and opportunities***

Socio-economic inequality and exclusion from opportunities and basic services form a large part of how poor children experience and describe poverty. Social exclusion and discrimination are key factors distancing poor families from job opportunities and access to quality education and healthcare, thus perpetuating their poverty.

Many children expressed **dissatisfaction with the health system**, and perceived that favouritism, clientelism and discrimination based on socio-economic status are key **determinants of access** to quality healthcare in Montenegro. They noted that children from better-off families and those with good social connections at health clinics receive more timely access to care. Those who cannot afford the cost of private health treatment and insurance (or bribes at public health facilities) may be left untreated, even in the event of serious illness. Similarly, respondents frequently described teachers in primary and secondary school as not paying enough attention to poorer students.

> My sister was in the health facility recently since she was injured in a car accident. We were waiting for a haematologist, and there were women jumping the line and going in before my sister because they knew the man who was calling out patients' names. My sister was kept waiting for a long time because she didn't know anybody.
> – Interview with a child from a single-parent family in Podgorica

> It's the education system. If they are a good student but their parents are not influential, then it's all in vain... but if they are a child of prominent parents who has poor grades they magically turn into good grades at the end of the year.
> – FGD with unemployed parents from Nikšić

Ethnicity and gender also identity heighten children's risk of discriminatory treatment in health and education. Girls of Roma and Egyptian origin in Nikšić reported that they and their families were not treated well in hospitals because of their ethnicity, which was independently corroborated by mothers in the same area. Boys from Roma and Egyptian populations in Podgorica also mentioned friends who lacked insurance or money to pay for medical services, leaving them vulnerable to untreated health problems. These children also reported discrimination by students and teachers, who were said to stereotype them as "underachievers" and assume that the girls are only interested in marrying young.

- ***Poor families' reliance on social welfare may trap children in intergenerational poverty***

Social welfare and its role in poverty reduction emerged as an important but complex topic in this study. In some contexts it has successfully lifted families and children out of poverty and alleviated financial deprivations, with positive implications for children's well-being. Several parents, particularly those of Roma or Egyptian origin, said that receiving social assistance (e.g., child allowance, school stipends or state-based schooling and childcare subsidies) offers important relief from poverty, enabling children's access to basic services, such as regular school attendance.

At the same time, however, reliance on social welfare can lead households to depend on external support creating a risk of being trapped by poverty. A key concern among unemployed recipients of social welfare was the potential risk of experiencing a reduced access to their benefit entitlements (e.g. subsidized electricity, free books, free kindergarten). As a result, many parents do not seek official employment, but rather combine social welfare with informal work to supplement their income.

> When parents become employed, they get removed [from employment agency lists], and so do their children. Well, that's the problem, really. For example, my daughter has to have regular check-ups, and when you are removed from the employment agency lists, you lose health insurance and everything.
>
> – FGD with unemployed parents from Podgorica

When asked about the way forward and possible solutions for their financial problems, many parents talked about increasing the social support and subsidies they receive from the state. This is partially a result of dissatisfaction with the labour market and a perceived dearth of decent work opportunities that offer adequate pay and social security benefits, such as those offered by the state employment sector. Yet this passive outlook leaves many entrenched in their reliance on state welfare and traps children in intergenerational poverty, as the following story illustrates:

> Well, for example there is a family which is a recipient of social assistance. Their daughter just turned 19 and she is already getting married. I think she is pregnant and she has already applied for social assistance with her husband. She doesn't have a plan of what to do, how to live, how to pay the bills... We have talked to them, but they are somehow used to it [...]. I believe that it is the same way his family lived, and his father's family before that.
>
> – Key informant interview with a representative of the Centre for Social Work

- ***Education and employment are a key avenue for overcoming poverty***

Most children believe that education is their 'way out' of hardship and a difficult life, and for many


Photo: Duško Miljanić / UNICEF Montenegro

this is the main motivation for going to school and continuing to study. In some situations, having temporary and physically demanding jobs leads them to place greater value on education, as a way of securing better quality jobs in the future. Another motivating factor is seeing parents with little education struggling to find employment. To avoid the fate of their parents, children seek to complete their own education. Some parents intentionally point to illiterate parents or spouses as negative examples for their children, to underline the lack of prospects resulting from an incomplete education.

> [Low-quality jobs] should be our driving force that makes us study more. I know how tired I'd get after doing this job, and when I come back home and start studying, I'd say to myself - as hard as it may be, better study now - because I know that if I didn't, I wouldn't be able to do the job I love one day instead of the one I did on a seasonal basis only.
>
> – FGD with adolescent boys from Nikšić

> We can observe the situation of our parents, some of them completed high school, some do not even have a high school degree, which makes it difficult to find employment and get a job at all, any job. And that is why education is very important for a person, so this is important to me personally, and I try to achieve some of my goals.
>
> – FGD with adolescent girls from rural Rožaje

Children also reported that although families generally perceive education as important, success often depends on context and direct support from parents. They distinguished between peers who are supported by their parents and live in contexts that provide an advantage and those who are not. Many Roma and Egyptian children reported that their parents 'make them study' to achieve a better future – but with important gender differences. For example, the pressure to do well in school was stressed in a focus group with girls from Nikšić; however, boys from Podgorica reported that families in their community discourage education in favour of starting to work. Some rural children were also reportedly discouraged from continuing their education into secondary school and college, as parents did not want them to attend school in larger cities.

> They're on the verge of poverty, so parents allow them to skip school and they seem not to recognize its importance. [...] We often communicate with the school and the Centre for Social Work as soon as we notice that they are not attending school. We report this and what concerns us is that schools allow them many days of absence without reporting it to the Centre.
>
> – Key informant interview with an NGO representative from Nikšić

6.2. Adolescent poverty and safe transition to adulthood

Adolescence is a period of considerable physiological, cognitive and emotional development.[40] It is also a time when key decisions are made related to schooling, skills' acquisition, partnerships, sexual debut, pregnancy and marriage – all of which have lasting effects on an individual's well-being and transition to adulthood. For poor adolescents these decisions and transitions often must be made in a context of multidimensional risks and deprivations. Yet in Montenegro there is only limited understanding of the link between poverty and adolescent wellbeing. This study sheds light on how adolescents experience poverty and overlapping deprivations and how these circumstances create barriers to their safety, health and future productivity. This section draws on information gathered during focus group discussions with adolescents aged 15–17 years.

Mental health issues emerged as a critical concern for adolescents. Adolescents in Montenegro reported being vulnerable to a wide-ranging spectrum of mental health issues, including disruptive behaviour, anxiety and mood disorders. There is a perception that the many depressed young people rarely seek help from peers or professionals. This is partly because mental health issues continue to be stigmatized among adolescents and communities more broadly – and thus are rarely discussed. Several adolescent respondents praised counselling services that had assisted them to overcome their psychological problems (e.g., Savjetovalište za mlade Bijelo Polje). They saw open discussion as a means to help young people who are otherwise disempowered to seek professional help.

Violence, peer pressure and substance abuse are common among adolescents, particularly in school settings. Peer pressure can take the form

40 N. Balvin and P. Banati (2017), "The Adolescent Brain: A second window of opportunity," Introduction, UNICEF Office of Research - Innocenti, Florence.

of verbal or physical abuse, belittling and social exclusion and is often the root cause of adolescent mental health issues. Adolescent respondents believe that peer pressure is one of the main reasons why young people start smoking, drinking or taking drugs early. Widespread use and accessibility in some areas also encourage this behaviour. Substance abuse appears to affect some groups of adolescents more than others. For example, adolescents from Bijelo Polje and Rožaje whose parents receive social welfare agreed that more than 90 per cent of their peers smoke and drink regularly. Some drink alcohol and others drink too many energy drinks, while some young people start smoking as early as 13. Exposure to substance abuse, such as smoking and drinking, was perceived by respondents to be greater among adolescents living in urban areas and in Roma settlements. Although schools warn about the dangers of smoking, the information is largely ignored. The young people also said that authority figures often turn a blind eye to such behaviour and punishments are too mild.

> It is easy to score some [drugs] in Rožaje. Because every other person, God forbid, is using [drugs].
>
> – FGD with adolescent boys from Rožaje

Awareness of sexual and reproductive health (SRH) is relatively limited, particularly among adolescents in rural areas who lack access to SRH education. Adolescent respondents said that some young people lack knowledge about sexual and reproductive health and engage in sexually risky behaviour (resulting in disease or unwanted pregnancy). Parents from Rožaje complained that there are no educational services to address sexual and reproductive health in their municipality; adolescents obtain this information mainly through NGO workshops or from their parents. In addition, access to information on SRH depends on gender. For example, respondents believe that adolescent girls are better informed than adolescent boys, since girls are viewed as more vulnerable to sexually transmitted diseases and early pregnancy.

> It's most often that parents teach us girls lessons about sex. Especially when I tell my parent that I'm going out with a boy in the evening. The lessons come immediately because we are female, so we need to be more careful.
>
> – Interview with a girl from a single-parent family from Podgorica

> You may have heard of the local NGO "Development Centre". I volunteer there. They do workshops on sexual and reproductive health. I really learned a lot and they taught us well.
>
> – Interview with a girl from a single-parent family from Podgorica

Adolescent use of social media and the internet, as reported by adolescent respondents in the study, has mixed effects on their opportunities for development and psycho-social wellbeing. The internet was widely mentioned as a critical aspect of daily life in Montenegro: adolescents and children with access said they spent most of their day online and are preoccupied with phones and other gadgets. Although adolescent respondents initially noted that they use the internet mostly for studying, further discussion showed that they dedicate most of their free time to social networking profiles, movies, YouTube videos and video games.

Respondents said they are taught about the internet's positive and negative aspects in school (via workshops, student parliaments, etc.). Nevertheless, children and adolescents continue to over-use the internet, to the detriment of their physical and psychological health. It also takes up time previously spent learning and undertaking healthier pursuits; for example, respondents reported a decline in reading.

> The internet has both good and bad sides, for example, it informs us. As far as learning is concerned, we can find a lot of things there. And there are also the bad sides, such as social networks. [...] A lot of bad things get circulated there.
>
> – FGD with adolescent girls from Nikšić

> My best friend watches TV series all day long. She could spend this time in a much better way, for example, working out, reading a book, going for a walk. She's wasting a lot of time.
>
> – FGD with adolescent girls from Podgorica

> When I was younger I used to read a lot. But now I do it rarely.
>
> – FGD with adolescent boys from Bijelo Polje

Access to extra-curricular activities is limited among poor adolescents, constraining their learning outcomes and opportunities for self-development. In addition to paying for school excursions and materials required for projects, parents frequently mentioned the need to pay for additional classes to improve students' learning outcomes. This particularly affects Roma and Egyptian adolescents, whose parents are generally unable to afford additional classes, but who often feel 'blackmailed' into doing so due to the shame of refusing to do this.


Photo: Duško Miljanić / UNICEF Montenegro

Extra-curricular activities (such as socio-cultural events, sport competitions, art and dance), though very popular, are also often too expensive for poorer adolescents, negatively affecting their opportunity for development and personal growth. Adolescents who could not afford to continue hobbies reported having to pretend that they had lost interest in them to avoid stigma. Adolescents with particular talents or passions also shared their experience of giving up their aspirations due to financial difficulties. Parents from northern cities were especially concerned about the lack of activities and spaces for adolescents to enjoy in their free time. One parent felt sad that his children had never been inside a theatre or museum, as they did not have the money for transport and food. Other parents were afraid their children would spend all their time in cafes and betting shops due to a lack of safe spaces for young people.

> I attended the course for one and half years or two. I always had a talent for drawing and painting. I had my own exhibition for a while but since my dad lost his job, I did not have any income so I could not carry on.
>
> – FGD with adolescent girls from Podgorica

> My older girl is talented in volleyball. At the sports centre you need to pay €25 for membership. I asked a trainer if there was any discount, since we were receiving social assistance. He said that there is no discount. [...] When [my daughter] was supposed to go to Sarajevo last year to compete, we couldn't let her go because we could not afford the costs.
>
> – FGD with unemployed parents from Nikšić

Box 2: Do adolescent boys and girls experience differences in deprivation and if so, how?

The qualitative research sought to explore gender-specific dimensions of deprivation and differences in the experience of multidimensional poverty between girls and boys. The findings suggest that gender dynamics, compounded by ethnic identity, play a key role in girls and boys to being exposed differing risks of poverty and vulnerability in three main ways.

First, adolescent girls from Roma communities remain vulnerable to the practice of child marriage. Although respondents confirmed that early marriages do occur, most parents and children do not support it. Adolescent Roma girls agreed that a girl shouldn't marry before 22, and boys said no one should marry before the age of 20. Poverty was identified as a key driver of early marriage. Financially deprived boys expressed frustration with their inability to 'buy a bride', leading girls to marry richer suitors. Indeed, some girls in the north were said to marry too early in order to escape poverty; however, upon entering a new family some found themselves in precarious situations – exposed to intimate partner violence without rights or support. Others, however, saw child marriage as a practice sustained by deeply entrenched discriminatory gender norms, social customs and expectations related to the cultural importance of a girl's virginity when entering marriage, as well as by weakly enforced measures and policies to address the problem.

> I think that's where the violence originates from. When a wealthier guy marries a poor girl, he says I paid for this, I bought everything you're wearing. He feels like he owns her, and that's where violence starts from. – FGD with adolescent girls from Bijelo Polje

> Virginity…is the most important thing for us. You marry a virgin, you know who to give your virginity to, do not just give it away to anyone.
> – FGD with parents of Roma and Egyptian origin from Nikšić

> Arranged marriages of Roma and Egyptian children are a part of the custom, but it was also a custom for Montenegrin women 100 years ago, but the society tried and changed that. The society is doing nothing to eradicate this in the Roma and Egyptian communities.
> – Key informant interview with NGO representative

Second, methods of discipline and control over children tend to be gender-specific. Boys are more likely to be subject to violent psychological or physical behaviour than girls, while girls are typically brought up in a more restrictive environment. Girls' mobility is often curtailed by social and gender norms. Parents are more likely to be stricter with girls and limit their ability to leave the house, hindering their prospects for pursuing education, employment or other development opportunities. Also, in some parts of the country extra-curricular activities are generally very limited, especially for girls.

> For example, I like football, kickboxing, and karate, and all of that, but I am not allowed to practice. – FGD with adolescent girls from Rožaje

> **Parent 1:** I'm pretty strict with my daughter, and she can see it and says - how come he [her brother] is allowed to do something, and I'm not? **Moderator:** And what is it that he can do and she can't? **Parent 1:** Well, I don't even know myself, I guess the environment, so that a girl does not do something wrong, and somehow a male is allowed to do anything, I do not even know myself. […] **Parent 2:** If a young man has a girlfriend then he's [seen as] a 'hot shot' here, but if a girl has a boyfriend, then she's immoral. That's the difference, I say.
> – FGD with parents from Rožaje

Finally, work patterns mirror typical gender roles, with girls generally expected to undertake domestic chores, while boys perform heavier labour. Girls noted that they are expected to undertake various domestic chores and fulfil care responsibilities at home, which can at times be overwhelming.

> **Participant 1:** Mainly, male jobs are harder. Women are helping in the house mainly. **Moderator:** Do boys clean their rooms? **Participant 2:** No. **Moderator:** Then your sister or mother cleans it? **Participant 2:** Sister. (laughing)
> – FGD with boys from Podgorica

> I am female, pots and dishes are my destiny. – FGD with girls from Bijelo Polje

Photo: Duško Miljanić / UNICEF Montenegro

Photo: Duško Miljanić / UNICEF Montenegro

7. Coping strategies

This section outlines the key coping strategies that children and parents/caregivers employ to manage or overcome their poverty or deprivation. The information was drawn from data collected through individual interviews and focus groups with both children and parents. Overall, it revealed that poor families rely on a broad range of informal and formal coping mechanisms. Although the mechanisms generally provide effective temporary relief from material hardship, the coping strategies adopted rarely lift families out of poverty and instead may deepen deprivations and poverty traps for children over time.

Reducing expenses and falling into debt is the most common approach for coping with material deprivation. Many children cope with scarcity by reducing their consumption of basic items, such as food and clothing. Children may claim not to be hungry when parents cannot afford to give them money to buy lunch and snacks on school break. Also, as noted above, some children feign a loss of interest in hobbies and passions to conceal their lack of resources to pay for these them.

> **Parent 1:** There are kids who have no snack money, trust me, I watch out for these things. They can't afford it. They say – I already ate, I brought some from home. They say - I've put it in my backpack– but you can't see it.

> **Moderator:** And how do you know these things? **Parent 2:** I am with them all the time. I spend as much time as I can with them. I hear things, children confide in me and so on. – FGD with single parents from Bijelo Polje

Often, parents try to teach their children to live with what they have. On several occasions, parents told us that they steer their children away from the things they cannot afford. This sometimes even includes cutting out expenditures essential for children's welfare. One father explained how he managed to convince his son that he doesn't need boots in winter:

> He [my son] told me to buy him boots. I tell him that...it is not a problem, but we should sit down [to talk] first. How much do you walk to school, if your feet get wet, I'll buy you two pairs. From the house to the road, how far it is? He says: 100 meteres. Ok, do you step into the water? No. When the van drops you off, do you get your sneakers wet? No. So, what do you need the boots for? He says: 'You are right'. You just need to talk to them.
> – FGD with unemployed parents from Podgorica

Parents said that they often must prioritize expenses among their children since they cannot pay all costs for all of them. This was mentioned in regard to expenses such as taking the bus to school (while siblings must walk), going on school fieldtrips (while siblings remain at home) and attending musical events (while others can go 'the next year').

> One time, they [our daughters] needed to go on an excursion to Budva on Monday. Because we couldn't afford it we sent one girl last time and now another one. So it happens alternately.
> – FGD with unemployed parents from Nikšić

Nearly all respondents reported being in debt. Some take out bank loans, some use the 'overdraft' on their accounts and debit cards and others borrow money from relatives and friends. In some communities it is possible to buy items '*na veresiju*' (on long-term credit, or 'layaway'). Parents admit that they buy necessities from such places and pay when social assistance transfers arrive or they somehow come into possession of the money.

Respondents reported the use of several different strategies to overcome the problem of sourcing adequate food for their families. Some have their own gardens, while others try to buy food (e.g., meat) 'on credit' and others still shop in neighbouring Serbia, where food is cheaper. Many families cannot afford to buy a variety of food, so children eat similar or the same food every day.

Child labour by adolescents is used to address the shortfalls in household income. As noted above, children from Roma families often must start work at an early age to help support their families, while others are discouraged by parents from pursuing their education. An NGO representative noted that some children are forced to beg; however, during focus groups with Roma participants begging was only mentioned once.

Families also rely on labour migration and external support. Many children dream about leaving Montenegro and starting a new life elsewhere. This is especially, but not exclusively, true of children from the northern municipalities. Many of these children live in families that depend on financial help (e.g., remittances) from family members or relatives working abroad to meet their basic needs for food, clothes, school support etc. While remittances help to meet children's basic material needs, those who are left behind without parental care are vulnerable to many other deprivations, including maltreatment – particularly neglect and abuse, psychological distress and stigma from peers. A parent that left the country looking for work shed light on a rarely discussed issue – parents who leave their children behind to provide for them financially.

> As I told you just now, I go away, I work hard [...] and I have to worry from abroad, to talk to him and see where he is at, top-up his phone so he's available and ready to answer when I call, make a recording so I can see who he is with, call home to confirm with the wife to see if he was lying to me, how truthful he was being. It's not so easy. You know, I became a father late in life, and my whole life belongs to them. And I have to go away when they need me the most, especially this older one, I need to make sure he doesn't wander off somewhere. Maybe he does smoke, I myself am a non-smoker; he may be lying to me.
> – FGD with parents from Rožaje

Finally, reliance on social welfare is common. Many parents mentioned social welfare, particularly government support, as a means to solve their problems, as noted earlier. This study also found that people also rely on more proactive, constructive coping mechanisms to address their poverty and hardship, including finding appropriate and additional employment, making arrangements with extended family to take care of children, seeking preventive advice from mental health professionals, writing complaints and proposals to different responsible institutions and taking their cases to court. Families that have children with a disability often must be self-reliant to cope with problems. Although institutional support (day care centres) is available, some parents try to overcome difficulties themselves.

> We tried to find money by ourselves. For example, we would take out a loan to make a parking ramp so that we can park closer to home and approach it easier. No one from the institutions was involved; we have tried to manage it by ourselves. Of course, it was hard and a slow, slow process of waiting and imagining all of it, but with God's help we have managed.
> – Interview with a child with disability from Podgorica

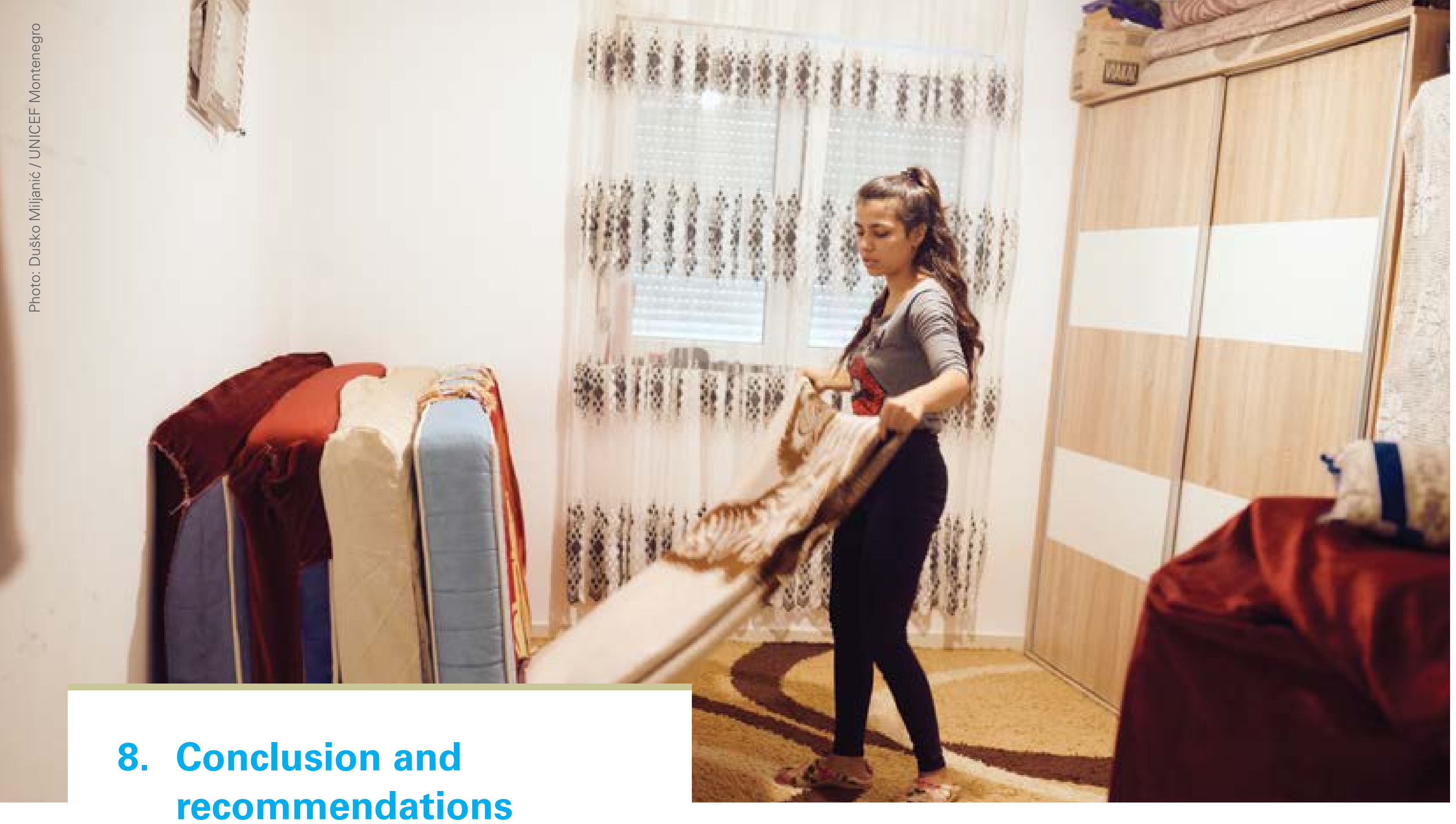


Photo: Duško Miljanić / UNICEF Montenegro

8. Conclusion and recommendations

The findings presented in this report offer a unique opportunity to deepen understanding of multidimensional child poverty in Montenegro. Data on relative poverty, issued annually in line with the EU-SILC methodology, can now be complemented by additional analysis of multiple overlapping deprivations faced by boys and girls living in Montenegro.

The mixed-method approach to multidimensional poverty analysis used in this report – combining secondary quantitative data from MICS 2018 with primary qualitative data, thus relying on both deductive and inductive approaches – is widely acknowledged to permit greater depth than the use of one method alone.[41] This report presents the analysis of single and multiple overlapping deprivations both in the general population and in Roma populations. In addition, qualitative research captures children's and caregivers' perceptions and experience with poverty, examines how various deprivations affect their material, psychological and social wellbeing and sheds light on the coping strategies available to poor families. Finally, it documents emerging issues related to childhood deprivations in Montenegro, beyond MODA dimensions and indicators, through its in-depth look at gender-specific vulnerabilities within the adolescent population. This mixed-methods MODA study thus makes an important contribution to the evolving debate on measuring, monitoring and addressing child poverty.

Poverty harms children both now and in the future. Children's needs are different from those of adults, and resources may not be equally distributed within the household. Therefore, income-based measures do not fully capture the non-monetary aspects of poverty that affect children's lives. The fact that nearly every third child in Montenegro is at risk of poverty sends an urgent signal of the need for concerted efforts to reduce child poverty, and thus support national human capital development.

In Montenegro, **more than 80 per cent of children experience deprivations in at least one domain of their well-being, regardless of age group; the percentage is even higher (96 per cent) for Roma children.** As indicated in the report, younger age groups exhibit higher levels of poverty, based on both single and multiple deprivation analysis. This holds true for both the general and Roma populations.

Analysis of single deprivations for the general population can be summarized as follows:

- For children 0-23 months old, the greatest deprivations are in nutrition and ECD (65.2 per cent and 56.3 per cent, respectively). Poor nutrition results primarily from inappropriate breastfeeding. Poor ECD outcomes are mostly related to a lack of suitable books at home.
- Children under 5 and 5-to-17-year-olds are most deprived in child protection. The children suffer from psychological aggression or physical punishment by a caregiver. Corporal punishment remains a dominant means of discipline at various child ages, according to qualitative evidence.
- Children under 5 years of age living in female-headed households are more deprived in all dimensions except child protection, a deprivation that is more common in male-headed households.
- Some 44 per cent of children in all age groups live in a household reporting housing problems, while one in three children live in a household that faces difficulties paying utility costs.
- Generally low deprivation levels are observed in information access, whereas qualitative evidence shows this to be an issue for poorer and rural households. An emerging risk is that e-learning school reform might widen existing gaps between children from different socio-economic backgrounds.
- Parents' education matters. In general, children living in households where the mother did not attend or complete secondary school are more deprived in all dimensions, especially water and sanitation (6.5 times as much) and ECD (3 times as much).
- Children's risk of being deprived across a majority of dimensions is closely tied to certain background characteristics: mother's education, household head's education and, to some extent, the sex of head of household.

Multiple deprivation analysis for the general population confirmed the relevance of demographic and geospatial factors as predictors of multi-dimensional poverty:

- Low education levels of mothers and heads of household significantly affects the likelihood that their child will be multi-dimensionally poor, for all age groups
- Children under five in rural areas are more likely to be deprived in 4 or more dimensions. Living in the southern, central, or northern regions is also a factor affecting deprivation levels
- The head of household's gender significantly affects the likelihood of multidimensional poverty for children; living in a household headed by a female family member increases the likelihood of children being deprived in four or more dimensions.

Qualitative evidence from multiple deprivation analysis sheds further insight on real-life experiences of multidimensional poverty:

- Several deprivations are closely linked to monetary poverty; monetary constraints affect children's access to healthcare and education
- Substandard housing closely links with poor water quality, health deterioration, quality of education and access to services and information
- Distance from public services in rural locations negatively impacts access to health care and educational facilities and expose children to safety risks. Unaffordable transportation costs exacerbate the concern, especially for younger children
- Poor children's nutrition outcomes spill over to adverse education and health outcomes.

Analysis of data from Roma settlements in Montenegro revealed:

- Extremely high levels of deprivation were registered in almost all dimensions. Over 85 per cent of children in all age groups live in sub-standard living conditions. Informant interviews pointed to poor quality and illegal housing with inadequate sanitation. Deprivations in the domains of ECD (95.8 per cent) and nutrition (77 per cent) rank high for the 0–23 months age group.
- Difficulties in access to education and poor education outcomes, although not ranking at the top of deprivations, are widespread. Only 80 per cent of Roma children (5–17 years) attend school, and three out of 10 lag behind in academic achievements. As seen by adult Roma, language barriers – combined with a lack of support for acquiring official language skills and mastering school material - prevent Roma children from keeping up in class. On a positive note, qualitative analysis shows that Roma mothers are eager for their children to access ECD, as they recognize its importance for their socialization and integration into mainstream society.

41 P. Shaffer (2013), "Ten Years of 'Q-Squared': Implications for Understanding and Explaining Poverty," *World Development* 45, pp. 269–85; K. Roelen (2017), "Monetary and multidimensional child poverty: A contradiction in terms?", *Development and Change*, 48, 3, pp. 502-533.

- Children living in rural areas are generally worse off than their urban peers. For these children, the parent's level of education had no impact on any of the deprivation domains. Health is the only dimension where children in urban areas fare worse than their rural counterparts.
- The risk of deprivation in each domain is more likely to be related to the region of residence than to the demographic characteristics of Roma households. Demographic factors, such as mothers' (and parents') education level, appear to be less significant in determining children's deprivation levels in the Roma community. Geographical factors are a better predictor: Roma children in the northern and southern regions are significantly more likely to be deprived in most dimensions.

Multiple overlapping deprivation analysis for Montenegro's Roma settlements revealed:

Almost all children are deprived in two or more dimensions. Even at the higher threshold of four or more dimensions, more than half of the children 0–59 months and 5–17-years of age remain deprived.

For children aged 0-23 months, nutrition most frequently overlaps with other dimensions. Most frequent overlapping deprivations are for nutrition, housing and ECD, which affect more than two-thirds of all children.

For the under-5 and 5-17 age group, a lack of ECD and poor housing most frequently overlap with other deprivations. Utilities, housing and ECD deprivations overlap for 53 per cent of young children. Utilities, housing and protection deprivations overlap for 35 per cent of children aged 5–17 years.

Qualitative evidence highlights how poverty, segregation and discrimination influence Roma children's experience of multidimensional deprivation:

- Water and sanitation deprivation are associated with poor health outcomes. Stigma related to poverty (e.g., the impact of a lack of running water on personal hygiene and washing of clothes) and related experiences of psychological aggression drive children away from school.
- The Roma population's prolonged isolation and segregation from the mainstream society is one of the factors contributing to the survival of harmful social norms (child labour, child marriage), negatively affecting their development across the lifecycle and exposing them to the risk of inter-generational poverty.
- Roma children are particularly at risk of child labour. Early debut into work often leads to missed classes and school dropout. Organizations that work with the Roma also point to begging as a significant problem in urban settings.

Main considerations for policy formulation

In Montenegro, more than 80 per cent of children experience deprivations in at least one domain of their well-being regardless of their age group; the percentage is even higher (96 per cent) for Roma children. As detailed in the report, younger age groups exhibit higher levels of poverty, based on both single and multiple deprivation analysis. This holds true for both the general and Roma populations.

The examination of multidimensional poverty allowed the authors to understand which dimensions of deprivation overlap the most, and thus serve as evidence to inform multi-sectoral solutions:

Children under 5 are extremely vulnerable to deprivations. Interventions to improve conditions in any of the domains of care for this age group (nutrition, health, protection, early childhood education) will have positive, long-lasting effects later in life. Apart from focusing only on age groups, interventions should consider profiles of poor children in relation to relevant background characteristics.

In addition to identifying overlapping dimensions of poverty, this study was also able to highlight the critical links between material poverty and psychosocial dimensions of deprivation, such as shame, stigma and discrimination. Some evidence of discrimination by teachers and healthcare workers has emerged, especially affecting Roma children. These compounding aspects of poverty not only damage children's psychological wellbeing and identity, but can also cause their isolation from peer networks and lead to exclusion from critical resources and opportunities necessary for children's holistic development.

Poor families rely on a broad range of coping mechanisms, including social welfare, borrowing money and/or engaging their children in economic activities to meet their basic needs. While such strategies generally provide effective short-term relief from financial hardship, in the long run they can deepen deprivations and poverty traps for children.

General population	Roma population
0–23 months	
Nutrition Early childhood development Housing **23 per cent of children deprived in all three dimensions**	Nutrition Early childhood development Housing **65 per cent of children deprived in all three dimensions**
0–59 months	
Child protection Housing Health **11 per cent of children deprived in all three dimensions**	Early childhood development Housing Utilities **53 per cent of children deprived in all three dimensions**
5–17 years	
Child protection Housing Utilities **13 per cent of children deprived in all three dimensions**	Child protection Housing Utilities **35 per cent of children deprived in all three dimensions**

For example, a review of policies related to welfare payments in light of multidimensional poverty could influence decisions by recipients about whether to join the work force or continue in the informal sector in order to retain welfare benefits.

On a positive note both children and caregivers view education, including pre school, as a key avenue for overcoming poverty through its support for children's human capital accumulation and the strengthening of their social networks, which can expand their employment choices later in life.

Overall, the results of the study indicate the need to look for multi-sector solutions that address both the material and non-material facets and drivers of poverty and deprivation, instead of focusing only on the monetary aspects of children's well-being (such as family income). Multidimensional child poverty can only be addressed through an integrated, cross-sectoral response; poverty-reduction strategies need to be carefully crafted and to consider differing needs among different groups of children (i.e. according to age group, area/region of residence, household type, evolving developmental needs). Finally, policies informed by evidence are critical for effective poverty reduction.

Policy and programme recommendations

The present report provides the knowledge base required to launch a public debate on reducing child poverty, particularly since the last study of child poverty in Montenegro is nearly a decade old (2011). The following policy and programme recommendations were formulated based on the rich evidence presented in the report.

1: In both the general and Roma populations, younger age groups exhibit high levels of deprivation in individual sectors, when looking at single vulnerabilities. Since children under the age of 5 are extremely vulnerable to poverty, interventions tailored to address this age group must be taken into serious consideration. Particular attention should be given to nutrition, particularly for the 0-23 months age group.

2: Overall, given the many overlaps between the dimensions under analysis, the study shows the need to look at multi-sector solutions. Integrated, cross-sectoral solutions are recommended since such interventions have a considerably greater potential for reducing multi-dimensional poverty.

3: Both short- and long-term solutions are needed for all age groups. For example, for young children it will be important to step up inter-sector efforts to increase pre-school attendance, including nutritional support at pre-school institutions (e.g., nutritious school meals, nutrition education) to improve children's food security and nutrition outcomes. The provision of quality services during pre-school would yield many benefits for young children as they age and develop (i.e. increased social inclusion and improved early education, learning, nutrition and health outcomes).

4: Concerted efforts should be taken to: alter social norms on violence; support parents to adopt positive parenting techniques; ensure that children are empowered to report violations of their rights; and promote tolerance, empathy, diversity and equality among all Montenegrin children.

5: The concentration of Roma populations in predominantly Roma informal settlements or designated housing and the associated risk of segregation and social exclusion have multidimensional adverse effects on children's social and educational development and transition to adulthood. Greater attention should be paid to improving the housing conditions of the Roma and Egyptian populations, as well as their access to basic services.

6: Stepped-up efforts to prevent drop-out from primary/secondary education are recommended. Integrated policies for early childhood development and inclusive, quality primary and secondary education should take into consideration the specific needs of children from minority ethnic groups.

7: Mental health issues emerged as a critical concern for adolescents, while violence and peer pressure are common among adolescents, particularly in school settings. Therefore, adequate services for mental health support should be rolled out, and adolescents should participate in their design.

8: Social policy plays an important role in improving the quality of life of poor families with children. When social protection programmes are well-designed and integrated with complementary services for child protection, health, education and employment, they can also empower and enhance the capacity of poor families to lead independent and productive lives. Ensuring that social protection programmes are based on evidence will help to achieve this goal.

9: Stigma and discrimination related to poverty is a salient issue. Community mobilization and media campaigns to change discriminatory social norms and attitudes of communities should be implemented to promote values of tolerance, inclusion and diversity (e.g. through schools, parental associations, youth groups). Stigma regarding poverty is often compounded with stigma regarding ethnic affiliation (e.g. belonging to the Roma minority) or disability. An integrated approach is required to holistically and systemically address combined sources of stigma.


Photo: Duško Miljanić / UNICEF Crna Gora

9. References

Alkire, S. and J. Foster, 'Counting and Multidimensional Poverty Measurement', *Journal of Public Economics*, 95, 7, 2011.

Alkire, S., et al., *Multidimensional Poverty Measurement and Analysis*, Oxford, Oxford University Press, 2015.

Balvin, N. and P. Banati, "The Adolescent Brain: A second window of opportunity," Introduction, UNICEF Office of Research - Innocenti, Florence, 2017.

Carraro, A. and L. Ferrone, 'Measurement of Multidimensional Child Poverty', in W. Leal Filho et al. (Eds.), *No Poverty — Encyclopedia of the UN Sustainable Development Goals*, Springer, 2020.

Chzhen, Y. and L. Ferrone, 'Multidimensional child deprivation and poverty measurement: Case study of Bosnia and Herzegovina,' *Social Indicators Research*, 131, 3, 2017.

de Neubourg et al., 'Step-by-step guidelines to the multiple overlapping deprivation analysis (MODA)', working paper, UNICEF Office of Research WP-2012-10, Florence, 2012.

'Wellbeing of Children in Kosovo: Poverty and Deprivation among Children using the Multiple Overlapping Deprivation Analysis', UNSCR 1244, UNICEF, 2015.

Duclos, J.Y. and L. Tiberti, 'Multidimensional poverty indices: A critical assessment', in M. Adler and M. Fleurbaey (Eds.), *The Oxford handbook of well-being and public policy*, Oxford University Press, Oxford, 2016.

Ferrone, L. and Y. Chzhen, 'How to reach the sustainable development goal 1.2: Simulating different strategies to reduce multidimensional child poverty in two middle-income countries,' *Child Indicators Research*, 11, 3, 2018.

'Child Poverty in Armenia', working paper, UNICEF Office of Research WP-2016-24, Florence, 2016.

Gordon, D., et al., *The distribution of child poverty in the developing world*, University of Bristol, Bristol, UK, 2003.

Main, G., and J. Bradshaw, 'Child poverty in the UK: Measures, prevalence and intra-household sharing,' *Critical Social Policy*, 36, 1, 2016.

Roelen, K., 'Monetary and multidimensional child poverty: A contradiction in terms?', *Development and Change*, 48(3), 2017.

Shaffer, P., "Ten Years of 'Q-Squared': Implications for Understanding and Explaining Poverty," *World Development* 45, 2013.

United Nations Children's Fund, *Ethical standards in research, analysis and data collection,* UNICEF, New York, 2015.

World Bank, https://data.worldbank.org/country/montenegro.

'Monitoring Global Poverty: Report of the Commission on Global Poverty', Washington, DC: World Bank, 2017.

Annex A

Background characteristics	Age group		
	0–23 months	0–59 months	5-17 years
Girl	44.60	45.83	50.43
Boy	55.40	54.17	49.57
Child is under 3 years old	100.00	59.20	0.00
Child is over 3 years old	0.00	40.80	0.00
Female-headed household	10.26	11.20	11.38
Male-headed household	89.74	88.80	88.62
Mother has less education	20.66	20.97	20.97
Mother has more education	79.34	79.03	79.03
Head has less education	21.33	21.70	22.04
Head has more education	78.67	78.30	77.96
Household size	5.47	5.46	5.33
Age of household head	42.95	43.97	47.46
Three or more children aged 5-18	11.18	10.89	30.91
One or two children aged 5-18	88.82	89.11	69.09
Caregiver has functional difficulty	0.65	0.71	1.61
Caregiver has no functional difficulty	99.35	99.29	98.39
South	18.52	19.67	19.44
Centre (excluding Podgorica)	23.99	23.38	21.72
Podgorica	35.14	33.97	33.84
North	22.35	22.99	25.00
Rural	33.24	32.36	33.05
Urban	66.76	67.64	66.95

Table A.1 Summary of general population background variables, by age group (%)

Source: Authors' calculations based on MICS6 2018. Children's sex and age are measured for the individual child; the remaining variables are measured for the entire household. Parents' education level is represented by a binary variable assuming a value of '0' if the parent has primary or no education and a value of '1' if education levels are secondary or higher. Regions of residence include three macro-areas (north, centre and south). Figures for the central region are noted as "Centre (excluding Podgorica)" and "Podgorica".

Background characteristics	Age group		
	0–23 months	0–59 months	5-17 years
Girl	47.66	48.06	49.79
Boy	52.34	51.94	50.21
South	9.81	10.25	11.91
Centre	71.50	72.26	73.62
North	18.69	17.49	14.47
Podgorica	0.00	0.00	0.00
Rural	27.10	25.44	22.55
Urban	72.90	74.56	77.45
Female-headed household	13.55	12.90	17.66
Male-headed household	86.45	87.10	82.34
Mother has less education	47.66	49.65	55.53
Mother has more education	52.34	50.35	44.47
Head has less education	40.65	38.16	35.11
Head has more education	59.35	61.84	64.89
Household size	7.10	7.21	6.54
Age of household head	35.10	34.94	40.30
Child is under 3 years old	100.00	60.42	-
Child is over 3 years old	-	39.58	-
Three or more children aged 5–18	39.25	44.17	53.19
One or two children aged 5–18	60.75	55.83	46.81
Caregiver has functional difficulties	1.87	2.47	4.68
Caregiver has no functional difficulties	98.13	97.53	95.32

Table A. 2 Summary of background variables in Roma settlements, by age group (%)

Source: Authors' calculations based on MICS6 Roma Settlements 2018.

Backgound characterist.	Dimension	Age group								
		0–23 months			0–59 months			5–17 years		
		1=Yes	0= Otherw.		1=Yes	0= Otherw.		1=Yes	0= Otherw.	
Child is under 3 years old										
Child is under 3 years old					37.07	36.00				
Child is under 3 years old					35.21	71.87	***			
Child is under 3 years old					45.58	21.66	***			
Child is under 3 years old					10.34	9.47				
Child is under 3 years old					44.69	44.69				
Child is under 3 years old					30.74	34.47				
Child is under 3 years old										
Child is under 3 years old										
Female-headed house-hold	Nutrition	68.95	64.76							
Female-headed household	Health	54.46	30.14	***	50.87	34.83	***			
Female-headed household	Protection	13.29	25.34	*	36.32	51.92	***	49.40	49.33	
Female-headed household	Education	63.25	55.50		44.17	34.77	**	7.46	6.88	
Female-headed household	Wat/San	32.41	7.94	***	30.33	7.42	***	10.82	9.71	
Female-headed household	Housing	61.59	42.38	**	54.42	43.47	**	44.88	43.72	
Female-headed household	Utilities	52.50	30.56	***	44.80	30.68	***	42.08	32.49	**
Female-headed household	Information							14.12	19.04	
Female-headed household	Dom. chores							7.60	7.82	
Girl	Nutrition	59.05	70.14	**						
Girl	Health	36.31	29.68		37.78	35.66				
Girl	Protection	22.68	25.24		53.28	47.54	*	48.65	50.03	
Girl	Education	59.52	53.70		36.15	35.55		7.00	6.89	
Girl	Wat/San	15.96	6.02	***	11.05	9.09		9.74	9.94	
Girl	Housing	46.00	43.02		44.48	44.87		40.07	47.70	***
Girl	Utilities	33.89	31.94		29.11	34.92	**	33.83	33.32	
Girl	Information							16.04	20.96	**
Girl	Dom. chores							12.38	3.12	***
Head has less education	Nutrition	81.92	60.66	***						
Head has less education	Health	55.06	26.56	***	43.89	34.62	***			
Head has less education	Protection	24.33	24.04		46.44	51.21		58.00	46.89	***
Head has less education	Education	89.02	47.42	***	73.99	25.24	***	24.39	2.01	***
Head has less education	Wat/San	26.37	6.14	***	27.66	5.09	***	27.62	4.81	***
Head has less education	Housing	89.29	32.16	***	78.18	35.41	***	68.05	37.01	***
Head has less education	Utilities	54.52	26.93	***	48.81	27.67	***	53.29	28.00	***
Head has less education	Information							39.04	12.67	***
Head has less education	Dom. chores							5.51	8.44	
Mother has less education	Nutrition	84.85	60.07	***						
Mother has less education	Health	51.51	27.72	***	38.81	36.05				
Mother has less education	Protection	18.32	25.61		43.18	52.03	**	57.13	47.27	***
Mother has less education	Education	93.20	46.69	***	77.67	24.72	***	28.19	1.31	***

Mother has less education	Wat/San	28.95	5.64	***	30.16	4.63	***	30.13	4.46	***
Mother has less education	Housing	88.93	32.74	***	80.70	35.14	***	69.80	36.97	***
Mother has less education	Utilities	47.75	28.92	***	45.73	28.69	***	51.83	28.73	***
Mother has less education	Information							38.46	13.18	***
Mother has less education	Dom. chores							9.96	7.21	
Rural	Nutrition	61.02	67.27							
Rural	Health	27.79	35.06		28.52	40.51	***			
Rural	Protection	20.63	25.83		46.50	51.93	*	49.02	49.49	
Rural	Education	66.00	51.46	***	45.76	31.07	***	6.24	7.29	
Rural	Wat/San	20.88	5.26	***	17.65	6.32	***	18.07	5.77	***
Rural	Housing	47.42	42.82		46.61	43.78		45.91	42.84	
Rural	Utilities	36.87	30.79		36.32	30.32	**	34.18	33.28	
Rural	Information							27.23	14.16	***
Rural	Dom. chores							8.73	7.33	
South	Nutrition	62.42	65.82							
South	Health	34.15	32.30		27.97	38.75	***			
South	Protection	28.34	23.14		55.56	48.85	*	54.67	48.05	*
South	Education	49.42	57.86		28.54	37.61	**	6.20	7.12	
South	Wat/San	10.61	10.42		7.81	10.52		10.57	9.66	
South	Housing	41.12	45.08		41.65	45.44		35.85	45.79	***
South	Utilities	31.18	33.18		27.86	33.34		35.48	33.12	
South	Information							5.76	21.55	***
South	Dom. chores							6.41	8.12	
Three or more children 5–18	Nutrition	64.41	71.44							
Three or more children 5–18	Health	30.27	51.43	***	35.70	44.25	*			
Three or more children 5–18	Protection	25.74	11.07	**	51.63	38.24	***	47.14	54.24	**
Three or more children 5–18	Education	52.35	87.60	***	31.74	69.27	***	4.16	13.16	***
Three or more children 5–18	Wat/San	8.92	22.63	***	8.71	20.41	***	6.28	17.79	***
Three or more children 5–18	Housing	40.40	75.77	***	41.67	69.42	***	40.44	51.48	***
Three or more children 5–18	Utilities	30.60	50.36	***	30.36	47.81	***	30.71	39.99	***
Three or more children 5–18	Information							16.80	22.25	**
Three or more children 5–18	Dom. chores							7.89	7.56	
Caregiver has functional diff.	Nutrition	35.81	65.38							
Caregiver has functional diff.	Health	100.00	32.20	**	69.51	36.39	*			
Caregiver has functional diff.	Protection	0.00	24.26		63.56	50.08		29.87	49.65	*
Caregiver has functional diff.	Education	78.91	56.15		62.54	35.63		17.49	6.77	*
Caregiver has functional diff.	Wat/San	0.00	10.52		0.00	10.06		18.40	9.70	
Caregiver has functional diff.	Housing	57.82	44.26		66.86	44.53		81.64	43.24	***
Caregiver has functional diff.	Utilities	35.81	32.79		55.88	32.09		59.65	33.15	**
Caregiver has functional diff.	Information							23.72	18.40	
Caregiver has functional diff.	Dom. chores							24.11	7.52	***

Table A.3 Proportion of deprivations and t-tests for mean differences, by dimensions, background characteristics and age group for the general population (%)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Background characteris.	Dimension	Age group								
		0–23 monts			0–59 months			5–17 years		
		1=Yes	0= Otherw.		1=Yes	0= Otherw.		1=Yes	0= Otherw.	
Child is under 3 years old	Nutrition	77.10								
Child is under 3 years old	Health	41.59			43.67	37.20				
Child is under 3 years old	Protection	25.23			50.00	48.55				
Child is under 3 years old	Education	95.79			38.97	45.92	*			
Child is under 3 years old	Wat/San	44.86			84.48	85.43				
Child is under 3 years old	Housing	85.98			72.22	63.51	*			
Child is under 3 years old	Utilities	66.82			65.20	66.52				
Child is under 3 years old	Information									
Child is under 3 years old	Dom. chores									
Female-headed household	Nutrition	89.66	75.14	*						
Female-headed household	Health	41.38	41.62		39.73	40.97				
Female-headed household	Protection	37.93	23.24	*	51.60	45.61		57.83	54.78	
Female-headed household	Education	93.10	96.22		93.52	88.86	*	54.22	41.09	**
Female-headed household	Wat/San	48.28	44.32		44.30	40.40		40.96	36.69	
Female-headed household	Housing	79.31	87.03		85.71	85.33		89.16	85.53	
Female-headed household	Utilities	62.07	67.57		63.01	66.13		75.90	70.03	
Female-headed household	Information							54.22	51.94	
Female-headed household	Dom. chores							20.48	10.34	
Girl	Nutrition	77.45	76.79							
Girl	Health	36.27	46.43		38.60	42.86				
Girl	Protection	20.59	29.46		43.10	49.21		50.85	59.75	*
Girl	Education	96.08	95.54		93.75	89.57		41.45	45.34	
Girl	Wat/San	37.25	51.79	**	52.05	41.18	*	38.03	36.86	
Girl	Housing	86.27	85.71		83.99	86.67		84.62	87.71	
Girl	Utilities	66.67	66.96		65.28	66.00		72.22	69.92	
Girl	Information							50.00	54.66	
Girl	Dom. chores							20.09	4.24	
Head has less education	Nutrition	75.86	77.95							
Head has less education	Health	36.78	44.88		39.35	41.71				
Head has less education	Protection	25.29	25.20		46.52	51.20		56.97	54.43	
Head has less education	Education	97.70	94.49		85.71	90.76		46.06	41.97	
Head has less education	Wat/San	48.28	42.52		85.66	85.03		42.42	34.75	*
Head has less education	Housing	85.06	86.61		58.62	66.54		87.88	85.25	
Head has less education	Utilities	64.37	68.50		65.28	66.00		67.88	72.79	
Head has less education	Information							58.79	48.85	**
Head has less education	Dom. chores							11.52	12.46	
Mother has less education	Nutrition	82.35	72.32	*						
Mother has less education	Health	36.27	46.43		37.72	43.86				
Mother has less education	Protection	28.43	22.32		46.30	50.00		53.64	57.42	
Mother has less education	Education	98.04	93.75		90.19	91.20		52.11	32.54	***
Mother has less education	Wat/San	50.00	40.18		28.57	42.93		39.46	34.93	

Mother has less education	Housing	82.35	89.29		65.07	66.33		86.21	86.12	
Mother has less education	Utilities	66.67	66.96		64.77	66.67		71.65	70.33	
Mother has less education	Information							54.79	49.28	
Mother has less education	Dom. chores							12.26	11.96	
Rural	Nutrition	81.03	75.64							
Rural	Health	24.14	48.08	***	27.08	45.50	***			
Rural	Protection	18.97	27.56		58.90	47.06	*	53.77	55.77	
Rural	Education	96.55	95.51		94.66	86.67	***	44.34	43.13	
Rural	Wat/San	60.34	39.10	***	51.85	36.86	***	45.28	35.16	*
Rural	Housing	89.66	84.62		85.13	85.60		93.40	84.07	**
Rural	Utilities	68.97	66.03		57.14	65.94		75.47	69.78	
Rural	Information							72.64	46.43	***
Rural	Dom. chores							9.43	12.91	**
South	Nutrition	90.48	75.65							
South	Health	19.05	44.04	**	29.31	42.13	*			
South	Protection	23.81	25.39		45.83	49.53		60.71	54.59	
South	Education	90.48	96.37		90.41	90.67		33.93	44.69	
South	Wat/San	47.62	44.56		46.26	38.95	*	42.86	36.71	
South	Housing	80.95	86.53		87.50	84.00		94.64	85.02	*
South	Utilities	57.14	67.88		64.56	67.20		64.29	71.98	
South	Information							55.36	51.93	
South	Dom. chores							7.14	12.80	
Three or more children 5–18	Nutrition	74.62	80.95							
Three or more children 5–18	Health	42.31	40.48		28.57	41.12				
Three or more children 5–18	Protection	26.15	23.81		88.97	92.18		52.27	58.00	
Three or more children 5–18	Education	96.15	95.24		51.72	41.54		45.91	41.20	
Three or more children 5–18	Wat/San	43.85	46.43		90.28	83.65	*	37.27	37.60	
Three or more children 5–18	Housing	84.62	88.10		63.01	66.13		86.36	86.00	
Three or more children 5–18	Utilities	64.62	70.24		64.56	67.20		70.00	72.00	
Three or more children 5–18	Information							53.64	51.20	***
Three or more children 5–18	Dom. chores							13.64	10.80	
Caretaker has functional diff.	Nutrition	75.00	77.14							
Caregiver has functional diff.	Health	50.00	41.43		47.43	49.66				
Caregiver has functional diff.	Protection	25.00	25.24		87.93	90.94		36.36	56.25	*
Caregiver has functional diff.	Education	100.00	95.71		58.33	37.20	***	45.45	43.30	
Caregiver has functional diff.	Wat/San	25.00	45.24		87.67	84.99		45.45	37.05	
Caregiver has functional diff.	Housing	75.00	86.19		64.77	66.67		95.45	85.71	
Caregiver has functional diff.	Utilities	50.00	67.14		57.14	65.94		68.18	71.21	
Caregiver has functional diff.	Information							13.64	54.24	***
Caregiver has functional diff.	Dom. chores							18.18	11.83	

Table A.4 Proportion of deprivations and t-tests for mean differences by dimensions, background characteristics, age group for Roma settlements (%)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Annex B / B1 – Regressions outcomes

Researchers incorporated the background characteristics simultaneously in a logistic regression framework and derived the marginal effects of each predictor for each given dimension. The marginal effects will reflect the estimated change in the probability of being deprived in the specific dimension, for a unit increase in each predictor, ceteris paribus. If the predictor is a dummy variable, e.g., childsex=1 (girl), then the predicted probability of deprivation is calculated with respect to its base category, i.e. childsex=0 (boy).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Nutr.	Hlth.	Protc.	ECD	W&S	Hsng.	Utls.
Sex: Female	-0.125**	0.029	0.008	0.017	0.064*	-0.014	-0.010
	[-2.25]	[0.50]	[0.14]	[0.29]	[1.95]	[-0.26]	[-0.17]
Rural	-0.062	0.002	-0.008	0.084	0.149***	-0.002	0.018
	[-1.01]	[0.04]	[-0.14]	[1.45]	[4.35]	[-0.04]	[0.30]
Age h/h head	0.078	-0.190	-0.026	-0.125	-0.236*	-0.049	-0.067
	[0.33]	[-0.87]	[-0.10]	[-0.53]	[-1.83]	[-0.22]	[-0.28]
Female head	0.003	0.111	-0.175*	0.036	0.175***	0.093	0.209**
	[0.03]	[1.38]	[-1.71]	[0.30]	[3.48]	[0.91]	[2.37]
Mother less educ.	0.230**	-0.050	-0.144	0.393***	0.108***	0.282***	-0.054
	[2.13]	[-0.48]	[-1.30]	[2.96]	[2.94]	[3.69]	[-0.46]
Head less educ.	0.067	0.157*	0.216**	0.137	0.008	0.320***	0.269***
	[0.63]	[1.76]	[2.14]	[1.05]	[0.23]	[4.18]	[2.91]
Household size	0.735	1.508	-3.716	2.542	1.337*	0.900	2.106
	[0.40]	[0.92]	[-1.51]	[1.08]	[1.65]	[0.55]	[1.08]
Three or more children under 18	-0.062	0.054	-0.018	0.125	-0.036	0.100	0.062
	[-0.51]	[0.49]	[-0.11]	[0.96]	[-0.68]	[0.84]	[0.53]
Caretaker has functional difficulties	-0.283			0.020		0.067	0.013
	[-0.84]			[0.06]		[0.27]	[0.07]
Region: South	0.030	0.095	0.001	-0.063	-0.035	-0.010	-0.113
	[0.36]	[1.25]	[0.01]	[-0.70]	[-0.66]	[-0.12]	[-1.46]
Region: Centre	0.096	0.283***	0.008	-0.030	0.030	0.020	-0.281***
	[1.00]	[4.12]	[0.10]	[-0.32]	[0.55]	[0.24]	[-2.82]
Region: North	0.031	-0.159**	-0.178**	0.004	0.046	-0.051	-0.140*
	[0.44]	[-2.10]	[-2.49]	[0.05]	[1.09]	[-0.78]	[-1.94]
Observations	431	427	427	431	427	431	431
r2							

Table B.1 Marginal effects, general population, children aged 0–23 months

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

t statistics in brackets

	(1)	(2)	(3)	(4)	(5)	(6)
	Health	Protect'n	ECD	Wat/San	Housing	Utilities
Sex: Female	0.030	0.063	-0.031	-0.003	-0.025	-0.067*
	[0.71]	[1.45]	[-0.93]	[-0.13]	[-0.65]	[-1.66]
Rural	-0.014	-0.051	0.092***	0.112***	-0.001	0.062
	[-0.33]	[-1.16]	[2.75]	[5.08]	[-0.03]	[1.56]
Age h/h head	-0.141	-0.209	-0.182	-0.091	-0.081	-0.096
	[-0.80]	[-1.18]	[-1.40]	[-1.22]	[-0.52]	[-0.63]
Female head	0.107	-0.161**	0.034	0.163***	0.031	0.137**
	[1.64]	[-2.32]	[0.58]	[5.93]	[0.51]	[2.23]
Mother less educ.	-0.092	-0.076	0.228***	0.103***	0.269***	0.012
	[-1.42]	[-1.10]	[4.79]	[3.70]	[4.75]	[0.20]
Head less educ.	0.058	0.096	0.167***	0.052**	0.200***	0.149***
	[0.98]	[1.44]	[3.46]	[2.01]	[3.60]	[2.74]
H/hold size	-0.667	0.093	3.734***	0.852*	1.919	2.195*
	[-0.45]	[0.06]	[3.20]	[1.79]	[1.63]	[1.85]
Three or more children under 18	0.124	-0.105	-0.011	-0.016	0.022	0.015
	[1.54]	[-1.14]	[-0.16]	[-0.47]	[0.27]	[0.19]
Caregiver has functional difficulties	0.377*	0.229	0.193*		0.219	0.176
	[1.84]	[1.46]	[1.65]		[1.30]	[1.08]
Region: South	-0.066	-0.001	-0.031	-0.067**	-0.014	-0.115**
	[-1.19]	[-0.01]	[-0.59]	[-2.17]	[-0.27]	[-2.25]
Region: Centre	0.249***	-0.122*	0.058	-0.033	0.010	-0.135**
	[4.37]	[-1.73]	[1.14]	[-0.99]	[0.17]	[-1.97]
Region: North	-0.135***	-0.117**	0.004	-0.005	-0.089*	-0.107**
	[-2.75]	[-2.33]	[0.09]	[-0.18]	[-1.90]	[-2.33]
Observations	1138	1138	1138	1129	1138	1138
r2						

Table B.2 Marginal effects, general population, children aged 0–59 months

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

t statistics in brackets

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Educ.	Protec.	Domestic chores	Wat/San	Housing	Utilities.	Information
Sex: Female	0.011	-0.024	0.100***	-0.001	-0.079*	0.015	-0.042
	[0.81]	[-0.48]	[3.22]	[-0.02]	[-1.89]	[0.34]	[-1.25]
Rural	0.001	-0.009	0.026	0.088***	0.032	-0.008	0.102***
	[0.09]	[-0.19]	[1.05]	[3.75]	[0.79]	[-0.18]	[3.15]
Age h/h head	-0.097*	-0.962***	0.024	0.041	-0.198	-0.157	-0.070
	[-1.65]	[-4.82]	[0.23]	[0.36]	[-1.02]	[-0.85]	[-0.44]
Female head	0.007	0.055	0.033	0.019	0.036	0.094	-0.046
	[0.40]	[0.70]	[1.05]	[0.53]	[0.49]	[1.47]	[-0.83]
Mother less educ.	0.085***	-0.028	0.055*	0.103***	0.186***	0.097*	0.121***
	[5.17]	[-0.44]	[1.78]	[3.58]	[3.38]	[1.71]	[2.74]
Head less educ.	0.024	0.077	-0.089**	0.080**	0.158**	0.156**	0.155***
	[1.54]	[1.03]	[-2.21]	[2.45]	[2.51]	[2.36]	[2.84]
H/hold size	1.430***	5.610***	1.183	-0.475	1.990	1.428	-2.139
	[3.36]	[2.80]	[1.46]	[-0.55]	[1.45]	[0.88]	[-1.50]
Three + children under 18	-0.045**	-0.066	-0.049	0.045	-0.004	-0.006	0.018
	[-2.44]	[-1.05]	[-1.34]	[1.20]	[-0.07]	[-0.11]	[0.43]
Caregiver has functional difficulties	0.046*	-0.249	0.125**	0.035	0.353*	0.161	0.037
	[1.92]	[-1.16]	[2.05]	[0.57]	[1.86]	[0.83]	[0.31]
Region: South	0.054**	0.033	-0.015	0.021	-0.105*	0.001	-0.183***
	[2.01]	[0.53]	[-0.44]	[0.52]	[-1.80]	[0.01]	[-2.91]
Region: Centre	0.093***	-0.067	0.018	0.064	0.075	-0.052	0.078
	[3.70]	[-0.82]	[0.47]	[1.31]	[1.05]	[-0.68]	[1.33]
Region: North	0.042*	-0.072	-0.043	0.039	-0.068	-0.031	0.074*
	[1.75]	[-1.34]	[-1.39]	[1.10]	[-1.27]	[-0.60]	[1.68]
Observations	1155	1155	1155	1155	1155	1155	1155

Table B.3 *Marginal effects, general population, children aged 5–17 years*

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

t statistics in brackets

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Nutrition	Health	Protect'n	ECD	Wat/San	Housing	Utilities
Sex: Female	0.021	-0.098	-0.079	0.003	-0.134**	-0.020	0.016
	[0.36]	[-1.51]	[-1.31]	[0.12]	[-2.10]	[-0.42]	[0.25]
Rural	-0.081	-0.156	-0.056	0.025	0.140	-0.026	-0.067
	[-0.79]	[-0.90]	[-0.28]	[1.03]	[0.82]	[-0.32]	[-0.46]
Age h/h head	-0.050	0.068	-0.233	-0.040	-0.325	0.004	0.612**
	[-0.20]	[0.23]	[-0.97]	[-0.30]	[-1.03]	[0.02]	[2.13]
Female head	0.177*	-0.032	0.112	-0.027	0.046	-0.039	-0.061
	[1.72]	[-0.32]	[1.46]	[-0.75]	[0.47]	[-0.68]	[-0.72]
Mother less educ.	0.097	-0.071	0.077	0.042	0.066	-0.087*	0.002
	[1.55]	[-1.03]	[1.24]	[1.26]	[0.93]	[-1.82]	[0.03]
Head less educ.	-0.074	-0.063	-0.032	0.022	-0.013	-0.006	-0.077
	[-1.19]	[-0.90]	[-0.51]	[0.65]	[-0.19]	[-0.12]	[-1.15]
H/hold size	-2.526**	0.721	-0.165	0.533	-1.956	3.276*	-3.818***
	[-2.38]	[0.50]	[-0.13]	[0.73]	[-1.25]	[1.90]	[-2.87]
Three+ child. <18yo	0.175**	-0.013	-0.010	-0.044	0.086	-0.079	0.197**
	[2.23]	[-0.15]	[-0.13]	[-1.15]	[0.89]	[-0.97]	[2.18]
Caregiver has funct. difficulties	-0.090	0.003	-0.010		-0.172	-0.067	-0.237
	[-0.44]	[0.01]	[-0.04]		[-0.60]	[-0.47]	[-1.05]
Region: South	0.274**	-0.196	0.019	-0.058*	-0.029	-0.006	-0.036
	[2.17]	[-1.25]	[0.10]	[-1.82]	[-0.17]	[-0.07]	[-0.24]
Region: North	0.067	-0.024	-0.060		0.076	0.185	0.160
	[0.60]	[-0.13]	[-0.28]		[0.41]	[1.59]	[1.00]
Observations	214	214	214	214	214	214	214
r2							

Table B.4 *Marginal effects, Roma Settlements, children aged 0–23 months*

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

t statistics in brackets

Note: Caregiver functional difficulties and north variables for ECD omitted due to collinearity.

	(1)	(2)	(3)	(4)	(5)	(6)
	Health	Protect'n	ECD	Wat/San	Housing	Utilities
Sex: Female	-0.038	-0.029	-0.033	-0.051	-0.003	-0.005
	[-0.92]	[-0.70]	[-1.35]	[-1.26]	[-0.12]	[-0.13]
Rural	-0.215**	0.007	-0.092**	-0.084	-0.086	-0.087
	[-2.06]	[0.06]	[-2.41]	[-0.79]	[-1.48]	[-0.88]
Age h/h head	0.379**	-0.242	-0.112	-0.222	0.009	0.636***
	[1.99]	[-1.21]	[-1.04]	[-1.13]	[0.06]	[3.32]
Female head	-0.034	0.134**	0.000	0.103	0.052	-0.038
	[-0.54]	[2.13]	[0.01]	[1.63]	[1.04]	[-0.66]
Mother less educ.	-0.036	0.062	0.071***	0.023	-0.052*	-0.034
	[-0.82]	[1.38]	[2.58]	[0.53]	[-1.69]	[-0.83]
Head less educ.	-0.000	-0.073	0.023	0.115***	0.021	-0.025
	[-0.01]	[-1.58]	[0.83]	[2.75]	[0.65]	[-0.58]
H/hold size	-0.914	-0.027	0.617	-0.934	2.688***	-2.885***
	[-0.93]	[-0.03]	[1.11]	[-1.02]	[2.58]	[-3.46]
Three or more children under 18	-0.025	0.055	-0.028	-0.016	-0.088*	0.106**
	[-0.48]	[1.01]	[-0.84]	[-0.29]	[-1.95]	[2.04]
Caregiver has functional difficulties	-0.155	-0.022	-0.001	-0.064	0.039	-0.105
	[-1.14]	[-0.15]	[-0.01]	[-0.44]	[0.43]	[-0.80]
Region: South	-0.030	-0.070	0.065*	0.210**	0.086*	0.012
	[-0.31]	[-0.65]	[1.93]	[2.17]	[1.75]	[0.13]
Region: North	0.066	-0.049	0.215***	0.313***	0.258***	0.301***
	[0.57]	[-0.39]	[3.07]	[2.74]	[3.14]	[2.71]
Observations	566	566	566	566	566	566
r2						

Table B.5 *Marginal effects, Roma settlements, children aged 0–59 months*

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

t statistics in brackets

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Educat'n	Protect'n	Domestic chores	Wat/San	Housing	Utilities	Info.
Sex: Female	-0.044	-0.088**	0.167***	0.014	-0.036	0.021	-0.027
	[-1.00]	[-1.99]	[4.78]	[0.31]	[-1.14]	[0.51]	[-0.63]
Rural	-0.089	-0.001	-0.015	-0.001	-0.053	-0.058	0.094
	[-0.90]	[-0.01]	[-0.24]	[-0.01]	[-0.94]	[-0.60]	[0.95]
Age h/h head	0.264	-0.835***	0.248**	-0.273	0.051	0.049	-0.323*
	[1.34]	[-4.30]	[2.00]	[-1.37]	[0.33]	[0.27]	[-1.65]
Female head	0.143**	0.067	0.083**	0.025	0.039	0.078	-0.003
	[2.40]	[1.11]	[2.41]	[0.42]	[0.84]	[1.33]	[-0.06]
Mother less educ.	0.209***	-0.045	0.018	0.017	-0.013	0.013	-0.011
	[4.53]	[-0.91]	[0.53]	[0.34]	[-0.40]	[0.28]	[-0.23]
Head less educ.	-0.082	0.017	-0.036	0.066	0.017	-0.075	0.089*
	[-1.55]	[0.33]	[-0.98]	[1.32]	[0.47]	[-1.59]	[1.72]
H/hold size	1.489	2.001*	0.824	-0.346	1.369	-1.847*	-3.771***
	[1.34]	[1.69]	[1.42]	[-0.30]	[1.35]	[-1.85]	[-3.48]
Three or more children under 18	-0.111**	0.014	-0.049	-0.003	-0.045	0.059	0.059
	[-2.00]	[0.25]	[-1.37]	[-0.06]	[-1.10]	[1.19]	[1.11]
Caregiver has functional difficulties	0.027	-0.191*	0.028	0.108	0.165	-0.042	-0.426***
	[0.23]	[-1.66]	[0.43]	[1.04]	[1.41]	[-0.46]	[-3.28]
Region: South	-0.004	0.040	-0.051	0.085	0.197***	-0.025	-0.020
	[-0.05]	[0.43]	[-0.77]	[0.83]	[3.13]	[-0.29]	[-0.21]
Region: North	0.158	-0.098	0.018	0.108	0.193***	0.244**	0.282**
	[1.44]	[-0.95]	[0.27]	[0.95]	[2.67]	[2.11]	[2.48]
Observations	470	470	470	470	470	470	470
r2							

Table B.6 *Marginal effects, Roma settlements, children aged 5–17 yrs.*

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

t statistics in brackets

B.2 – Sensitivity analysis

General population							
Cutoff		0–59 months			5–17 years		
		Excluded dimension			Excluded dimension		
		Protection	Housing	Health	Protection	Housing	Health
HC	2+	47%	53%	54%	33%	35%	41%
	3+	23%	20%	25%	17%	12%	15%
	4+	7%	4%	8%	4%	3%	4%
A	2+	54%	49%	52%	53%	49%	50%
	3+	68%	64%	67%	66%	66%	67%
	4+	85%	82%	83%	85%	85%	83%
M^0	2+	25%	26%	28%	18%	17%	20%
	3+	15%	13%	17%	11%	8%	10%
	4+	6%	3%	7%	3%	3%	3%
Roma							
Cutoff		0–59 months			5–17 years		
		Excluded dimension			Excluded dimension		
		ECD	Housing	Utilities	Protection	Housing	Utilities
HC	2+	93%	90%	93%	87%	83%	86%
	3+	77%	64%	84%	66%	56%	61%
	4+	44%	29%	58%	37%	29%	32%
A	2+	68%	62%	75%	67%	62%	64%
	3+	75%	71%	78%	76%	73%	74%
	4+	86%	84%	86%	88%	85%	86%
M^0	2+	64%	56%	69%	58%	52%	55%
	3+	57%	45%	66%	50%	41%	45%
	4+	37%	24%	50%	33%	25%	28%

Tabele B.7 *Sensitivity of HC, A, M^0 to the exclusion of a dimension.*

Annex C

C.1 - Indicators and dimensions of deprivation

Water and sanitation: A child is considered to be deprived if:

- Household access to drinking water is through unimproved sources: unprotected dug well; unprotected spring; rainwater; tanker truck; surface water (river, dam, pond); unimproved bottled water
- Dwelling lacks a shower unit or bathtub
- Toilet facility does not flush to piped sewer system or to a septic tank.

Housing: A child is considered to be deprived based on a combination of the following two indicators.

- Lives in a dwelling suffering from at least one of the following three problems: (i) leaking roof; (ii) damp walls, floors or foundation; (iii) rot in window frames or floor.
- The home is overcrowded; i.e., more people live within a given space than is considered tolerable from a health and safety perspective. For the general population and Roma communities in Montenegro, a child is deprived in this indicator if he/she lives in a household where the ratio between the number of people per bedroom exceeds three. Household size is adjusted by counting children as 0.5 and adults as 1. If a household member does not sleep in the home, he/she is considered "absent," and not counted for purposes of measuring overcrowding.

Utilities: Material deprivation refers to inability to pay utilities. A child is considered as deprived if he/she lives in a household where the adults are facing problems in either:

- Paying the bills
- Keeping the home sufficiently warm.

Information: This dimension aims to capture children's and household's access to information and ability to communicate. Therefore, children are considered to be deprived if:

- No member of the household has a mobile telephone
- The home lacks internet access.

Child protection:

Children are considered to be deprived of protection from *physical punishment* if they experience:

- Being shaken; being spanked, hit or slapped on the bottom with bare hand; being hit on the bottom or elsewhere with belt, brush, stick, etc; being hit or slapped on the hand, arm; being hit or slapped on the face, head or ears; being beaten with force. (Data is available for children aged from 12 months to 14 years old.)

Deprivation in child protection due to *psychological aggression* occurs when:

- A child is either shouted, yelled or screamed at, or is called dumb, lazy or another name. (Data is available for children aged 12 months to 14 years old.)

Deprivation in child protection due to *neglect* occurs when:

- A child has been left alone for more than one hour, or left in the care of another child younger than 10 years old, at least once in the past week. (Data on neglect is available for children 0–59 months of age)

The child protection (neglect/discipline) dimension is constructed combining neglect and violent behaviour (psychological aggression & physical punishment) for the three age groups, as follows:

- Children 0–23 months: Neglect (0-11 months) and violent behaviour (12–23 months)
- Children 0–59 months: Neglect (0-11 months) and violent behaviour (12–59 months).
- Children 5–17 years: Violent behaviour (5-14). Deprivation is set to zero for children age 15-17.

Health: Deprivation in the health domain is calculated using two indicators:

- Immunization hesitancy; meaning that a child has experienced either (a) a delay in acceptance of vaccines; or (b) refusal of vaccines by its parents or caregivers, despite the availability of vaccination services, for reasons such as: vaccines not available, occupied with other tasks, doubts about the vaccines, other reasons. (Data on immunization hesitancy is available for the 0–23 months and 0–59 months age groups.)
- Child has no chosen paediatrician. (Data available for 0-23 months and 0-59 months age groups.)

Nutrition: Data is available only for children aged 0–23 months. The nutrition dimension is the result of a combination of indicators on exclusive breastfeeding (0–5 months) and frequency of infant and young child feeding (IYCF) (6–23 months). Specifically:

– Between 0 and 5 months, a child is considered to be deprived if not exclusively breastfed

– Children between 6–23 months are considered deprived if the IYCF indicator falls short of one or both of the following dietary guidelines:

- *Minimum dietary diversity*: meaning, the child during the past 24 hours received foods from fewer than five of these food groups: 1) grains, roots, and tubers; 2) legumes and nuts; 3) dairy products (milk, yogurt, cheese, infant formula); 4) meat/ meat products; 5) fish, fresh or dried; 6) poultry and liver/ organ meats); 7) eggs; 8) vitamin-A rich fruits and vegetables; 9) leafy vegetables; 10) cereals, bread and pasta; 11) other fruits and vegetables.
- *Minimum acceptable diet*: refers to a child under the age of six months who is not exclusively breast-fed; a child aged 6–8 months (who is currently breastfed) receiving solid or semi-solid food less than twice a day; a child aged 9–23 months (who is currently breast-fed) receiving solid or semi-solid food less than three times a day; non-breast-fed children aged 6–23 months receiving solid, semi-solid or soft foods and milk fewer than four times a day; a child aged 24–59 months receiving fewer than five meals a day.

ECD/education. A child's experiences during the early stages of life lays a critical foundation for the rest of his or her life. The quality of care in children's home environment plays a key role in determining their chances for survival and development.

For the 0–23 months age group deprivation in early child development is the result of a combination of two criteria, based mostly on deprivation in learning material:

– A child aged up to 23 months is identified as deprived in ECD if he/she either

a. has no children's books or picture books to look at, or

b. lacks access to either homemade toys (such as dolls, cars or other toys made at home), or toys from a shop or manufactured toys, or household items (e.g., bowls or pots), or items found outside (e.g., sticks, rocks, leaves).

For children 36–59 months ECD deprivation is defined as the combination of lack of learning materials, and support for learning. Specifically:

– A child aged 36–59 months is identified as deprived in ECD if he/she either:

a. has no children's books or picture books to look at, or

b. no adult has engaged with the child in four or more activities to promote learning and school readiness in the past three days (e.g., read books or looked at picture books; told stories; sang songs, including lullabies; took outside the house; played together; named, counted or drew things for or with the child).

For children 5–17 years-old, deprivation levels for education were measured through a combination of school attendance and school performance.

Specifically, a child aged 5–14 years is considered to be deprived if:

a. he/she was not attending school in the current school year, or

b. was in a classroom two or more grades below the age-appropriate grade.

Children aged 15–17 years are considered as being deprived in education if:

a. they are not participating in employment, education or training, or

b. was in a classroom two or more grades below the age-appropriate grade.

Child labour (domestic chores). The criterion for establishing deprivation levels related to child labour for children aged 5–17 years old was:

– A child is considered as deprived in this dimension if he/she was involved in five or more domestic chores during the week. These activities include: fetching water or collecting firewood for household needs, shopping for the household, cooking, washing dishes or cleaning around the house, washing clothes, caring for children, caring for someone elderly and sick, other household tasks.

C.2 Multidimensional Poverty Indices

For the MODA methodology, the **multidimensional deprivation headcount ratio (*H*)** shows the proportion of children experiencing different numbers of deprivations at the same time.

Once indicators are aggregated into dimensions using the union approach, it becones possible to compute the total number of deprivations each child experiences. The countring of deprivations takes place for each child ***i*** separately to reveal his/her breadth of deprivation, so let ***j = 1,2,... d*** index dimensions[42] and let ***i = 1,2,... n*** identify the children.

$$ND_i = \sum_{j=1}^{d} y_j \qquad (1)$$

where

ND_i = the total number of dimensions ***j*** in which each child i is deprived

y_j = 1 if child i is deprived in the dimension ***j***;

y_j = 0 if child i is not deprived in dimension ***j.***

Next, the total number of deprivations per child was used to identify those children who are multidimensionally deprived, depending on the chosen cut-off point ***k*** (two, three, four or more dimensions). This means that child ***i*** is considered to be multidimensionally deprived if the number of dimensions in which he/she is deprived ***(ND_i)*** is equal to or larger than ***k***. This can be defined as follows:

$$D_k = 1 \text{ if } ND_i \geq k$$
$$D_k = 0 \text{ if } ND_i < k \qquad (2)$$

with ***k*** representing the cut-off point.

Simply put, the **deprivation headcount ratio** (3) is represented as follows:

42 MODA uses the union approach to aggregate indicators into dimensions in order to capture all children who are deprived in any of the chosen indicators. This approach implies that every child who is deprived in at least one indicator in a given dimension will be considered deprived in the said dimension. (de Neubourg et al. 2012).

$$H = \frac{nch_k}{tch} \qquad (3)$$

with

$$nch_k = \sum_{i=1}^{n} D_k \qquad (4)$$

where:

H represents the multidimensional child deprivation headcount ratio according to cut-off point ***k***;

nch_k is the number of children affected by at least ***k*** deprivations;

tch is the total number of children in specified age group;

y_k is the deprivation status of a child ***I***, depending on the cut-off point ***k***.

The **average intensity of multidimensional deprivation *A*** measures the extent of child deprivation among those children identified as multidimensionally deprived. It is defined as the sum of all of a child's deprivations as a share of the sum of all possible deprivations among those deprived in at least ***k*** dimensions. The average intensity of deprivation is represented by the following equation:

$$A = \frac{\sum_1^{nch_k} c_k}{nch_k * d} \qquad (5)$$

Where:

A represents the average intensity of multidimensional deprivation according to the cut-off point ***k*** for the specific age group; ***nch_k*** represents the number of children affected by at least ***k*** deprivations in that age group; ***d*** represents the total number of dimensions considered per child in the relevant age group; ***c_k*** represents the number of deprivations each multidimensionally deprived child ***i*** experiences, with ***$c_k = ND_i * D_k$***.

M^0 represents both the multidimensional child deprivation headcount ratio and the average intensity of deprivations. The multidimensional child deprivation headcount ratio uses the following formula:

$$\mathbf{M0} = \mathrm{H} * A = \frac{\sum_1^{q_k} c_k}{tch * d}$$

Annex D. Overview of qualitative instruments

D.1 Focus group discussions[1]

Target group	Participants	Urban/rural	Approach
Unemployed	Parents/Caregivers of children aged 0–8	Urban	Semi-structured
	Parents/Caregivers of children aged 0–8	Rural	Exploratory
	Parents/Caregivers of children aged 9–17	Urban	Semi-structured
	Parents/Caregivers of children aged 9–17	Rural	Exploratory
	Children, aged 15–17	Urban	Semi-structured
	Children, aged 15–17	Rural	Exploratory
Social assistance beneficiaries	Parents/Caregivers of children aged 0–8	Urban	Exploratory
	Parents/Caregivers of children aged 0–8	Rural	Semi-structured
	Parents/Caregivers of children aged 9–17	Urban	Exploratory
	Parents/Caregivers of children aged 9–17	Rural	Semi-structured
	Children, aged 15–17	Urban	Exploratory
	Children, aged 15–17	Rural	Semi-structured
Extended families[2]	Parents/Caregivers of children aged 0–8	Rural	Semi-structured
	Parents/Caregivers of children aged 0–8	Urban	Exploratory
	Parents/Caregivers of children aged 9–17	Urban	Semi-structured
	Parents/Caregivers of children aged 9–17	Rural	Exploratory
	Children, aged 15–17	Rural	Semi-structured
	Children, aged 15–17	Rural	Semi-structured
Single parent families	Parents/Caregivers of children aged 0–17	Urban	Semi-structured
	Parents/Caregivers of children aged 0–17	Rural	Exploratory
	Children, aged 15–17	Urban	Exploratory
	Children, aged 15–17	Rural	Semi-structured
Roma and Egyptian families	Parents/Caregivers of children aged 6–14 (men)		Semi-structured
	Parents/Caregivers of children aged 6–14 (women)		Semi-structured
	Children, aged 15–17 (male)		Semi-structured
	Children, aged 15–17 (female)		Semi-structured

D.2 Individual in-depth interviews

Target group	Number of interviews
Foster parents	2
Children in foster care	2
Parent of child with a disability	2
Children with a disability	2

D.3 Key informant interviews

Ministry of Labour and Social Welfare

Ministry of Education

Red Cross

Centre for Social Work, Nikšić

Centre for Social Work, Rožaje

Centre for Social Work, Podgorica

NGO Roditelji.me

NGO Centre for Roma Initiatives

Local government, Rožaje

Local government, Nikšić

Local government, Podgorica

NGO Igrackoteka

[1] Each focus group was implemented by two people, a moderator and an assistant. The main task of the moderator was to facilitate the discussion, while the assistant made sure that there were no interruptions and took notes. Focus groups lasted, on average, between 1.5 and 2 hours.